

Delocalization of One-Dimensional Random Band Matrices

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Abstract

Consider an $N \times N$ Hermitian one-dimensional random band matrix with bandwidth $W > N^{1/2+\mathfrak{c}}$ for some fixed $\mathfrak{c} > 0$. In the bulk of the spectrum and in the large N limit, we obtain the following results: (i) The semicircle law holds up to the scale $N^{-1+\varepsilon}$ for any $\varepsilon > 0$. (ii) All L^2 -normalized eigenvectors are delocalized, meaning their L^∞ norms are simultaneously bounded by $N^{-\frac{1}{2}+\varepsilon}$ with overwhelming probability, for any $\varepsilon > 0$. (iii) Quantum unique ergodicity holds in the sense that the local L^2 mass of eigenvectors becomes equidistributed with high probability. (iv) Universality of eigenvalue statistics holds, i.e., the local eigenvalue statistics of these band matrices coincide with those of the Gaussian Unitary Ensemble (GUE).

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1 Introduction

The localization–delocalization transition has been a central question in mathematical physics since Anderson’s seminal work on the tight-binding model, which describes a discrete random Schrödinger operator. Localization for this operator was rigorously established more than four decades ago by Fröhlich and Spencer [43] using a multiscale analysis. Subsequently, Aizenman and Molchanov [4] gave an alternative proof based on the fractional moment method. Since then, many remarkable results on the localization of the Anderson model have been obtained (see, e.g., [44, 12, 13, 25, 24, 46, 53]). Nevertheless, despite decades of intensive research, the existence of delocalized states remains unproved.

A prominent model that interpolates between the random Schrödinger operator and mean-field Wigner matrices is the random band matrix ensemble. In these matrices, the entries H_{ij} become negligible once the distance between the lattice sites i and j exceeds a parameter W , called the *bandwidth*. Here $i, j \in \mathbb{Z}_N^d$ denote lattice sites in d dimensions. As W varies from order one to N , random band matrices interpolate between local random Schrödinger-type models and mean-field Wigner matrices. They therefore provide a natural framework for studying the localization–delocalization transition.

Band matrices may be either real symmetric or complex Hermitian. Although the two symmetry classes are expected to exhibit the same qualitative behavior, the complex Hermitian case is technically simpler because of its more tractable diagrammatic structure. Accordingly, we restrict our attention to complex Hermitian band matrices.

In the one-dimensional case ($d = 1$), it was conjectured [15, 14, 42], and supported by non-rigorous supersymmetric arguments [45], that band matrices undergo a localization–delocalization transition, accompanied by a corresponding transition in the local eigenvalue statistics. More precisely, the conjecture predicts:

- (i) For $W \gg \sqrt{N}$, the bulk eigenvectors are delocalized, and the local eigenvalue statistics are governed by the Gaussian Unitary Ensemble (GUE).
- (ii) For $W \ll \sqrt{N}$, the bulk eigenvectors are localized, and the local eigenvalue statistics converge to those of a Poisson point process.

Similar conjectures hold for edge eigenvalues, with the transition expected at $W = N^{5/6}$. While the edge eigenvalue statistics (but not delocalization of eigenvectors) have been established for several band-matrix models by Sodin [65] using moment methods, the bulk transition at $W = \sqrt{N}$ remains open. Nevertheless, substantial partial progress has been made [5, 23, 16, 40, 35, 33, 47, 11, 10, 9, 74, 57, 55, 65, 59, 62, 64, 60, 63]. Still, the localization–delocalization transition in the bulk of one-dimensional band matrices remains one of the central open problems in random matrix theory.

When the covariance profile of the Gaussian matrix entries has a special form, supersymmetric techniques become available [5, 59, 62, 64, 61, 26]; see [31, 68] for surveys. Using these methods, Disertori, Pinson and Spencer [26] first obtained precise estimates for the density of states in three dimensions. In one dimension, a transition at $W = N^{1/2}$ was established for moments of characteristic polynomials in [59, 62], while the considerably more difficult analysis of two-point functions was carried out in [61]. Partial delocalization results are also available in dimensions $d > 1$. In particular, delocalization and quantum diffusion were established for $d \geq 7$ in [71, 72, 70] using intricate graphical expansion techniques.

In all the works mentioned above, the central object of study is the Green’s function (or resolvent)

$$G(z) = (H - z)^{-1}$$

of the random matrix H . A fundamental reason why the analysis of band matrices is considerably more difficult than that of Wigner matrices can be traced to Ward’s identity. In its simplest form, it reads

$$\sum_j |G_{ij}(z)|^2 = \frac{1}{\eta} \operatorname{Im} G_{ii}(z), \quad \eta = \operatorname{Im} z.$$

This identity provides an optimal ℓ_2 bound (with respect to the index j) on the Green’s function entries. For Wigner matrices, the entries $|G_{ij}(z)|$ are of the same order for all j . Combined with the mean-field nature of the model, Ward’s identity ultimately yields sharp estimates on every entry $G_{ij}(z)$, leading to the celebrated local law for Wigner matrices; see, for example, [39].

For band matrices, however, $G_{ij}(z)$ is expected to be exponentially localized in the distance $|i - j|$, with localization length ℓ_{loc} . Consequently, it is essential to understand the spatial profile of $|G_{ij}(z)|$. Moreover, the ℓ_2 estimate provided by Ward's identity cannot by itself be converted into a sharp pointwise bound on $|G_{ij}(z)|$ without quantitative control of the localization length. To address this issue, the fundamental T -observable was introduced in [32]:

$$T_{xy} \sim \sum_a S_{xa} |G_{ay}|^2, \quad S_{xa} = \mathbb{E} |H_{xa}|^2, \quad (1.1)$$

where H is the band matrix. This observable was analyzed either through the time-independent T -equation introduced in [32] or by graphical expansion methods [71, 72, 70].

We now briefly review the stochastic flow approach for Wigner matrices. Consider the standard complex matrix Brownian motion

$$dH_{t,ij} = \sqrt{S_{ij}} dB_{t,ij}, \quad H_0 = 0,$$

where $B_{t,ij}$ are independent standard complex Brownian motions for $1 \leq i \leq j \leq N$, with $B_{ji} = \bar{B}_{ij}$. Pastur [54] observed that, under this matrix Brownian motion, the normalized trace of the Green's function,

$$\tilde{m}_t(z) := N^{-1} \text{Tr} \tilde{G}_t, \quad \tilde{G}_t := (H_t - z)^{-1},$$

satisfies the complex Burgers equation

$$\partial_t \tilde{m}_t(z) = \tilde{m}_t(z) \partial_z \tilde{m}_t(z).$$

This naturally suggests following the characteristics of the Burgers equation by introducing a time-dependent spectral parameter z_t (Definition 2.7) and considering the Green's function

$$G_t := (H_t - z_t)^{-1},$$

whose dynamics are renormalized at leading order. This idea has appeared in many previous works; see, for example, [28, 52, 1, 2, 8, 48, 49, 50, 69]. Applying this framework to Wigner matrices, Sooster and Warzel [67, 66] gave a new derivation of the local law. This framework was also applied to other mean-field type matrix models in [22, 18, 69].

Under the stochastic flow described above, Itô's formula yields an evolution equation for the T -observable. This equation involves products of more than two Green's functions and therefore is not closed. Nevertheless, Dubova and Yang [28] obtained a sufficient condition for delocalization, formulated in terms of the bandwidth W , by linearizing the stochastic flow of the T -observable. Unfortunately, the resulting condition on W is not optimal.

This naturally leads us to introduce higher-order generalized T -observables, defined, for every n , by

$$T_{x_1, \dots, x_n} = \sum_{a_1, \dots, a_n} \left(\prod_{i=1}^n S_{x_i a_i} \right) \prod_{i=1}^n G_{a_i a_{i+1}}(z_i), \quad a_{n+1} := a_1, \quad z_i \in \{z, \bar{z}\}. \quad (1.2)$$

Throughout this paper, we work with random band matrices possessing a block structure. More precisely, we partition the interval $[[1, N]]$ into L consecutive blocks of size W , so that $N = LW$. The covariance matrix S then admits the decomposition

$$S = S^{(B)} \otimes S_W,$$

where $S^{(B)}$ is an $L \times L$ matrix and S_W is a $W \times W$ matrix. Their entries are given by

$$S_{ab}^{(B)} = \frac{1}{3} (\text{dist}_{\mathbb{Z}_L}(a, b) \leq 1), \quad (S_W)_{\alpha\beta} = W^{-1}, \quad a, b \in \mathbb{Z}_L, \quad \alpha, \beta \in \{1, \dots, W\}.$$

The block structure is introduced only for convenience of exposition and is not essential for our arguments. We shall return to this point later.

With this block structure, the following G -loops are technically more convenient than the generalized T -observables. By definition, the n - G -loop (or simply the n -loop) is

$$\mathcal{L}_{t,\sigma,\mathbf{a}} = \left\langle \prod_{i=1}^n (G_t(\sigma_i) E_{a_i}) \right\rangle, \quad \text{where} \quad \langle A \rangle = \text{Tr} A, \quad (1.3)$$

where E_a is the normalized block projection defined in (2.5), namely, $(E_a)_{ij} = \delta_{ij} \cdot W^{-1} \cdot \mathbf{1}(aW \leq i < aW + W)$, $1 \leq i, j \leq N$ and $G_t(\sigma)$ denotes the Green's function with spectral parameter z_t or $\bar{z}_t$, depending on whether $\sigma = \pm$; see Definition 2.9. We note that traces of products of Green's functions with traceless operator insertions (i.e., replacing E_{a_i} in (1.3) by traceless operators A_i) were already considered in [19, 20].

A direct application of Itô's formula shows that the G -loops satisfy a system of evolution equations, which we call the *loop hierarchy*; see (2.45). The evolution of an n -loop depends on the $(n+1)$ -loop as well as on a martingale term whose quadratic variation involves $2n+2$ loops. The lack of a closed equation at any fixed level is a common feature of many-body systems. It is analogous to the classical BBGKY hierarchy, which governs the evolution of n -point correlation functions. The analysis of such hierarchies typically relies on truncation procedures, but controlling the associated errors in higher-order correlation functions is notoriously difficult. Consequently, rigorous results on the BBGKY hierarchy have remained rare since its introduction more than a century ago.

As mentioned earlier, the local law is already available for mean-field (Wigner-type) matrices and provides optimal estimates for the Green's function entries. Partly based on these estimates, sharp bounds for traces of products of Green's functions with operator insertions were established in [19, 20]. Subsequently, Cipolloni, Erdős, and Henheik [17] studied the evolution of these observables under the Ornstein–Uhlenbeck flow in their analysis of edge spectral statistics for Wigner matrices. Up to suitable reinterpretations, the resulting evolution equations are analogous to the loop hierarchy (2.45). We will discuss the relation between these two hierarchies in more detail at the end of Section 2.4.

For band matrices, the hierarchical nature of the loop hierarchy presents a fundamental challenge. The first step toward analyzing this hierarchy is to identify the correct scale of the n -loops. A naive estimate suggests

$$|\mathcal{L}_{t,\sigma,\mathbf{a}}| \sim (\max_{x \neq y} |G_{xy}(z)|)^n. \quad (1.4)$$

However, their actual size is

$$|\mathcal{L}_{t,\sigma,\mathbf{a}}| \sim (\max_{x \neq y} |G_{xy}(z)|)^{2n-2}. \quad (1.5)$$

To illustrate these estimates, consider the Wigner case with even n and alternating charges,

$$\sigma = (+, -, +, -, \dots, +, -).$$

Let λ_k denote the eigenvalues of H . Then

$$\mathcal{L}_{t,\sigma,\mathbf{a}} = N^{-n} \sum_k \frac{1}{|\lambda_k - z|^n}.$$

Assuming that the eigenvalue density is bounded, the right-hand side is bounded by $C(N\eta)^{-(n-1)}$, where $\eta = \text{Im } z$. On the other hand, the local law for Wigner matrices asserts that

$$\max_{x \neq y} |G_{xy}(z)| \leq C(N\eta)^{-1/2}.$$

Hence the estimate (1.5) follows immediately from matrix multiplication together with the local law. In contrast, by the local law, the naive bound (1.4) gives $(N\eta)^{-n/2}$, which is substantially weaker than (1.5) whenever $n \geq 3$ and $\eta \gg N^{-1}$. The improvement from (1.4) to (1.5) therefore reflects delicate cancellation arising from the summation over the internal indices. For band matrices, it is particularly striking that

the estimate (1.5) continues to hold uniformly in the entire delocalized regime, despite the fact that each summation index in an n -loop ranges over only W sites rather than all N sites.

One approach to analyzing the loop hierarchy for high-dimensional band matrices is based on graphical expansions [71, 70]. The underlying heuristic is that high-dimensional band matrices behave similarly to mean-field models when the dimension d is sufficiently large. In this paper, we avoid graphical expansions entirely. Our starting point is the observation that the loop observables can be accurately approximated by *primitive loops*. These are defined by retaining only the quadratic terms on the right-hand side of the loop hierarchy; the resulting evolution is called the *primitive equation*. We denote the primitive loop of size n by $\mathcal{K}_{t,\sigma,\mathbf{a},n}$. The primitive equation takes the form (see (2.48) for the precise definition)

$$\frac{d}{dt} \mathcal{K}_{t,\sigma,\mathbf{a},n} = W \cdot \sum_{2 \leq j \leq n} \sum_{a,b} \mathcal{K}_{t,\sigma,\mathbf{a},j} S_{ab}^{(B)} \mathcal{K}_{t,\sigma,\mathbf{a},n-j+2}. \quad (1.6)$$

Notice that the right-hand side depends only on primitive loops of size at most n . Consequently, given suitable initial data, the primitive equation admits a unique solution. We also remark that the primitive equation is quadratic in $\mathcal{K}$ and may be viewed as a *multidimensional analogue of the Burgers equation*.

Our analysis of the loop hierarchy and primitive equation is based on three key observations:

- The primitive loop admits an explicit solution.
- The estimate (1.5) for the primitive loops is a consequence of this explicit representation.
- The primitive loops provide remarkably accurate approximations to the G -loops.

A key ingredient in our analysis is the *sum-zero property*, first discovered in the study of high-dimensional band matrices in [71, 70]. In those works, the sum-zero property was established for the self-energy associated with two-point functions, whereas here we extend it to the self-energies of multi-point functions. At its core, this property captures subtle cancellation effects among the self-energy terms. These cancellations originate from the Ward identities (see Lemma 3.6). They play a crucial role in establishing the correct scaling of the primitive loops in our analysis of the primitive equation; see Section 3 for details.

Using the primitive approximation, we will prove delocalization for one-dimensional band matrices whenever $W \gg \sqrt{N}$. In addition, we will establish quantum diffusion, quantum unique ergodicity, and universality of the local eigenvalue statistics. We believe that the primitive loop approximation introduced in this paper can be extended to a much broader class of non-mean-field models, including band matrices with general variance profiles, higher-dimensional band matrices, and random Schrödinger operators with random band potentials. An extension of the n -loop to the n -chain, which will be needed for general variance profiles, will be discussed in Section 2.6. Some of these directions have since been pursued in subsequent works.

Since the first arXiv version of this paper appeared, there has been substantial progress on random band matrices and related non-mean-field random matrix models, including Wegner orbital models and Anderson block models. We briefly summarize these developments for the reader's convenience. For one-dimensional band matrices, Drogin proved localization whenever $W^2 \ll N$ [27]. Combined with the delocalization result proved here, this establishes the localization–delocalization transition at $W = \sqrt{N}$, up to N^ϵ factors. The methods developed in the present paper, together with the primitive loop approximation, were extended to edge eigenvalues in [73]. Assuming $W \geq N^\epsilon$, analogous results for two-dimensional band matrices were established in [30], and for all dimensions $d \geq 3$ in [29]. Furthermore, Erdős and Riabov [36] extended the results of this paper to one-dimensional band matrices with arbitrary single-entry distributions (not necessarily Gaussian) and general variance profile in both the Hermitian and symmetric symmetry classes. Finally, extensions of our results to general power-law random band matrices were obtained in [41].

2 The model and main results

2.1 Band matrix model

In this paper, we focus on the block band-matrix model, which is defined as follows. Although our methods can be extended to a larger class of band matrices, this specific model allows us to avoid many technical

complications that are not central to the main results. The model is described by a complex Hermitian random band matrix H whose entries, subject to the Hermitian condition $H_{ij} = \overline{H_{ji}}$, are independent Gaussian random variables with complex off-diagonal and real diagonal entries:

$$H_{ij} \sim \mathcal{N}_{\mathbb{C}}(0, S_{ij}) \cdot \mathbf{1}_{i \neq j} + \mathcal{N}(0, S_{ij}) \cdot \mathbf{1}_{i=j}, \quad i, j \in \mathbb{Z}_N,$$

where $\mathbb{Z}_N$ denotes the set of integers modulo N , with periodic boundary conditions. The matrix H has block structure, where the block size is given by an integer $W \in \mathbb{N}$, and the number of blocks is denoted by $L \in \mathbb{N}$ with $L \geq 3$. The total matrix size is $N \times N$, with

$$N = W \cdot L.$$

Let $\mathcal{I}_a$ denote the block

$$\mathcal{I}_a := \{aW, aW + 1, \dots, aW + W - 1\}, \quad a \in \mathbb{Z}_L.$$

The elements of the matrix S are defined by

$$S_{ij} = \frac{1}{3W} \sum_a \mathbf{1}(i \in \mathcal{I}_a) \mathbf{1}(j \in \mathcal{I}_a \cup \mathcal{I}_{a+1} \cup \mathcal{I}_{a-1}),$$

where $\mathbf{1}(\cdot)$ denotes the indicator function. Clearly, $S = S^T$, and it satisfies

$$\sum_j S_{ij} = 1.$$

We will follow the convention that indices $a, b, c, \dots$ are elements of $\mathbb{Z}_L$, while $i, j, k, \dots$ represent indices in $\mathbb{Z}_N$. We can express the matrix S as a Kronecker product:

$$S = S^{(B)} \otimes S_W,$$

where $S^{(B)}$ is an $L \times L$ matrix, and S_W is a $W \times W$ matrix. The entries of these matrices are given by

$$S_{ab}^{(B)} = \frac{1}{3} (\text{dist}_{\mathbb{Z}_L}(a, b) \leq 1), \quad (S_W)_{\alpha\beta} = W^{-1}, \quad a, b \in \mathbb{Z}_L, \quad \alpha, \beta \in \{1, \dots, W\}.$$

Let $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$ be the eigenvalues of H . Denote by $(\psi_k)_{k=1}^N$ the corresponding normalized eigenvectors so that

$$H\psi_k = \lambda_k \psi_k, \quad k = 1, 2, \dots, N.$$

It is well known that the empirical spectral measure $\frac{1}{N} \sum_{k=1}^N \delta_{\lambda_k}$ converges almost surely to the Wigner semicircle law with density

$$\rho_{\text{sc}}(x) = \frac{1}{2\pi} \sqrt{(4 - x^2)_+}.$$

Moreover, it is believed that the resolvent of H , i.e., $(H - z)^{-1}$, exhibits quantum diffusion, meaning that

$$\mathbb{E} |(H - z)^{-1}|_{xy}^2 \sim \left(\frac{|m|^2}{1 - |m|^2 S} \right)_{xy},$$

where $m = m_{\text{sc}}(z)$ is the Stieltjes transform of the semicircle density ρ_{sc} , defined by

$$m(z) := m_{\text{sc}}(z) := \int_{\mathbb{R}} \frac{\rho_{\text{sc}}(x)}{x - z} dx.$$

Using the definitions of S , $S^{(B)}$, and S_W , we can express

$$(1 - |m|^2 \cdot S)^{-1} = I + \frac{|m|^2 S^{(B)}}{1 - |m|^2 \cdot S^{(B)}} \otimes S_W.$$

We define the diffusion length at the block level by

$$\ell(z) := \min\left(\eta^{-1/2}, L\right) + 1, \quad \eta = \text{Im}(z). \quad (2.1)$$

In Lemma 2.14, we show that the kernel $(1 - |m|^2 S^{(B)})^{-1}$ decays exponentially on the scale $\ell(z)$:

$$\left(\frac{1}{1 - |m|^2 S^{(B)}}\right)_{ab} = O\left((\eta\ell)^{-1} \cdot e^{-c \frac{|a-b|}{\ell}}\right),$$

where $c > 0$ is a constant.

Stochastic Domination

In this paper, we adopt the convention of stochastic domination introduced in [34]. This framework will be used throughout the analysis to control the behavior of random matrices and their spectral properties.

Definition 2.1 (Stochastic domination and high probability event). (i) *Let*

$$\xi = \left(\xi^{(N)}(u) : N \in \mathbb{N}, u \in U^{(N)}\right), \quad \zeta = \left(\zeta^{(N)}(u) : N \in \mathbb{N}, u \in U^{(N)}\right),$$

be two families of non-negative random variables, where $U^{(N)}$ is a possibly N -dependent parameter set. We say ξ is stochastically dominated by ζ , uniformly in u , if for any fixed (small) $\tau > 0$ and (large) $D > 0$,

$$\mathbb{P}\left(\bigcup_{u \in U^{(N)}} \left\{\xi^{(N)}(u) > N^\tau \zeta^{(N)}(u)\right\}\right) \leq N^{-D}$$

for large enough $N \geq N_0(\tau, D)$, and we will use the notation $\xi \prec \zeta$. If for some complex family ξ we have $|\xi| \prec \zeta$, then we will also write $\xi \prec \zeta$ or $\xi = O_{\prec}(\zeta)$.

(ii) *As a convention, for two deterministic non-negative N -dependent quantities ξ and ζ , we write $\xi \prec \zeta$ if $\xi \leq N^\tau \zeta$ for every $\tau > 0$ and all sufficiently large N .*

(iii) *Let A be a family of random matrices and ζ be a family of non-negative random variables. Then, we use $A = O_{\prec}(\zeta)$ to mean that $\|A\| \prec \zeta$, where $\|\cdot\|$ denotes the operator norm.*

(iv) *We say an event Ξ holds with high probability (w.h.p.) if for any constant $D > 0$, $\mathbb{P}(\Xi) \geq 1 - N^{-D}$ for large enough N . More generally, we say an event Ω holds w.h.p. in Ξ if for any constant $D > 0$, $\mathbb{P}(\Xi \setminus \Omega) \leq N^{-D}$ for large enough N .*

2.2 Main results

The following theorems are our main results on delocalization, local semicircle law, quantum unique ergodicity and quantum diffusion.

Theorem 2.2 (Delocalization). *Fix any constant $\mathfrak{c} > 0$, and suppose that*

$$W \geq N^{1/2+\mathfrak{c}}. \quad (2.2)$$

For the band matrix defined in this subsection, we have the following estimate. For any (small) constants $\kappa, \tau > 0$ and (large) $D > 0$, there exists N_0 such that for all $N \geq N_0$ we have

$$\mathbb{P}\left(\max_k \|\psi_k\|_{\infty}^2 \cdot \mathbf{1}(\lambda_k \in [-2 + \kappa, 2 - \kappa]) \leq N^{-1+\tau}\right) \geq 1 - N^{-D}$$

Theorem 2.3 (Local semicircle law). *Suppose that the assumptions of Theorem 2.2 hold. We denote the Green's function by $G(z) = (H - z)^{-1}$. Then for any fixed small constants $\kappa, \tau > 0$, large $D > 0$, and*

$$z = E + i\eta, \quad |E| \leq 2 - \kappa, \quad 1 \geq \eta \geq N^{-1+\tau},$$

there exists N_0 such that for all $N \geq N_0$ we have the local law (with $\ell = \ell(z)$ defined in (2.1))

$$\mathbb{P} \left(\max_{x,y \in \mathbb{Z}_N} \left| (G(z) - m(z) \cdot I)_{xy} \right| \leq \frac{W^\tau}{(W\ell\eta)^{1/2}} \right) \geq 1 - N^{-D}, \quad \ell = \ell(z) \quad (2.3)$$

and the partial tracial local law

$$\mathbb{P} \left(\max_a \left| W^{-1} \sum_{x \in \mathcal{I}_a} G_{xx}(z) - m(z) \right| \leq \frac{W^\tau}{W\ell\eta} \right) \geq 1 - N^{-D}, \quad \ell = \ell(z). \quad (2.4)$$

The partial tracial local law implies the standard tracial local law

$$\mathbb{P} \left(\left| N^{-1} \text{Tr} G(z) - m(z) \right| \leq \frac{W^\tau}{W\ell\eta} \right) \geq 1 - N^{-D}, \quad \ell = \ell(z).$$

Theorem 2.4 (Quantum diffusion). Denote by E_a the block identity matrix

$$(E_a)_{ij} = \delta_{ij} \cdot W^{-1} \cdot \mathbf{1}(i \in \mathcal{I}_a), \quad 1 \leq i, j \leq N \quad (2.5)$$

Let $G = G(z)$. We denote by $G^\dagger$ the Hermitian adjoint of G ; equivalently,

$$G^\dagger = G(z)^* = G(\bar{z}).$$

Under the assumptions of Theorem 2.3 and the notation $m = m(z)$ we have

$$\mathbb{P} \left(\max_{a,b} \left| \text{Tr} G E_a G^\dagger E_b - W^{-1} \left(\frac{|m|^2}{1 - |m|^2 S^{(B)}} \right)_{ab} \right| \leq \frac{W^\tau}{(W\ell\eta)^2} \right) \geq 1 - N^{-D}, \quad (2.6)$$

$$\mathbb{P} \left(\max_{a,b} \left| \text{Tr} G E_a G E_b - W^{-1} \left(\frac{m^2}{1 - m^2 S^{(B)}} \right)_{ab} \right| \leq \frac{W^\tau}{(W\ell\eta)^2} \right) \geq 1 - N^{-D}, \quad (2.7)$$

and a stronger bound on the expectation value

$$\max_{a,b} \left| \mathbb{E} \text{Tr} G E_a G^\dagger E_b - W^{-1} \left(\frac{|m|^2}{1 - |m|^2 S^{(B)}} \right)_{ab} \right| \leq (W\ell\eta)^{-3} \cdot W^\tau, \quad (2.8)$$

$$\max_{a,b} \left| \mathbb{E} \text{Tr} G E_a G E_b - W^{-1} \left(\frac{m^2}{1 - m^2 S^{(B)}} \right)_{ab} \right| \leq (W\ell\eta)^{-3} \cdot W^\tau. \quad (2.9)$$

Theorem 2.2 is a simple consequence of Theorem 2.3.

Proof of Theorem 2.2. Following a standard delocalization argument, we have

$$|\psi_k(x)|^2 \leq \sum_l \frac{\eta^2 |\psi_l(x)|^2}{(\lambda_k - \lambda_l)^2 + \eta^2} \leq \eta \text{Im} G_{xx}(\lambda_k + \eta i) \quad (2.10)$$

By assumption (2.2), $\eta = N^{-1+\tau}$ with small enough $\tau > 0$ satisfying $\eta \leq (W/N)^2$. With this choice of η , we have $\ell \sim L$. Applying (2.3) and the fact that $\ell \sim L$, we obtain $G - m \ll 1$ and thus $G = O(1)$ with high probability. Substituting it into (2.10), we obtain

$$|\psi_k(x)|^2 \leq C\eta \leq N^{-1+\tau}$$

with high probability. This completes the proof of Theorem 2.2. $\square$

Quantum diffusion immediately implies the following probabilistic version of quantum unique ergodicity (QUE). Historically, QUE was introduced by Rudnick and Sarnak [56] as a deterministic notion in geometric analysis. Its probabilistic counterpart for Wigner matrices was first established in [7]. Since then, several stronger forms of probabilistic QUE for Wigner matrices have been obtained; see, for example, [21, 20, 6, 3, 17].

Theorem 2.5 (Generalized quantum unique ergodicity). *Suppose that the assumptions of Theorem 2.2 hold and*

$$0 < \tau^* < \frac{\mathfrak{c}}{2}, \quad (2.11)$$

where $\mathfrak{c}$ was defined in (2.2). Then generalized quantum unique ergodicity holds with probability tending to one at the polynomial rate stated below. More precisely, for any small constant $\kappa > 0$, there exists a large N_0 such that for all $N \geq N_0$,

$$\max_{E: |E| < 2-\kappa} \max_{a \in \mathbb{Z}_L} \mathbb{P} \left(\max_{\lambda_i, \lambda_j \in \mathcal{J}_E} |N(\psi_i^* (E_a - N^{-1}I) \psi_j)|^2 \geq N^{-\tau^*/6} \right) \leq N^{-\tau^*/6}. \quad (2.12)$$

Here

$$\mathcal{J}_E := \left\{ x : |x - E| \leq N^{-1-\tau^*} (W^2/N)^{1/3} \right\}.$$

Furthermore, for any subset $A \subset \mathbb{Z}_L$,

$$\max_{E: |E| < 2-\kappa} \max_{A \subset \mathbb{Z}_L} \mathbb{P} \left(\max_{\lambda_k \in \mathcal{J}_E} \left| \sum_{a \in A} \sum_{x \in \mathcal{I}_a} |\psi_k(x)|^2 - \frac{|A| \cdot W}{N} \right| > \frac{|A| \cdot W}{N^{1+\tau^*/12}} \right) \leq N^{-\tau^*/6} \quad (2.13)$$

Remark: We briefly comment on the optimality of our main results stated above. The high-probability local laws in Theorem 2.3 and the high-probability quantum diffusion estimates in Theorem 2.4 have the expected optimal dependence on the relevant scales W , ℓ , and η , up to small losses of the form W^τ with τ any positive constant. The expectation bounds (2.8)–(2.9) exploit an additional cancellation, but the error bound may not be optimal. The generalized QUE estimate in Theorem 2.5 is not optimal. Optimal QUE estimates, together with related high-probability eigenvector estimates, were established in a subsequent work [36].

Returning to the proof of Theorem 2.5, we note that its conclusion is a simple consequence of Theorem 2.4.

Proof of Theorem 2.5. We first choose $z = E + \eta i$, $\eta = N^{-1-\tau^*} (W^2/N)^{1/3}$. By definition of $\mathcal{J}_E$, we have

$$\begin{aligned} & \sum_{i,j} \mathbf{1}(\lambda_i, \lambda_j \in \mathcal{J}_E) \cdot |(\psi_i^* (E_a - N^{-1}I) \psi_j)|^2 \\ & \leq C\eta^4 \sum_{i,j} \frac{|(\psi_i^* (E_a - N^{-1}I) \psi_j)|^2}{|\lambda_i - z|^2 |\lambda_j - z|^2} \\ & \leq C\eta^2 \text{Tr} (\text{Im } G(z) (E_a - N^{-1}I) \text{Im } G(z) (E_a - N^{-1}I)) \end{aligned} \quad (2.14)$$

We can bound the last term by

$$\begin{aligned} & \mathbb{E} \text{Tr} (\text{Im } G(z) (E_a - N^{-1}I) \text{Im } G(z) (E_a - N^{-1}I)) \\ & = \frac{1}{L^2} \sum_{b,b'} \mathbb{E} \text{Tr} (\text{Im } G(z) (E_a - E_b) \text{Im } G(z) (E_a - E_{b'})) \\ & \leq C \max_{a,b,a',b'} \left| \mathbb{E} \text{Tr} (\text{Im } G(z) E_a \text{Im } G(z) E_b) - \mathbb{E} \text{Tr} (\text{Im } G(z) E_{a'} \text{Im } G(z) E_{b'}) \right| \end{aligned} \quad (2.15)$$

By (2.11), we have $\eta^{-1/2} \geq L$ and hence $\ell(z) = L$. Then from (2.8) and (2.9) in Theorem 2.4, we have

$$\begin{aligned} & \eta^2 \mathbb{E} \text{Tr} (\text{Im } G(z) (E_a - N^{-1}I) \text{Im } G(z) (E_a - N^{-1}I)) \\ & \leq \eta^2 (W\ell\eta)^{-3} \cdot W^\delta + \eta^2 W^{-1+\delta} \max_{a,b,a',b'} \max_{\xi=|m|^2 \text{ or } m^2} \left| \left(\frac{1}{1-\xi \cdot S^{(B)}} \right)_{ab} - \left(\frac{1}{1-\xi \cdot S^{(B)}} \right)_{a'b'} \right| \\ & \leq W^\delta N^{-2} [N^{-1}\eta^{-1} + N^2\eta^{3/2}W^{-1}] = W^\delta N^{-2} [N^{\tau^*-2\epsilon/3} + N^{-3\tau^*/2}] < N^{-2} N^{-\tau^*/3}, \end{aligned}$$

provided that δ is small enough and (2.11) is satisfied. Here we have used the $\ell(z) = L$,

$$|m(z)|^2 \leq 1 - c\eta, \quad (2.16)$$

and the estimate on $(1 - |m|^2 S^{(B)})^{-1}$ in Lemma 2.14. Inserting the above estimate back to the right hand side of (2.14), we have

$$\mathbb{E} \eta^4 \sum_{i,j} \frac{|N(\psi_i^*(E_a - N^{-1}I)\psi_j)|^2}{|\lambda_i - z|^2 |\lambda_j - z|^2} \leq N^{-\tau^*/3}. \quad (2.17)$$

Therefore with probability $1 - O(N^{-\tau^*/6})$, the right hand side of (2.14) is bounded by $N^{-\tau^*/6}$. Together with (2.14), we obtain (2.12). For (2.13), we only need to replace E_a in (2.14) with $|A|^{-1} \sum_{a \in A} E_a$ and use the same argument. This completes the proof of Theorem 2.5 $\square$

2.3 Universality

Recall that the k -point correlation functions of H are defined by

$$\rho_H^{(k)}(\alpha_1, \alpha_2, \dots, \alpha_k) := \int_{\mathbb{R}^{N-k}} \rho_H^{(N)}(\alpha_1, \alpha_2, \dots, \alpha_N) d\alpha_{k+1} \cdots d\alpha_N,$$

where $\rho_H^{(N)}(\alpha_1, \alpha_2, \dots, \alpha_N)$ is the joint density of all unordered eigenvalues of H .

Theorem 2.6 (Bulk universality). *For any small fixed $\kappa > 0$, suppose that the assumptions of Theorem 2.2 hold. For any fixed $k \in \mathbb{N}$, the k -point correlation functions of H converge to those of the GUE in the following sense. For any $|E| \leq 2 - \kappa$ and smooth test function $\mathcal{O}$ with compact support, we have*

$$\lim_{N \rightarrow \infty} \int_{\mathbb{R}^k} d\alpha \mathcal{O}(\alpha) \left\{ \left(\rho_H^{(k)} - \rho_{\text{GUE}}^{(k)} \right) \left(E + \frac{\alpha}{N} \right) \right\} = 0 \quad (2.18)$$

where we use the notation

$$\left(E + \frac{\alpha}{N} \right) := \left(E + \frac{\alpha_1}{N}, \dots, E + \frac{\alpha_k}{N} \right), \quad \alpha = (\alpha_1, \dots, \alpha_k) \in \mathbb{R}^k.$$

This universality result demonstrates that the local eigenvalue statistics at a fixed energy level for band matrices match those of Wigner matrices. However, the distributions of individual eigenvalues of band matrices differ significantly from those of Wigner matrices. As suggested by Theorem 1 of [58], the fluctuations of bulk eigenvalues in band matrices are of order at least $(WN)^{-1/2}$, whereas the fluctuations for the eigenvalues of Wigner matrices are of order N^{-1} [39].

Proof of Theorem 2.6. We first give the structure of the proof. We apply the standard three-step strategy with three inputs proved above: the local law, delocalization, and the generalized QUE estimate. First, with the local law, the standard short-time Dyson Brownian motion ergodic theorem gives bulk universality for H_{t_*} for $t_* = N^{-1+\tau_*}$ with any $\tau_* > 0$. Second, a Green's function comparison estimate shows that, for a sufficiently small time $t_U = N^{-1+\tau_U}$, the local correlation functions of H_0 and H_{t_U} have the same limit. The bulk universality follows by choosing $t_* = t_U$. The only model-dependent part of the argument is the verification of the comparison estimate (2.23); all other comparison inputs are standard.

Our proof adopts the strategy employed in Section 1.2 of [70], which proved bulk universality of certain high-dimensional band matrices by using the local law (Theorem 2.3 in our setting), delocalization (Theorem 2.2) and QUE estimates (Theorem 2.5). The proof follows the standard three step strategy with the following key components:

- By the local law and the delocalization estimates, the standard short-time Dyson Brownian motion ergodic theorem gives bulk universality for H_{t_*} for $t_* = N^{-1+\tau_*}$ with any $\tau_* > 0$.

- A Green's function comparison estimate shows that, for a sufficiently small time $t_U = N^{-1+\tau_U}$, the local correlation functions of H_0 and H_{t_U} have the same limit. The bulk universality follows by choosing $t_* = t_U$. Notice that this step was relatively trivial in the mean-field setting. For band matrices, the key observation of [70] is that the standard comparison proof in the mean-field setting yields the correlation comparison result for band matrices if the QUE estimates (Theorem 2.5) are applied to the second order terms in the comparison proof. We will give the details of this step in the following formal proof.

We now start the formal proof. Define the matrix Ornstein-Uhlenbeck process H_t as the solution to

$$dH_t = -\frac{1}{2}H_t dt + \frac{1}{\sqrt{N}} dB_t, \quad \text{with } H_0 = H. \quad (2.19)$$

By definition, $H_\infty = H_{GUE}$. We aim to show that

$$\lim_{N \rightarrow \infty} \int_{\mathbb{R}^k} d\alpha \mathcal{O}(\alpha) \left\{ \left(\rho_{H_0}^{(k)} - \rho_{H_\infty}^{(k)} \right) \left(E + \frac{\alpha}{N} \right) \right\} = 0 \quad (2.20)$$

The first external input is the short-time universality theorem for Dyson Brownian motion. Combined with the local law from Theorem 2.3, this theorem implies that, after time $t_* = N^{-1+\tau_*}$, the correlation functions of H_{t_*} converge to those of the GUE in the limit $N \rightarrow \infty$.

Step 1: Using the local semicircle law (Theorem 2.3) as input and applying Theorem 2.2 from [51],¹ we establish the universality of the correlation functions of H_{t_*} at $t_* = N^{-1+\tau_*}$ for any fixed $\tau_* \in (0, 1)$, explicitly,

$$\lim_{N \rightarrow \infty} \int_{\mathbb{R}^k} d\alpha \mathcal{O}(\alpha) \left\{ \left(\rho_{H_{t_*}}^{(k)} - \rho_{H_\infty}^{(k)} \right) \left(E + \frac{\alpha}{N} \right) \right\} = 0, \quad t_* = N^{-1+\tau_*}. \quad (2.21)$$

Here (2.21) follows from [51, Theorem 2.2] together with the local law, the rescaling of the correlation functions by the density of the free convolution, and the fact that the limits of the rescaled GUE correlation functions near the energies E and 0 coincide.

The second input is a Green's function comparison criterion. It reduces the comparison of the correlation functions of H_0 and H_{t_U} to a bound on the evolution of products of the Stieltjes transforms $\text{Im } m_t(z_i)$ at spectral parameters with imaginary parts between $N^{-1-\tau_U}$ and $N^{-1+\tau_U}$.

Step 2: Analogously to Proposition 4.17 in [70], we assert that there exists a constant $c' > 0$ such that the following holds: Fix a sufficiently small parameter $\tau_U > 0$ (where the subscript U stands for universality) and assume $|E| \leq 2 - \kappa$. Define complex parameters $z_i = E_i + i\eta_i$ for $i = 1, \dots, n$, satisfying

$$|E_i - E| \leq CN^{-1}, \quad N^{-1-\tau_U} \leq \eta_i \leq N^{-1+\tau_U}, \quad t_U := N^{-1+\tau_U}. \quad (2.22)$$

Under the conditions of Theorem 2.6, we claim that for any fixed $n \in \mathbb{N}$, there exists $c' > 0$ such that for sufficiently small τ_U ,

$$\left| \mathbb{E} \prod_{i=1}^n \text{Im } m_0(z_i) - \mathbb{E} \prod_{i=1}^n \text{Im } m_{t_U}(z_i) \right| \leq N^{-c' + C_n \tau_U}, \quad t_U = N^{-1+\tau_U}, \quad (2.23)$$

where

$$m_t(z_i) = \frac{1}{N} \text{tr}(H_t - z_i)^{-1}, \quad t \in [0, t_U].$$

This estimate is the only point where the band-matrix structure enters the universality proof. Once (2.23) is available, the standard correlation-function comparison theorem converts it into (2.24). Thus the rest of the proof is devoted to verifying (2.23) along the Ornstein-Uhlenbeck flow for a short time with the local law and QUE estimates as inputs.

¹[51, Theorem 2.2] is stated for the real symmetric case; the complex Hermitian case, asserted in the abstract of [51], follows from the same proof, since the homogenization argument of [51, Section 3.2.1] uses only rigidity and the regularity of the free convolution.

Following the approach in [70], for each fixed $n \in \mathbb{N}$, we choose τ_U to be much smaller than c' . By applying the standard correlation function comparison theorem (Theorem 15.3 in [37]) and Proposition 4.17 from [70], we conclude that for every fixed integer k , there exists a sufficiently small $\tau_U > 0$ such that

$$\lim_{N \rightarrow \infty} \int_{\mathbb{R}^k} d\alpha \mathcal{O}(\alpha) \left\{ \left(\rho_{H_0}^{(k)} - \rho_{H_{t_U}}^{(k)} \right) \left(E + \frac{\alpha}{N} \right) \right\} = 0, \quad t_U = N^{-1+\tau_U}. \quad (2.24)$$

Combining (2.24) and (2.21) (choosing $t_* = t_U$), we obtain (2.20), which yields the universality result (2.18). The proof of the key claim (2.23) is given in the following step 3.

Step 3: By Lemma 4.18 of [70] we have for any fixed $n \in \mathbb{N}$ that

$$\begin{aligned} & \left| \mathbb{E} \prod_{i=1}^n \text{Im } m_0(z_i) - \mathbb{E} \prod_{i=1}^n \text{Im } m_{t_U}(z_i) \right| \\ & \leq C \cdot t_U \cdot \max_{0 \leq t \leq t_U} \mathbb{E} \left[\sum_u L_{1,t}(z_u) \prod_{i \neq u} \text{Im } m_t(z_i) + \sum_{u \neq v} L_{2,t}(z_u, z_v) \prod_{i \neq u, v} \text{Im } m_t(z_i) \right], \end{aligned} \quad (2.25)$$

where $C > 0$ is an absolute constant and

$$\begin{aligned} L_{1,t}(z) &:= \sum_{\mathbf{G}_1, \mathbf{G}_2 \in \{G_t, G_t^*\}} \left| \frac{1}{N} \sum_{a,b} (\mathbf{G}_1^2(z))_{aa} S_{ab}^\circ(\mathbf{G}_2(z))_{bb} \right|, \\ L_{2,t}(z_1, z_2) &:= \sum_{\mathbf{G}_1, \mathbf{G}_2 \in \{G_t, G_t^*\}} \left| \frac{1}{N^2} \sum_{a,b} (\mathbf{G}_1^2(z_1))_{ab} S_{ab}^\circ(\mathbf{G}_2^2(z_2))_{ba} \right|, \\ S_{xy}^\circ &= S_{xy} - N^{-1}. \end{aligned}$$

Similar to [70], we claim that the following weaker versions of Theorems 2.3 and 2.5 hold for H_t defined in (2.19).

1. (Weak version of Theorem 2.3): Under the assumptions of Theorem 2.2, for sufficiently small τ_U (depending on $\mathbf{c}$ in (2.2)), and parameters satisfying

$$t \in [0, t_U], \quad t_U = N^{-1+\tau_U}, \quad z = E + i\eta, \quad |E| < 2 - \kappa, \quad \eta = N^{-1+2\tau_U},$$

we have

$$\max_x |(G_t)_{xx}| \prec 1, \quad G_t := (H_t - z)^{-1}. \quad (2.26)$$

For the purposes of the present proof, it suffices to establish (2.26) for $\eta = N^{-1+4\tau_U}$; we therefore restrict attention to this case.

2. (Weak version of Theorem 2.5): For sufficiently small fixed τ_U (depending on $\mathbf{c}$ in (2.2)), we have

$$\text{Theorem 2.5 holds for } H_t, \text{ with } \tau^* = \mathbf{c}/3, \text{ and } 0 \leq t \leq N^{-1+\tau_U}. \quad (2.27)$$

We remark that the results above remain valid without the extra condition on η . However, this simplified form is sufficient for our universality proof and significantly streamlines the argument.

We postpone the proof of (2.26) and (2.27) to subsection 7.2. Assuming these two results, we complete the proof of Theorem 2.6.

It is well known that, for any fixed t and any bulk energy E ,

$$\eta \leq \tilde{\eta} \implies \eta \text{Im } m_t(E + i\eta) \leq \tilde{\eta} \text{Im } m_t(E + i\tilde{\eta}).$$

By choosing $\tilde{\eta} = N^{-1+2\tau_U}$ and (2.26), we obtain that for z_i 's in (2.22),

$$\max_{0 \leq t \leq t_U} \text{Im } m_t(z_i) \prec N^{2\tau_U} \cdot (N \text{Im } z_i)^{-1} \prec N^{3\tau_U}.$$

Substituting it into (2.25), we obtain that

$$\left| \mathbb{E} \prod_{i=1}^n \text{Im } m_0(z_i) - \mathbb{E} \prod_{i=1}^n \text{Im } m_{t_U}(z_i) \right| \prec N^{-1+C_n \tau_U} \cdot \max_{0 \leq t \leq t_U} \max_{u \neq v} \mathbb{E} [L_{1,t}(z_u) + L_{2,t}(z_u, z_v)]. \quad (2.28)$$

To bound L_1 and L_2 , in the following we will prove that, for $\mathbf{G}_1, \mathbf{G}_2 \in \{G_t, G_t^*\}$, and z_i 's in (2.22)

$$\max_i \max_y \mathbb{E} \left| \sum_x (\mathbf{G}_1^2(z_i))_{xx} S_{xy}^\circ (\mathbf{G}_2(z_i))_{yy} \right| \leq N^{1-\epsilon/36+C\tau_U} \quad (2.29)$$

$$\max_{i \neq j} \max_y \mathbb{E} \left| \sum_x (\mathbf{G}_1^2(z_i))_{xy} S_{xy}^\circ (\mathbf{G}_2^2(z_j))_{yx} \right| \prec N^{2-\epsilon/36+C\tau_U}. \quad (2.30)$$

By Lemma 4.20 of [70] (or use the eigen-decomposition of G)

$$\left| \sum_x (\mathbf{G}_1^2(z))_{xx} S_{xy}^\circ (\mathbf{G}_2(z))_{yy} \right| \prec N^{-2} \sum_{\alpha, \beta} |p_\alpha(z)|^2 \cdot |p_\beta(z)| \cdot |M_{y, \alpha}|, \quad (2.31)$$

where

$$p_\alpha(z) := (\lambda_\alpha - z)^{-1}$$

and

$$M_{y, \alpha} := N \sum_x |u_\alpha(x)|^2 S_{xy}^\circ = N \sum_{a: |a-a_0| \leq 1} \frac{1}{3} u_\alpha^*(E_a - N^{-1}I) u_\alpha, \quad y \in \mathcal{I}_{a_0}.$$

With the local law estimate (2.26) at the scale $N^{-1+2\tau_U}$, and the eigenvector L_∞ bound in (2.10), one can obtain that

$$\max_{\alpha, y} |M_{y, \alpha}| \prec N^{C\tau_U}, \quad \sum_\alpha |p_\alpha(z)|^2 \prec N^{2+C\tau_U}, \quad \sum_\beta |p_\beta(z)| \prec N^{1+C\tau_U} \quad (2.32)$$

On the other hand, $M_{y, \alpha}$ has a sharper bound with probability $1 - o(1)$. For each fixed y , define the bad event $\mathcal{B} = \mathcal{B}_y$ by

$$\mathcal{B} = \mathcal{B}_y := \left\{ \exists \alpha \text{ such that } |\lambda_\alpha - E| \leq N^{-1+\epsilon/6} \quad \text{and} \quad |M_{y, \alpha}| \geq N^{-\epsilon/36} \right\}.$$

Using the QUE estimate of H_t in (2.27) and (2.12) with $\tau^* = \mathbf{c}/3$, we have

$$\mathbb{P}(\mathcal{B}) = O(N^{-\epsilon/18}).$$

Therefore, by (2.32),

$$\mathbb{E} \left(\mathbf{1}_{\mathcal{B}} \cdot \sum_{\alpha, \beta} |p_\alpha(z)|^2 \cdot |p_\beta(z)| \cdot |M_{y, \alpha}| \right) = O(N^{3-\epsilon/18+C\tau_U}).$$

On the good event $\mathcal{B}^c$, we split the sum over α into

$$\sum_\alpha = \sum_\alpha^{\leq} + \sum_\alpha^{>}, \quad \sum_\alpha^{\leq} := \sum_{\alpha: |\lambda_\alpha - E| \leq N^{-1+\epsilon/6}}.$$

Then

$$\mathbf{1}_{\mathcal{B}^c} \cdot \sum_\alpha^{\leq} |p_\alpha(z)|^2 \cdot |M_{y, \alpha}| \prec \mathbf{1}_{\mathcal{B}^c} \cdot N^{-\epsilon/36} \cdot \sum_\alpha^{\leq} |p_\alpha(z)|^2 \leq N^{2-\epsilon/36+C\tau_U} \quad (2.33)$$

$$\mathbf{1}_{\mathcal{B}^c} \cdot \sum_{\alpha}^> |p_{\alpha}(z)|^2 \cdot |M_{y,\alpha}| \prec \mathbf{1}_{\mathcal{B}^c} \cdot N^{C\tau_U} \cdot \sum_{\alpha}^> |p_{\alpha}(z)|^2 \leq N^{2-\epsilon/6+C\tau_U}.$$

Combining them with the last term in (2.32), we obtain

$$\mathbb{E} \left(\mathbf{1}_{\mathcal{B}^c} \cdot \sum_{\alpha,\beta} |p_{\alpha}(z)|^2 \cdot |p_{\beta}(z)| \cdot |M_{y,\alpha}| \right) \prec N^{3-\epsilon/36+C\tau_U},$$

which implies (2.29).

To prove (2.30), we use Lemma 4.20 of [70] to have

$$\left| \sum_x (\mathbf{G}_1^2(z_i))_{xy} S_{xy}^{\circ} (\mathbf{G}_2^2(z_j))_{yx} \right| \prec N^{-2} \sum_{\alpha,\beta} |p_{\alpha}(z_i)|^2 |p_{\beta}(z_j)|^2 |M_{y,\alpha,\beta}|,$$

where

$$M_{y,\alpha,\beta} := N \sum_x u_{\alpha}(x) \overline{u_{\beta}(x)} \cdot S_{xy}^{\circ} = \sum_{a:|a-a_0|\leq 1} \frac{N}{3} u_{\beta}^*(E_a - N^{-1}I) u_{\alpha}, \quad y \in \mathcal{I}_{a_0}.$$

Similar to (2.32), we have

$$|M_{y,\alpha,\beta}| \prec N^{C\tau_U}.$$

For each fixed y , define the bad event

$$\tilde{\mathcal{B}} = \tilde{\mathcal{B}}_y := \left\{ \exists \alpha_1, \alpha_2 \text{ such that } \max_i |\lambda_{\alpha_i} - E| \leq N^{-1+\epsilon/6} \text{ and } |M_{y,\alpha_1,\alpha_2}| \geq N^{-\epsilon/36} \right\}.$$

Again using (2.27) and (2.12) with $\tau^* = \epsilon/3$, we have $\mathbb{P}(\tilde{\mathcal{B}}) = O(N^{-\epsilon/18})$ and

$$\mathbb{E} \left(\mathbf{1}_{\tilde{\mathcal{B}}} \cdot \sum_{\alpha,\beta} |p_{\alpha}(z_i)|^2 |p_{\beta}(z_j)|^2 |M_{y,\alpha,\beta}| \right) = O(N^{4-\epsilon/18+C\tau_U}).$$

On the good event, we have

$$\mathbf{1}_{\tilde{\mathcal{B}}^c} \cdot \sum_{\alpha,\beta} |p_{\alpha}(z_i)|^2 |p_{\beta}(z_j)|^2 |M_{y,\alpha,\beta}| \leq N^{4-\epsilon/36+C\tau_U}.$$

We have thus proved (2.30). Combining it with (2.29) and (2.28) (and the definitions of L_1 and L_2), we obtain (2.23) (with $c' = \epsilon/36$). This completes the proof of Theorem 2.6 under the assumption that (2.27) and (2.26) hold. $\square$

2.4 Stochastic flow and G -loops

In the remainder of Section 2, we introduce the loop hierarchy and primitive loops, which are the main tools used to prove the results stated in the preceding subsections. In this subsection, we state a fundamental estimate for the G -loops that is a key input for Theorems 2.3 and 2.4. To this end, recall the matrix Brownian motion defined by

$$dH_{t,ij} = \sqrt{S_{ij}} dB_{t,ij}, \quad H_0 = 0, \quad (2.34)$$

where $B_{t,ij}$ are independent standard complex Brownian motions for all $1 \leq i \leq j \leq N$ and $B_{ji} = \overline{B_{ij}}$ for all i, j . We will consider the resolvent with a time dependent spectral parameter z_t given by the following definition.

Definition 2.7 (The z_t flow). For fixed $E \in \mathbb{R}$, denote by $m^{(E)} := \lim_{\epsilon \rightarrow 0+} m_{sc}(E + i\epsilon)$. Define the linear flow z_t ($0 \leq t \leq 1$) by

$$z_t^{(E)} = E + (1 - t)m^{(E)}, \quad 0 \leq t \leq 1.$$

The imaginary part of $z_t^{(E)}$ is given by

$$\eta_t = \text{Im } z_t^{(E)} = (1 - t) \text{Im } m^{(E)}. \quad (2.35)$$

Denote the resolvent of H_t at $z_t^{(E)}$ by

$$G_t := G_t^{(E)} := (H_t - z_t^{(E)})^{-1}. \quad (2.36)$$

By Ito's formula, $G_t := G_t^{(E)}$ satisfies the SDE

$$dG_t = -G_t dH_t G_t + G_t \{ \mathcal{S}[G_t] - m^{(E)} \} G_t dt,$$

where $\mathcal{S} : M_N(\mathbb{C}) \rightarrow M_N(\mathbb{C})$ is the linear operator defined by

$$\mathcal{S}[X]_{ij} := \delta_{ij} \sum_{k=1}^N S_{ik} X_{kk}.$$

Notice that G_t depends on E and we will use $G_t^{(E)}$ to emphasize the E dependence. For any spectral parameter z , we are interested in the resolvent $G(z) = (H - z)^{-1}$. This function can be related to $G_t^{(E)}$ by the following lemma.

Lemma 2.8. For any $z \in \mathbb{C}$ with $0 < \text{Im } z \leq 1$ and $|\text{Re } z| \leq 2 - \kappa$ for some $\kappa > 0$, there exists (E, t) satisfying $0 < t < 1$ and $|E| \leq 2 - c\kappa$ for some $c > 0$ such that

$$z = t^{-1/2} \cdot z_t^{(E)}. \quad (2.37)$$

Denote by $m^{(E)}$ the solution of $m(m + E) = -1$ with $\text{Im } m^{(E)} > 0$. Then E and t are explicitly given by

$$E = -2 \frac{\text{Re } m_{sc}(z)}{|m_{sc}(z)|}, \quad t = m_{sc}(z)^2 / (m^{(E)})^2.$$

For z , E , and t satisfying (2.37), we have

$$m_{sc}(z) = t^{1/2} \cdot m^{(E)} \quad (2.38)$$

and

$$G(z) \sim t^{1/2} \cdot G_t^{(E)} \quad (2.39)$$

in the sense that they have the same distribution. Furthermore, there exists $c_\kappa > 0$ (depending only on κ) such that

$$t \geq c_\kappa \quad \text{and} \quad c_\kappa \text{Im } z \leq \text{Im } z_t \leq c_\kappa^{-1} \text{Im } z \quad (2.40)$$

Proof. For (2.37), we wish to solve

$$z = t^{-1/2} [(E + m^{(E)}) - tm^{(E)}] = \frac{E + m^{(E)}}{\sqrt{t}} - \sqrt{t} m^{(E)} = -\frac{1}{\sqrt{t} m^{(E)}} - \sqrt{t} m^{(E)}$$

This equation can be solved if we can find t and m so that $\sqrt{t} m^{(E)} = m_{sc}(z)$. The last equation is solved by $\sqrt{t} = |m_{sc}(z)|$ with E solving $m^{(E)} = m_{sc}(z)/|m_{sc}(z)|$. Explicitly, $E = -2 \frac{\text{Re } m_{sc}(z)}{|m_{sc}(z)|}$. The other properties can be checked directly. $\square$

For the rest of the paper, we will consider only $G_t^{(E)}$. We now define the G loops.

Definition 2.9 (G - Loop). For fixed E and $\sigma \in \{+, -\}$, define

$$G_t^{(E)}(\sigma) := G_{t,\sigma}^{(E)} = \begin{cases} (H_t - z_t)^{-1}, & \sigma = + \\ (H_t - \bar{z}_t)^{-1}, & \sigma = - \end{cases}$$

By definition, $G_{t,+}^{(E)} = (G_{t,-}^{(E)})^\dagger$. For simplicity of notation, we sometimes write $G_t = G_{t,+}^{(E)}$, in agreement with the notation in (2.36). Recall that E_a is defined in (2.5). For

$$\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n), \quad \mathbf{a} = (a_1, a_2, \dots, a_n), \quad \sigma_i \in \{+, -\}, \quad a_i \in \mathbb{Z}_L, \quad 1 \leq i \leq n$$

we define the n - G loop by

$$\mathcal{L}_{t,\sigma,\mathbf{a}} = \left\langle \prod_{i=1}^n (G_t(\sigma_i) E_{a_i}) \right\rangle, \quad \text{where} \quad \langle A \rangle = \text{Tr } A. \quad (2.41)$$

We also denote by

$$m(\sigma) := m^{(E)}(\sigma) = \begin{cases} m^{(E)}, & \sigma = + \\ \overline{m^{(E)}}, & \sigma = - \end{cases}, \quad (2.42)$$

and

$$\tilde{G}_t(\sigma_k) := G_t(\sigma_k) - m(\sigma_k). \quad (2.43)$$

With these notations, we can express the quantity $\text{Tr } G(z) E_a G^\dagger(z) E_b$ in Theorem 2.4 by (we drop the superscript E) $\text{Tr } G(z) E_a G^\dagger(z) E_b \sim t \cdot \text{Tr}(G_{t,+} E_a G_{t,-} E_b) = t \cdot \mathcal{L}_{t,(+,-),(a,b)}$. Here and below, $\sim$ denotes equality in distribution. In order to derive the loop hierarchy, Lemma 2.11, we need the following notations.

Definition 2.10 (Loop, Cut, and Glue). Recall the G loops $\mathcal{L}_{t,\sigma,\mathbf{a}}$ defined in Eq. (2.41) of Definition 2.9. Here we define some basic operators: cutting and gluing for these G loops. Assume that

$$\mathcal{L}_{t,\sigma,\mathbf{a}} = \left\langle \prod_{i=1}^n (G_t(\sigma_i) E_{a_i}) \right\rangle, \quad \sigma_i \in \{+, -\}, \quad 1 \leq i \leq n \quad (2.44)$$

1. For $1 \leq k \leq n$, we define the first cut and glue operator $\mathcal{G}_k^{(a)}$ such that $\mathcal{G}_k^{(a)} \circ \mathcal{L}_{t,\sigma,\mathbf{a}}$ is the G loop obtained by replacing $G_t(\sigma_k)$ by $G_t(\sigma_k) E_a G_t(\sigma_k)$. If we consider $\mathcal{L}_{t,\sigma,\mathbf{a}}$ as a loop, then the operator $\mathcal{G}_k^{(a)}$ cuts the k -th G edge $G_t(\sigma_k)$ and glues the two new ends with E_a . We also view $\mathcal{G}_k^{(a)}$ as an operator on the indices $\sigma, \mathbf{a}$ so that

$$\mathcal{L}_{t,\mathcal{G}_k^{(a)}(\sigma,\mathbf{a})} := \mathcal{G}_k^{(a)} \circ \mathcal{L}_{t,\sigma,\mathbf{a}}.$$

The new loop $\mathcal{G}_k^{(a)} \circ \mathcal{L}_{t,\sigma,\mathbf{a}}$ is one unit longer than the original $\mathcal{L}_{t,\sigma,\mathbf{a}}$. For example, in Figure 1, for $n = 4$:

$$\mathcal{G}_2^{(a)} \circ \mathcal{L}_{t,\sigma,\mathbf{a}} = \mathcal{L}_{t,\sigma',\mathbf{a}'}, \quad \mathcal{G}_2^{(a)} \circ (\sigma, \mathbf{a}) = (\sigma', \mathbf{a}')$$

and

$$\sigma = (\sigma_1, \sigma_2, \sigma_3, \sigma_4), \quad \mathbf{a} = (a_1, a_2, a_3, a_4), \quad \sigma' = (\sigma_1, \sigma_2, \sigma_2, \sigma_3, \sigma_4), \quad \mathbf{a}' = (a_1, a, a_2, a_3, a_4)$$

2. For $1 \leq k < l \leq n$, we define the cut and glue operator $\mathcal{G}_{k,l}^{(a),L}$ as follows: $\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{L}_{t,\sigma,\mathbf{a}}$ (where L stands for "left") is the G loop obtained by cutting the k -th and l -th G edges $G_t(\sigma_k)$ and $G_t(\sigma_l)$ (creating four end points and two "chains"), then gluing the two new ends of the chain that contains E_{a_n} and inserting a new E_a at the gluing point. The length of the new loop will be $k + n - l + 1$. For example in figure 2, for $n = 5$:

$$\mathcal{G}_{3,5}^{(a),L} \circ \mathcal{L}_{t,\sigma,\mathbf{a}} = \mathcal{L}_{t,\sigma',\mathbf{a}'}, \quad \mathcal{G}_{3,5}^{(a),L} \circ (\sigma, \mathbf{a}) = (\sigma', \mathbf{a}')$$

and

$$\sigma = (\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5), \quad \mathbf{a} = (a_1, a_2, a_3, a_4, a_5), \quad \sigma' = (\sigma_1, \sigma_2, \sigma_3, \sigma_5), \quad \mathbf{a}' = (a_1, a_2, a, a_4, a_5)$$

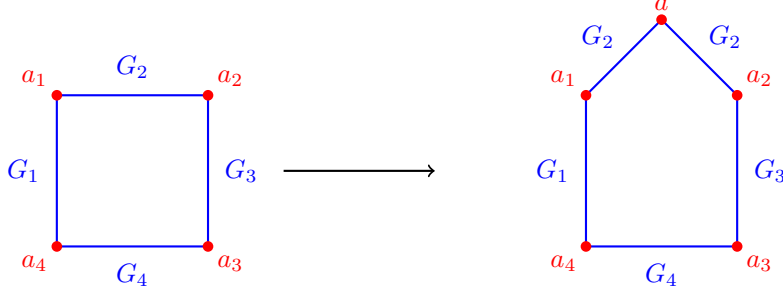


Figure 1: Illustration of operator $\mathcal{G}_k^{(a)}$

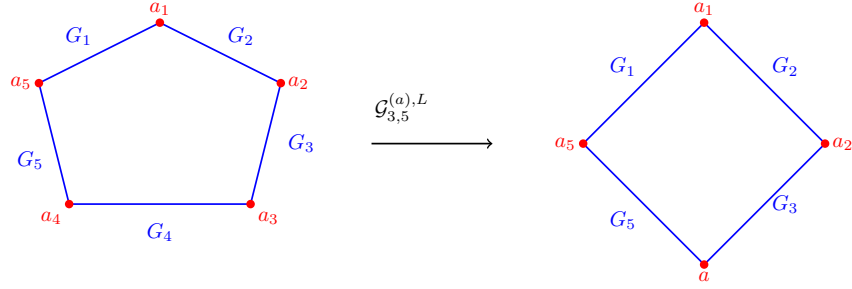


Figure 2: Illustration of operator $\mathcal{G}_{k,l}^{(a),L}$

Notice that a_n is always in $\mathcal{G}_{k,l}^{(a),L}$ for any k, l , and a_1 is in it for any $k \geq 2$.

3. For $1 \leq k < l \leq n$, we define the cut-and-glue operator $\mathcal{G}_{k,l}^{(a),R}$ (where R stands for "right") analogously to $\mathcal{G}_{k,l}^{(a),L}$, except that we glue the two new ends of the chain that does not contain E_{a_n} . The length of the new loop is $l - k + 1$. For example, in Figure 3, for $n = 5$:

$$\mathcal{G}_{3,5}^{(a),R} \circ \mathcal{L}_{t,\sigma,a} = \mathcal{L}_{t,\sigma',a'}, \quad \mathcal{G}_{3,5}^{(a),R} \circ (\sigma, a) = (\sigma', a')$$

and

$$\sigma = (\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5), \quad \mathbf{a} = (a_1, a_2, a_3, a_4, a_5), \quad \sigma' = (\sigma_3, \sigma_4, \sigma_5), \quad \mathbf{a}' = (a_3, a_4, a)$$

Notice that the loop containing the index a_n is the left loop, the other is the right loop.

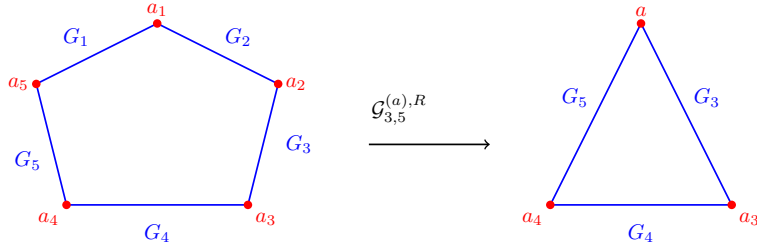


Figure 3: Illustration of operator $\mathcal{G}_{k,l}^{(a),R}$

Denote by $\partial_{(i,j)} := \partial_{H_t(i,j)}$, $1 \leq i \leq j \leq N$. By Itô's formula, we have the following lemma. We will use the convention that $\mathbf{a} = (a_1 \dots a_n)$ is a vector while a will be used as an index independent of $\mathbf{a}$. We will use this convention throughout the paper.

Lemma 2.11 (The loop hierarchy). *The G -loops satisfy the loop hierarchy*

$$d\mathcal{L}_{t,\sigma,a} = \mathcal{E}_{t,\sigma,a}^{(M)} + \mathcal{E}_{t,\sigma,a}^{(\tilde{G})} dt + W \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} \left(\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{L}_{t,\sigma,a} \right) S_{ab}^{(B)} \left(\mathcal{G}_{k,l}^{(b),R} \circ \mathcal{L}_{t,\sigma,a} \right) dt, \quad (2.45)$$

where the martingale term and the $\tilde{G}$ terms are defined by

$$\mathcal{E}_{t,\sigma,a}^{(M)} := \sum_{\alpha=(i,j)} (\partial_\alpha \mathcal{L}_{t,\sigma,a}) \cdot (S_\alpha)^{1/2} \cdot d(\mathcal{B}_t)_\alpha \quad (2.46)$$

$$\mathcal{E}_{t,\sigma,a}^{(\tilde{G})} := W \cdot \sum_{1 \leq k \leq n} \sum_{a,b} \left\langle \tilde{G}_t(\sigma_k) E_a \right\rangle \cdot S_{ab}^{(B)} \cdot \left(\mathcal{G}_k^{(b)} \circ \mathcal{L}_{t,\sigma,a} \right), \quad \tilde{G}_t = G_t - m \quad (2.47)$$

Notice that the factor W arises because E_a contains a factor W^{-1} .

In order to solve this hierarchy, we introduce the primitive loops, denoted by $\mathcal{K}$.

Definition 2.12 (The primitive equation). *For*

$$\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n), \quad \mathbf{a} = (a_1, a_2, \dots, a_n), \quad \sigma_i \in \{+, -\}, \quad a_i \in \mathbb{Z}_L, \quad 1 \leq i \leq n$$

and $m(\sigma)$ defined in (2.42), we define $\mathcal{K}_{t,\sigma,a}$ to be the unique solution to the equation

$$\frac{d}{dt} \mathcal{K}_{t,\sigma,a} = W \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} \left(\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{K}_{t,\sigma,a} \right) S_{ab}^{(B)} \left(\mathcal{G}_{k,l}^{(b),R} \circ \mathcal{K}_{t,\sigma,a} \right) \quad (2.48)$$

with the initial value

$$\mathcal{K}_{0,\sigma,a} = W^{-n+1} \cdot \prod_{k=1}^n m(\sigma_k) \cdot \mathbf{1}(a_1 = a_2 = \dots = a_n).$$

Here we define the $\mathcal{G}$ operator acting on $\mathcal{K}$ in the same way as it acts on $\mathcal{L}$, i.e.,

$$\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{K}_{t,\sigma,a} = \mathcal{K}_{t,\mathcal{G}_{k,l}^{(a),L}(\sigma,a)} \quad \forall t, k, \sigma, \mathbf{a}, \quad (2.49)$$

and similarly for $\mathcal{G}_{k,l}^{(b),R}$. For the special case $n = 1$, we define $\mathcal{K}_{t,+,a} = m$, $\mathcal{K}_{t,-,a} = \overline{m}$ for $0 \leq t \leq 1$.

Notice that $\mathcal{L}_{t,\sigma,\mathbf{a}}$ is an n -loop and $\partial_\alpha \mathcal{L}$ is a product of $n+1$ resolvents. Hence (2.45) is not an equation, but a hierarchy. However, the lengths of $\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{K}_{t,\sigma,\mathbf{a}}$ and $\mathcal{G}_{k,l}^{(a),R} \circ \mathcal{K}_{t,\sigma,\mathbf{a}}$ are no greater than the length of $\mathcal{K}_{t,\sigma,\mathbf{a}}$. Therefore, the system of equations for $\mathcal{K}$ can be solved inductively. In the next section, we will provide an explicit solution to the primitive equation.

Earlier mean-field work considered traces of products of resolvents with operator insertions [19, 20]. The evolution of these traces under a “renormalized” Ornstein–Uhlenbeck process was studied in connection with the edge spectral statistics of Wigner matrices [17]. The resulting flow equations (see (4.5) of [17]) are analogous to the loop hierarchy (2.45). In the mean-field setting, the leading terms of these traces are explicit expressions involving the insertion operators. After a suitable transformation, these leading terms satisfy a quadratic recursive relation analogous to (2.48), with $S^{(B)}$ replaced by a scalar corresponding to the one-block case and the primitive loop on the left-hand side replaced by the corresponding observable with multiple operator insertions.

Conceptually, the works [17, 19, 20] emphasize operator insertions, whereas the band structure enters only through the degenerate one-block setting, represented by a 1×1 covariance matrix. In contrast, the dependence on the band structure and the analysis of the primitive equation are central to the present work. For a more detailed discussion of the mean-field setting, we refer the reader to the original articles, especially [17].

2.5 Propagator $\Theta_\xi^{(B)}$ and examples of $\mathcal{K}$

Definition 2.13 (Propagator $\Theta_\xi^{(B)}$). Define the propagator $\Theta_\xi^{(B)}$ by

$$\Theta_\xi^{(B)} := \frac{1}{1 - \xi \cdot S^{(B)}}, \quad \xi \in \mathbb{C}, \quad |\xi| < 1 \quad (2.50)$$

where the superscript B indicates the block level matrix. Clearly,

$$\partial_\xi \Theta_\xi^{(B)} = \Theta_\xi^{(B)} \cdot S^{(B)} \cdot \Theta_\xi^{(B)}. \quad (2.51)$$

By definition, $\Theta_\xi^{(B)}$ is an $L \times L$ matrix at the block level. We will omit the superscript (B) for the rest of this paper. We remind the reader that Θ is often used to denote the $N \times N$ matrix $(1 - \xi S)^{-1}$ in the literature. In this paper, $\Theta_\xi = \Theta_\xi^{(B)}$. The following three special cases are often used in this paper:

$$\Theta_{tm^2}^{(B)}, \quad \Theta_{t\bar{m}^2}^{(B)}, \quad \text{and} \quad \Theta_{t|m|^2}^{(B)} = \Theta_t^{(B)}.$$

Here we have used $|m| = |m^{(E)}| = 1$ in our setting.

Here we state several properties of $\Theta^{(B)}$, whose proofs are deferred to the appendix.

Lemma 2.14. Suppose that $\xi \in \mathbb{C}$ and $|\xi| < 1$. Define

$$\hat{\ell}(\xi) := \min \left((|1 - \xi|)^{-1/2}, L \right), \quad \xi \in \mathbb{C}$$

Then $\Theta_\xi^{(B)}$ has the following properties:

1. Symmetric: $(\Theta_\xi^{(B)})_{xy} = (\Theta_\xi^{(B)})_{yx}$.
2. Translation invariant: $(\Theta_\xi^{(B)})_{xy} = (\Theta_\xi^{(B)})_{x+1, y+1}$.
3. Commutativity:

$$\forall \xi, \xi' \quad [S^{(B)}, \Theta_\xi^{(B)}] = [\Theta_{\xi'}^{(B)}, \Theta_\xi^{(B)}] = 0$$

4. Exponential decay at length scale $\hat{\ell}_\xi$:

$$\left| (\Theta_\xi^{(B)})_{xy} \right| \leq \frac{C \cdot e^{-c \cdot |x-y| / \hat{\ell}(\xi)}}{|1 - \xi| \cdot \hat{\ell}(\xi)}, \quad \hat{\ell}(\xi) := \min \left((|1 - \xi|)^{-1/2}, L \right) \quad (2.52)$$

5. The following random walk representation of $\Theta_\xi^{(B)}$ converges.

$$\Theta_\xi^{(B)} = \sum_{k=0}^{\infty} \xi^k \cdot \left(S^{(B)} \right)^k$$

6. Derivative bounds. Using the above random-walk representation, we obtain

$$\left| (\Theta_\xi^{(B)})_{x,y} - (\Theta_\xi^{(B)})_{x,y+1} \right| \prec \frac{1}{\hat{\ell}(\xi) \cdot |1 - \xi|^{1/2}} \quad (2.53)$$

$$\left| 2(\Theta_\xi^{(B)})_{x,y} - (\Theta_\xi^{(B)})_{x,y+1} - (\Theta_\xi^{(B)})_{x,y-1} \right| \prec \frac{1}{\|x - y\| + 1}. \quad (2.54)$$

The estimate (2.52) shows that $\Theta_\xi^{(B)}$ behaves very differently for $\xi = t|m|^2 = t$ or $\xi = tm^2$. In the former case, $\Theta_\xi^{(B)}$ decays exponentially on the length scale $(1-t)^{-1/2}$ (assuming $(1-t)^{-1/2} \ll L$), and we have the ℓ_1 bound

$$\sum_b \left| (\Theta_\xi^{(B)})_{ab} \right| \leq C(1-t)^{-1}.$$

In the case $\xi = tm^2$, $(\Theta_\xi^{(B)})_{ab}$ is a short-range kernel with ℓ_1 norm of order one. We refer to $\Theta_\xi^{(B)}$ as a long edge when $\xi = t$, and as a short edge when $\xi = tm^2$ or $\xi = t\bar{m}^2$.

We can use $\Theta_\xi^{(B)}$ to solve the primitive equation $\mathcal{K}_t$ in the special cases $n = 2$ or 3 .

Example 2.15. For $n = 2$, the primitive equation (2.48) can be written as

$$\frac{d}{dt} \mathcal{K}_{t,\sigma,(a_1,a_2)} = W \cdot \sum_{a,b} \mathcal{K}_{t,\sigma,(a_1,a)} \cdot S_{ab}^{(B)} \cdot \mathcal{K}_{t,\sigma,(b,a_2)} \quad (2.55)$$

Denote

$$m_i = m(\sigma_i). \quad (2.56)$$

Using the propagator $\Theta^{(B)}$ defined in equation (2.50) and the property (2.51), one can easily verify that

$$\mathcal{K}_{t,\sigma,\mathbf{a}} = W^{-1} m_1 m_2 \left(\Theta_{t \cdot m_1 m_2}^{(B)} \right)_{a_1 a_2}, \quad \mathbf{a} = (a_1, a_2) \quad (2.57)$$

Explicitly, we have

$$\mathcal{K}_{t,\sigma,(a,b)} = W^{-1} \cdot \begin{cases} |m|^2 \left[(1 - t|m|^2 S^{(B)})^{-1} \right]_{ab} & , \quad \sigma = (+, -) \\ m^2 \left[(1 - tm^2 S^{(B)})^{-1} \right]_{ab} & , \quad \sigma = (+, +) \end{cases} \quad (2.58)$$

Example 2.16. For $n = 3$, let $\sigma = (\sigma_1, \sigma_2, \sigma_3)$, $\mathbf{a} = (a_1, a_2, a_3)$. Then (2.48) becomes

$$\begin{aligned} \frac{d}{dt} \mathcal{K}_{t,\sigma,\mathbf{a}} = & W \cdot \sum_{b_1, c_1} \mathcal{K}_{t,(\sigma_1, \sigma_2), (a_1, b_1)} \cdot S_{b_1 c_1}^{(B)} \cdot \mathcal{K}_{t,\sigma, (c_1, a_2, a_3)} \\ & + W \cdot \sum_{b_2, c_2} \mathcal{K}_{t,(\sigma_2, \sigma_3), (a_2, b_2)} \cdot S_{b_2 c_2}^{(B)} \cdot \mathcal{K}_{t,\sigma, (a_1, c_2, a_3)} \\ & + W \cdot \sum_{b_3, c_3} \mathcal{K}_{t,(\sigma_3, \sigma_1), (a_3, b_3)} \cdot S_{b_3 c_3}^{(B)} \cdot \mathcal{K}_{t,\sigma, (a_1, a_2, c_3)} \end{aligned}$$

Using (2.57), we can rewrite it as

$$\begin{aligned} \frac{d}{dt} \mathcal{K}_{t,\sigma,\mathbf{a}} = & \sum_{c_1} \left(m_1 m_2 \Theta_{t m_1 m_2}^{(B)} \cdot S^{(B)} \right)_{a_1 c_1} \cdot \mathcal{K}_{t,\sigma, (c_1, a_2, a_3)} \\ & + \sum_{c_2} \left(m_2 m_3 \Theta_{t m_2 m_3}^{(B)} \cdot S^{(B)} \right)_{a_2 c_2} \cdot \mathcal{K}_{t,\sigma, (a_1, c_2, a_3)} \\ & + \sum_{c_3} \left(m_3 m_1 \Theta_{t m_3 m_1}^{(B)} \cdot S^{(B)} \right)_{a_3 c_3} \cdot \mathcal{K}_{t,\sigma, (a_1, a_2, c_3)} \end{aligned}$$

Recall $m_i = m(\sigma_i)$ defined in (2.56). Using the propagator $\Theta^{(B)}$ (2.50) and (2.51), one can easily verify that

$$\mathcal{K}_{t,\sigma,\mathbf{a}} = \sum_b \left(\Theta_{t \cdot m_1 m_2}^{(B)} \right)_{a_1 b} \left(\Theta_{t \cdot m_2 m_3}^{(B)} \right)_{a_2 b} \left(\Theta_{t \cdot m_3 m_1}^{(B)} \right)_{a_3 b} \cdot W^{-2} \cdot m_1 m_2 m_3, \quad \mathbf{a} = (a_1, a_2, a_3).$$

Alternatively, it can be expressed as

$$\mathcal{K}_{t,\sigma,\mathbf{a}} = \sum_{b_1 b_2 b_3} \left(\Theta_{t \cdot m_1 m_2}^{(B)} \right)_{a_1 b_1} \left(\Theta_{t \cdot m_2 m_3}^{(B)} \right)_{a_2 b_2} \left(\Theta_{t \cdot m_3 m_1}^{(B)} \right)_{a_3 b_3} \cdot \mathcal{K}_{0,\sigma,\mathbf{b}}, \quad \mathbf{b} = (b_1, b_2, b_3).$$

In the next section, we will prove in Lemma 3.4 that $\mathcal{K}$ can be expressed as a sum over certain tree graphs. This will be called the tree representation of $\mathcal{K}$. The previous explicit solution for $\mathcal{K}$ in the case $n = 3$ will be a special case of this representation. Using the primitive equation and the tree representation, we will establish the following estimate (2.59) for $\mathcal{K}$. This estimate in the special case $n = 3$ follows from the explicit expression of $\mathcal{K}$ and the estimate (2.52). However, starting from $n = 4$, the proof of bound (2.59) requires cancellation among contributions from different graphs in the tree representation of $\mathcal{K}$. To prove Lemma 2.17, we will have to use cancellations among summands in the tree representation. This key ingredient is the sum-zero property, (3.45), which serves as a critical input.

Lemma 2.17 (An upper bound on $\mathcal{K}$). *Recall $\hat{\ell}$ defined in (2.52). The primitive loop $\mathcal{K}_{t,\sigma,a}$ is bounded by*

$$\mathcal{K}_{t,\sigma,a} \prec (W\ell_t \cdot \eta_t)^{-n+1}, \quad \ell_t := \hat{\ell}(t) = \min\left(|1-t|^{-1/2}, L\right) \quad (2.59)$$

Finally, we note that for any fixed $|E| < 2 - \kappa$, it is straightforward to verify that

$$\text{Im } z_t = \text{Im } z_t^{(E)} = \text{Im } m^{(E)} \cdot (1-t) \sim (1-t).$$

Therefore, the $\ell(z)$ defined in (2.1) and ℓ_t defined in (2.59) are of the same order, as follows:

$$\ell(z_t) \sim \ell_t = \hat{\ell}(t).$$

Furthermore, for z , t , and E satisfying the conditions (2.37) and (2.40), we have

$$\text{Im } z \sim \text{Im } z_t^{(E)}, \quad \ell(z) \sim \ell(z_t^{(E)}) \sim \ell_t.$$

Notice that ℓ_t is just a simpler notation for $\hat{\ell}(t)$.

2.6 Estimates on loops

Our main estimates for the G -loops are given in the following lemmas (recall that $\prec$ and ℓ_t are defined in Definition 2.1 and (2.59), respectively).

Lemma 2.18 (loop estimates). *For any small constants $\kappa, \tau > 0$ and $E \in [-2+\kappa, 2-\kappa]$, $0 \leq t \leq 1 - N^{-1+\tau}$, we have*

$$\max_{\sigma,a} |\mathcal{L}_{t,\sigma,a} - \mathcal{K}_{t,\sigma,a}| \prec (W\ell_t \eta_t)^{-n}, \quad \eta_t := \text{Im } z_t, \quad (2.60)$$

$$\max_{\sigma,a} |\mathcal{L}_{t,\sigma,a}| \prec (W\ell_t \eta_t)^{-n+1}. \quad (2.61)$$

Lemma 2.19 (2-loop estimate). *With the notations and assumptions of the previous lemma, the expectation of a 2- G -loop is bounded by*

$$\max_{\sigma,a} |\mathbb{E} \mathcal{L}_{t,\sigma,a} - \mathcal{K}_{t,\sigma,a}| \prec (W\ell_t \eta_t)^{-3}, \quad \eta_t := \text{Im } z_t \quad (2.62)$$

for any choice of $\sigma \in \{+, -\}^2$. In case $\sigma = (+, -)$ and $\mathbf{a} = (a_1, a_2)$, we have

$$|\mathcal{L}_{t,\sigma,a} - \mathcal{K}_{t,\sigma,a}| \prec (W\ell_t \eta_t)^{-2} \exp\left(-\left|\frac{a_1 - a_2}{\ell_t}\right|^{1/2}\right) + W^{-D}. \quad (2.63)$$

This expectation estimate will be used below in the proofs of Theorems 2.3 and 2.4. Together with (2.66) and (2.57), it yields the expectation bounds (2.8) and (2.9) in Theorem 2.4.

Lemma 2.20 (Local law for G_t). *With the notations and assumptions of the previous lemma, we have*

$$\|G_{t,+}^{(E)} - m^{(E)}\|_{\max} \prec (W\ell_t \eta_t)^{-1/2}. \quad (2.64)$$

Proof of Theorems 2.3 and 2.4. For each z in Theorems 2.3 and 2.4, Lemma 2.8 shows that there exist E and t satisfying (2.37) and (2.40), along with the following conditions:

$$(1-t) \sim \text{Im } z_t \sim \text{Im } z \geq N^{-1+\tau}, \quad \eta_t \sim \eta, \quad \ell_t \sim \ell(z).$$

Combining (2.38) and (2.39), we obtain

$$G(z) - m_{sc}(z) \sim t^{1/2} \left(G_t^{(E)} - m^{(E)} \right). \quad (2.65)$$

This relation holds in distribution. Thus, the estimate on $G - m$ in (2.3) of Theorem 2.3 follows from the estimate on $G_t - m$ in (2.64) of Lemma 2.20.

Similarly, (2.4) follows from (2.60) for the 1- G -loop, with the definition of $\mathcal{K}$ in Definition 2.12:

$$\mathcal{K}_{t,+,a} = m^{(E)}, \quad 0 \leq t \leq 1.$$

Analogously to (2.65), for the 2- G terms we have:

$$\text{Tr } G(z) E_a G(z) E_b \sim t \cdot \mathcal{L}_{t,(+,+),(a,b)}, \quad \text{Tr } G(z) E_a G^\dagger(z) E_b \sim t \cdot \mathcal{L}_{t,(+,-),(a,b)}. \quad (2.66)$$

Using (2.57) for rank-2 $\mathcal{K}$, we derive:

$$\begin{aligned} t \cdot \mathcal{K}_{t,(+,+),(a,b)} &= W^{-1} m_{sc}^2(z) \cdot \left(\frac{1}{1 - m_{sc}^2(z) \cdot S^{(B)}} \right)_{ab}, \\ t \cdot \mathcal{K}_{t,(+,-),(a,b)} &= W^{-1} |m_{sc}(z)|^2 \cdot \left(\frac{1}{1 - |m_{sc}(z)|^2 \cdot S^{(B)}} \right)_{ab}. \end{aligned}$$

Therefore, (2.6) and (2.7) in Theorem 2.4 follow from (2.60) in the case $n = 2$, while (2.8) and (2.9) follow from (2.62). This completes the proof of Theorems 2.3 and 2.4. $\square$

We remark that there is a subtle difference between the 2-loop and the T -observable. By definition, the 2-loop is given by

$$\mathcal{L}_{t,(+,-),(a,b)} = \sum_{i,j} (G_{t,+})_{ij} E_a(j) (G_{t,-})_{ji} E_b(i) = \sum_{i,j} |(G_{t,+})_{ij}|^2 E_a(j) E_b(i).$$

Comparing $\mathcal{L}_{t,(+,-),(a,b)}$ with the T -observable (1.1), we find that there are two averages over indices in the loop observable, but only one average in T . Here we neglect the unimportant difference between the S and E_b operators. While an extra averaging might seem to be insignificant, we remind the reader that in the special case of Wigner matrices,

$$N^{-1} \sum_{xy} |G_{x,y}|^2 = (\text{Im } z)^{-1} N^{-1} \sum_x \text{Im } G_{xx}.$$

The averaging in the x index is critical for the local law of Wigner matrices asserting that the fluctuation of $N^{-1} \text{Tr } G$ is one order of magnitude smaller than that of G_{xx} . For similar reasons, our results for 2-loop will not hold for the T observable (1.1).

G -chains. The loop observables discussed above naturally lead to the following G -chains, which form a more general class of observables. For

$$\sigma = (\sigma_1, \dots, \sigma_n) \in \{+, -\}^n, \quad \mathbf{a} = (a_1, \dots, a_{n-1}) \in \mathbb{Z}_L^{n-1},$$

define the G -chain

$$\mathcal{C}_{t,\sigma,\mathbf{a}} := G_t(\sigma_1) E_{a_1} G_t(\sigma_2) E_{a_2} \cdots E_{a_{n-1}} G_t(\sigma_n).$$

The quantity $\mathcal{C}_{t,\sigma,\mathbf{a}}$ is an $N \times N$ matrix, whereas the loop observable $\mathcal{L}_{t,\sigma,\mathbf{a}}$ is a scalar. Under the assumptions of Lemma 2.18, for each fixed $n \in \mathbb{N}$, the diagonal and off-diagonal entries of a G -chain satisfy

$$\max_{\sigma \in \{+,-\}^n} \max_{\mathbf{a} \in \mathbb{Z}_L^{n-1}} \max_i |\mathcal{C}_{t,\sigma,\mathbf{a}}|_{ii} \prec (W \ell_t \eta_t)^{-n+1},$$

$$\max_{\sigma \in \{+, -\}^n} \max_{\mathbf{a} \in \mathbb{Z}_L^{n-1}} \max_{i \neq j} \left| (\mathcal{C}_{t, \sigma, \mathbf{a}})_{ij} \right| \prec (W \ell_t \eta_t)^{-n+1/2}.$$

These estimates, which parallel the loop estimates in Lemma 2.18, are proved in Lemma A.2 of Appendix A.

The exponents in the above estimates are compatible with those in Lemma 2.18 and are expected to be optimal. However, these bounds need not be sharp for every choice of charges and block indices, because additional cancellations or spatial separation may yield stronger estimates in some configurations.

The main results of this paper require only loop observables because of the special block structure $S = S^{(B)} \otimes S_W$. In this setting, the block-averaged loop observables form a closed hierarchy. For a general variance profile, this closure is no longer available: the evolution equations for loops also involve chains with diagonal insertions determined by the variance profile. The hierarchy must then be enlarged to a coupled loop-chain hierarchy, and the chains must be estimated in terms of loop observables. An extension of the above estimates has recently been used for power-law random band matrices with general variance profiles; see [41, Section 9]. This development illustrates how the loop-flow strategy can be adapted to general variance profiles and explains the importance of the chain estimates proved in Appendix A.

2.7 Proof strategy for the main lemmas

We now outline the proofs of Lemmas 2.18, 2.19 and 2.20. By Definitions 2.9 and 2.12,

$$G_0(+) = m \cdot I_{N \times N}, \quad \mathcal{L}_{0, \sigma, \mathbf{a}} = \mathcal{K}_{0, \sigma, \mathbf{a}}, \quad \forall \sigma, \mathbf{a},$$

where $I_{N \times N}$ is the identity matrix. It is easy to check that

$$\text{Lemmas 2.18, 2.19 and 2.20 hold at } t = 0 \text{ with no error.} \quad (2.67)$$

For $t > 0$, we will prove the following theorem.

Theorem 2.21. *Fix E and $s \in [0, 1]$, with $|E| \leq 2 - \kappa$, and assume that Lemmas 2.18, 2.19, and 2.20 hold at time s , namely, for $1 \leq n \in \mathbb{N}$ and large $D > 0$,*

$$\max_{\sigma, \mathbf{a}} |\mathcal{L}_{s, \sigma, \mathbf{a}} - \mathcal{K}_{s, \sigma, \mathbf{a}}| \prec (W \ell_s \eta_s)^{-n} \quad (2.68)$$

$$|\mathcal{L}_{s, \sigma, \mathbf{a}} - \mathcal{K}_{s, \sigma, \mathbf{a}}| \prec (W \ell_s \eta_s)^{-2} \exp \left(- \left| \frac{a_1 - a_2}{\ell_s} \right|^{1/2} \right) + W^{-D}, \quad \sigma = (+, -) \quad (2.69)$$

$$\|G_{s, +}^{(E)} - m^{(E)}\|_{\max} \prec (W \ell_s \eta_s)^{-1/2}, \quad (2.70)$$

$$\max_{\sigma, \mathbf{a}} |\mathbb{E} \mathcal{L}_{s, \sigma, \mathbf{a}} - \mathcal{K}_{s, \sigma, \mathbf{a}}| \prec (W \ell_s \eta_s)^{-3}, \quad \forall \sigma \in \{+, -\}^2. \quad (2.71)$$

Then for any $t > s$ with $t \leq 1 - N^{-1+\tau}$ for some fixed $\tau > 0$, satisfying

$$(W \ell_t \eta_t)^{-1} \leq \left(\frac{1-t}{1-s} \right)^{30} \quad (2.72)$$

we have that (2.68), (2.71), (2.69) and (2.70) hold with s replaced by t .

Proof of Lemmas 2.18, 2.19, and 2.20. For any fixed τ and $t \leq 1 - N^{-1+\tau}$, choose $\tau' > 0$ and $n_0 \in \mathbb{N}$ such that

$$(W \ell_t \eta_t)^{-1} \leq W^{-30\tau'}, \quad (1-t) = W^{-n_0\tau'}$$

Let

$$1 - s_k = W^{-k\tau'}, \quad k \leq n_0, \quad s_{n_0} = t$$

Since $W \ell_s \eta_s$ is decreasing in $s \in [0, 1]$, we have

$$(W \ell_{s_{k+1}} \eta_{s_{k+1}})^{-1} \leq (W \ell_t \eta_t)^{-1} \leq W^{-30\tau'} \leq \left(\frac{1-s_k}{1-s_{k+1}} \right)^{-30}$$

for all k such that $k + 1 \leq n_0$. We can now apply Theorem 2.21 from s_k to s_{k+1} for $k = 0$ until $k = n_0 - 1$ so that the conclusions of Theorem 2.21 hold for $s_{n_0} = t$. We have thus proved Lemmas 2.18, 2.19, and 2.20. Notice that for any τ fixed, n_0 is a finite number depending on τ . Thus we have only finitely many iterations. This is important because every time we apply Theorem 2.21 our inequalities deteriorate by a factor N^ε . At the end, we will have a factor $N^{n_0\varepsilon}$ at the time t . Since ε is arbitrarily small, this factor is still harmless for any τ fixed. $\square$

Theorem 2.21 will be proved in six steps, with their detailed proofs provided in Section 5. Throughout these steps, we assume that the conditions of Theorem 2.21 are satisfied. In addition, each step builds on the conclusions established in the preceding steps.

Step 1 (A priori loop bounds): The n -loop is bounded by

$$\mathcal{L}_{u,\sigma,\mathbf{a}} \prec (\ell_u/\ell_s)^{(n-1)} \cdot (W\ell_u\eta_u)^{-n+1}, \quad s \leq u \leq t. \quad (2.73)$$

Furthermore, the weak local law holds in the sense

$$\|G_u - m\|_{\max} \prec (W\ell_u\eta_u)^{-1/4}, \quad s \leq u \leq t. \quad (2.74)$$

Here the exponent is $1/4$ instead of $1/2$ in (2.70).

Step 2 (A priori 2-loop decay): The following local law holds for $u \in [s, t]$, namely,

$$\|G_u - m\|_{\max} \prec (W\ell_u\eta_u)^{-1/2}, \quad s \leq u \leq t. \quad (2.75)$$

Hence (2.70) in Theorem 2.21 holds. In addition, with $\sigma = (+, -)$, we have for any $s \leq u \leq t$ and $D > 0$ that

$$|\mathcal{L}_{u,\sigma,\mathbf{a}} - \mathcal{K}_{u,\sigma,\mathbf{a}}| \prec (\eta_s/\eta_u)^4 \cdot (W\ell_u\eta_u)^{-2} \exp\left(-\left|\frac{a_1 - a_2}{\ell_u}\right|^{1/2}\right) + W^{-D}. \quad (2.76)$$

Step 3 (Sharp loop bounds): The following sharp estimate on n - G -loop holds:

$$\max_{\sigma,\mathbf{a}} |\mathcal{L}_{u,\sigma,\mathbf{a}}| \prec (W\ell_u\eta_u)^{-n+1}, \quad s \leq u \leq t, \quad \forall n \in \mathbb{N}. \quad (2.77)$$

Step 4 (A sharp $\mathcal{L} - \mathcal{K}$ bound): The following sharp estimate on $\mathcal{L}_t - \mathcal{K}_t$ of length n holds:

$$\max_{\sigma,\mathbf{a}} |\mathcal{L}_{u,\sigma,\mathbf{a}} - \mathcal{K}_{u,\sigma,\mathbf{a}}| \prec (W\ell_u\eta_u)^{-n}, \quad s \leq u \leq t, \quad \forall n \in \mathbb{N} \quad (2.78)$$

This implies (2.68) in Theorem 2.21.

Step 5 (A sharp decay bound): For $\sigma = (+, -)$,

$$|\mathcal{L}_{u,\sigma,\mathbf{a}} - \mathcal{K}_{u,\sigma,\mathbf{a}}| \prec (W\ell_u\eta_u)^{-2} \exp\left(-\left|\frac{a_1 - a_2}{\ell_u}\right|^{1/2}\right) + W^{-D}. \quad (2.79)$$

The last bound implies (2.69) in Theorem 2.21.

Step 6 (A sharp $\mathbb{E}\mathcal{L} - \mathcal{K}$ bound): The following estimate on 2- G -loop with $\sigma \in \{+, -\}^2$ holds:

$$\max_{\sigma,\mathbf{a}} |\mathbb{E}\mathcal{L}_{u,\sigma,\mathbf{a}} - \mathcal{K}_{u,\sigma,\mathbf{a}}| \prec (W\ell_t\eta_t)^{-3}, \quad s \leq u \leq t, \quad \sigma \in \{+, -\}^2. \quad (2.80)$$

We will use Steps 1-5 to prove that (2.68), (2.69), and (2.70) of Theorem 2.21 hold with s replaced by t . Here (2.71) will not be needed for Steps 1-5, i.e., Theorem 2.21 holds if (2.71) were removed from both the assumption and statement.

2.8 Sum zero properties

Recall the loop hierarchy (2.45) of the n -loop is of the form

$$d\mathcal{L}_{t,\sigma,\mathbf{a}} = \mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)} + \mathcal{E}_{t,\sigma,\mathbf{a}}^{(\tilde{G})} + \text{quadratic terms.} \quad (2.81)$$

The first two terms are linear in the loops (assuming $\tilde{G}$ is given) and will be shown to be error terms. The primitive hierarchy drops these two error terms but keeps the quadratic terms. The term $\mathcal{E}_{t,\sigma,\mathbf{a}}^{(\tilde{G})}$ involves an $(n+1)$ -loop and the quadratic variation of $\mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)}$ depends on $2n+2$ loops. The quadratic term, however, involves only loops up to length n . Therefore, we can solve the primitive equation starting from $n = 1, 2, \dots$. This procedure clearly cannot be applied to the loop hierarchy.

It turns out that both $\mathcal{K}$ and $\mathcal{L}$ have similar singularities as $t \rightarrow 1$ in the form

$$\mathcal{L} \sim (1-t)^{-C_n} \sim \mathcal{K}, \quad \text{Im } z_t \sim 1-t.$$

This singularity at $t \rightarrow 1$ is difficult to control. It is a common phenomenon for quadratic differential equations which typically are unstable under perturbation. Since perturbations of quadratic differential equations are governed by a linear one, we consider a toy equation

$$\partial_t f = 2f - c \cdot t + 1, \quad f(0) = a.$$

This equation can be solved explicitly

$$f(t) = e^{2t} \left[a + \frac{1}{2} - \frac{c}{4} \right] + \frac{2ct + c - 2}{4}.$$

If $a + \frac{1}{2} - \frac{c}{4} = 0$ then $f(t) = O(t)$. The subtle condition

$$a + \frac{1}{2} - \frac{c}{4} = 0$$

changes the exponential growth of f to a linear growth! Without explicit solutions, it is not easy to prove the sub-exponential bound of the last toy equation. In our setting, $\mathcal{K}$ -loops can be solved by an explicit *tree representation formula* (Lemma 3.4) and the previous subtle condition will be implemented by a *sum-zero property* of the $\mathcal{K}$ -loops. We will show that both $\mathcal{L}$ and $\mathcal{K}$ -loops satisfy Ward's identity (Lemma 3.6). From these Ward's identities, we will prove a sum-zero property for the $\mathcal{K}$ -loops.

2.9 Notations

Here we summarize notation used throughout the paper.

- W is the bandwidth, L is the number of blocks, $N = W \times L$
- $\mathcal{I}_a$ is the a -th block, $[i]$ is the block where index i is. E_a is the following matrix only supported on $\mathcal{I}_a$.

$$i \in \mathcal{I}_{[i]}, \quad (E_a)_{ij} = W^{-1} \delta_{ij} \mathbf{1}(i \in \mathcal{I}_a)$$

- $S \in \mathbb{R}^{N \times N}$, $S^{(B)} \in \mathbb{R}^{L \times L}$, $S_W \in \mathbb{R}^{W \times W}$ are all related to the variances of the matrix entries, and

$$S = S^{(B)} \otimes S_W$$

- In the proof involving the stochastic flow, we typically omit the superscript (E) for simplicity. As a result, the notations z_t , G_t , and m are defined as follows:

$$z_t = z_t^{(E)}, \quad G_t := (\sqrt{t}H - z_t)^{-1}, \quad m = \lim_{\varepsilon \searrow 0} m_{sc}(E + i\varepsilon).$$

Additionally, we use G_t to denote $(H_t - z_t)^{-1}$, where H_t has the same distribution as $\sqrt{t}H$. The context will make it clear which interpretation of G_t is being applied in the proof.

- Associated with z_t , let $\eta_t = \text{Im } z_t$, and define ℓ_t by (2.59) as

$$\ell_t := \hat{\ell}(t) = \min(|1 - t|^{-1/2}, L)$$

Here $\ell_t \sim \ell(z_t) \sim \ell(z)$ is the decay length.

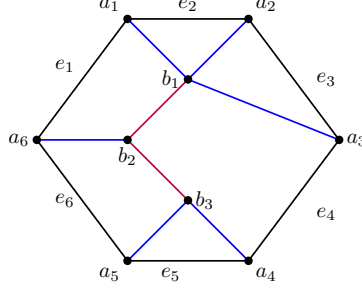
- $\mathcal{L}$ is for G -loop, $\mathcal{K}$ is for the deterministic partner of G -loop, and $\mathcal{C}$ is for G -chain.
- The propagator $\Theta_\xi = \Theta_\xi^{(B)}$ is defined in Definition 2.13. It is used to explicitly define the solution of $\mathcal{K}$ in Definition 3.3.
- The $\Xi^{(\mathcal{L})}$, $\Xi^{(\mathcal{L}-\mathcal{K})}$, $\Xi^{(\mathcal{C})}$ are ratios between these quantities and their heuristic size.

$$\begin{aligned} \Xi_{t,m}^{(\mathcal{L})} &:= \max_{\sigma, \mathbf{a}} |\mathcal{L}_{t,\sigma,\mathbf{a}}| \cdot (W \ell_t \eta_t)^{m-1} \cdot \mathbf{1}(\sigma \in \{+, -\}^m), \\ \Xi_{t,m}^{(\mathcal{L}-\mathcal{K})} &:= \max_{\sigma, \mathbf{a}} |(\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}}| \cdot (W \ell_t \eta_t)^m \cdot \mathbf{1}(\sigma \in \{+, -\}^m), \\ \Xi_{t,m}^{(\mathcal{C}, \text{diag})} &:= \max_{\sigma, \mathbf{a}} \max_i \left| \left(\mathcal{C}_{t,\sigma,\mathbf{a}}^{(m)} \right)_{ii} \right| \cdot (W \ell_t \eta_t)^{m-1} \cdot \mathbf{1}(\sigma \in \{+, -\}^m), \\ \Xi_{t,m}^{(\mathcal{C}, \text{off})} &:= \max_{\sigma, \mathbf{a}} \max_{i \neq j} \left| \left(\mathcal{C}_{t,\sigma,\mathbf{a}}^{(m)} \right)_{ij} \right| \cdot (W \ell_t \eta_t)^{m-1/2} \cdot \mathbf{1}(\sigma \in \{+, -\}^m). \end{aligned} \quad (2.82)$$

- The operators $\Theta_{t,\sigma}$ and $\mathcal{U}_{s,t,\sigma}$ are defined in Def. 5.2 as the linear operators for the integrated loop hierarchy for $\mathcal{L} - \mathcal{K}$ in Lemma 5.3.
- The $\mathcal{G}$ operators, such as $\mathcal{G}_k^{(a)}$, $\mathcal{G}_{k,l}^{(a),L}$, and $\mathcal{G}_{k,l}^{(a),R}$, are defined in Definition 2.10. These operators represent the cutting and gluing of Loop operators within the loop hierarchy.
- $\Gamma_{t,\sigma,\mathbf{a}}$ represents the tree graph used in the tree representation of $\mathcal{K}$.
- The sets $\mathcal{F}(\Gamma)$ and $\mathcal{F}_{\text{long}}(\Gamma)$ correspond to the non-neighboring internal edges and the neighboring long internal edges of Γ , respectively.
- $\mathcal{K}^{(\pi)}$ is defined in Definition 3.9 as the sum of certain tree graphs.
- $\Sigma^{(\pi)}$ is further defined as the self-energy of $\mathcal{K}^{(\pi)}$ in Definition 3.9. Certain specific $\Sigma^{(\pi)}$ exhibit the sum-zero property, as demonstrated in Lemma 3.10.
- The $\mathcal{E}$ terms represent the non-leading terms that arise in the (integrated) loop hierarchy (2.45) and (5.20). Specifically, $\mathcal{E}^{(M)}$ and $\mathcal{E}^{(\tilde{G})}$ are defined in (2.46) and (2.47), respectively. The term $\mathcal{E}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))}$ is defined in (5.13). Additionally, $(\mathcal{E} \otimes \mathcal{E})$ and $(\mathcal{E} \otimes \mathcal{E})^{(k)}$ are defined in Definition 5.4.
- The $\mathcal{T}_t^{(\mathcal{L}-\mathcal{K})}$ is defined as the tail function of $\mathcal{L} - \mathcal{K}$ in (5.26). The $\mathcal{T}_{t,D}$ represents the deterministic rough tail function, as defined in (5.27).
- The terms $\mathcal{J}_{u,D}$ and $\mathcal{J}^*$ are introduced in (5.28) and (5.29), respectively, to describe the ratio between $\mathcal{T}_t^{(\mathcal{L}-\mathcal{K})}$ and $\mathcal{T}_{t,D}$.
- The scale ℓ_t^* is defined as $\ell_t^* = (\log W)^{3/2} \ell_t$. In this scale Θ_t is exponentially small, while $\mathcal{T}_t$ is not.
- The operators $\mathcal{P}$ and $\mathcal{Q}_t$, along with the function ϑ , are used to define and construct a sum-zero tensor. Their definitions can be found in Definition 5.12.

3 Definition and properties of $\mathcal{K}$

The primitive loop $\mathcal{K}$ has an exact formula in terms of summation over tree graphs which we now present.



$$\mathcal{V}(\Gamma) = \{a_1, a_2, a_3, a_4, a_5, a_6, b_1, b_2, b_3\}$$

$$\mathcal{E}(\Gamma) = \{\{a_1, b_1\}, \{a_2, b_1\}, \{a_3, b_1\}, \{a_4, b_3\}, \{a_5, b_3\}, \{a_6, b_2\}, \{b_1, b_2\}, \{b_2, b_3\}, \}$$

$$\mathcal{F}(\Gamma) = \{\{1, 4\}, \{4, 6\}\}$$

Figure 4: Illustration of canonical partition of the polygons

3.1 Tree representation of $\mathcal{K}_{t,\sigma,a}$

Definition 3.1 (Canonical partition of polygon). Let $\mathcal{P}_a$ be an oriented n -gon with ordered vertices $\mathbf{a} = (a_1, \dots, a_n)$, where a_k and a_{k+1} are adjacent. We use the cyclic convention $a_0 = a_n$. The edge $\overline{a_{k-1}a_k}$ is called the k -th edge of $\mathcal{P}_a$. In particular, $\mathcal{P}_{(a,b,c)} \neq \mathcal{P}_{(b,c,a)}$, because the second edges of these polygons are $\overline{ab}$ and $\overline{bc}$, respectively. We now define three notions related to partitions of an oriented polygon.

I. Canonical (or standard) partitions. A partition of $\mathcal{P}_a$ is canonical if and only if

- Each sub-region in the partition is also a polygon.
- There is a one-to-one correspondence between the boundary edges of the polygon and its subregions. Each edge $e_k := \overline{a_{k-1}a_k}$ belongs to exactly one subregion, and each subregion contains exactly one edge e_k . We denote the subregion containing e_i by R_i .
- Each vertex a_i belongs to exactly two regions, i.e., R_i and R_{i+1} (with $R_1 = R_{n+1}$).

II. Equivalence of canonical partitions. Two canonical partitions P and $\tilde{P}$ are equivalent if

$$P \sim \tilde{P} \iff \left(\forall 1 \leq i, j \leq n, \quad R_i \cap R_j = \emptyset \iff \tilde{R}_i \cap \tilde{R}_j = \emptyset \right),$$

that is, the corresponding subregions in the two partitions have the same adjacency relations. We denote by $SP(\mathcal{P}_a)$ the set of equivalence classes of canonical partitions of $\mathcal{P}_a$:

$$SP(\mathcal{P}_a) := \{[P] : [P] \text{ is an equivalence class of canonical partitions of } \mathcal{P}_a\}.$$

III. The tree structure of a canonical partition. To each equivalence class of canonical partitions of an n -gon, we associate a tree obtained by removing the boundary edges of the polygon. We denote the resulting collection, namely, the set of trees of standard partitions, by $TSP(\mathcal{P}_a)$:

$$TSP(\mathcal{P}_a) := \{\Gamma : \Gamma \text{ is the tree associated with } [P] \in SP(\mathcal{P}_a)\}.$$

We divide edges of a tree Γ into two classes: 1. boundary edges consisting of any edge with a vertex in the polygon. 2. internal edges consisting of the rest. Finally, polygon edges are those edges in the original polygon.

Lemma 3.2 (Classification of canonical partitions). *Let $\Gamma_a \in TSP(\mathcal{P}_a)$ be the tree associated with a canonical partition of the polygon $\mathcal{P}_a$. We call two subregions R_i and R_j in Γ_a paired if $R_i \cap R_j \in \mathcal{E}(\Gamma_a)$ and $|i - j| > 1 \pmod n$, i.e., R_i and R_j are nonadjacent and share an internal edge. Denote by $\mathcal{F}(\Gamma_a)$ the collection of non-adjacent pairings of subregions, i.e.,*

$$\mathcal{F}(\Gamma_a) := \left\{ \{i, j\} : |R_i \cap R_j| \in \mathcal{E}(\Gamma_a), \quad |i - j| > 1 \pmod n \right\}. \quad (3.1)$$

Here $\pmod n$ implies that $|1 - n| = 1$. E.g., for the partition in Fig. 4, we have $\mathcal{F} = \{\{1, 4\}, \{6, 4\}\}$.

Then for $\Gamma'_a, \Gamma_a \in TSP(\mathcal{P}_a)$, we have

$$\Gamma'_a = \Gamma_a \iff \mathcal{F}(\Gamma_a) = \mathcal{F}(\Gamma'_a)$$

Furthermore, we say $\{i, j\}$ and $\{k, l\}$ (with $i < j, k < l$) are crossing pairs if they satisfy

$$i < k < j < l \quad \text{or} \quad k < i < l < j$$

where the ordering is on $\mathbb{Z}$ instead of $\mathbb{Z}_n$. Then we have the following properties

1. $\mathcal{F}(\Gamma_a)$ contains no crossing pairs.
2. If $\mathcal{F}^*$ is a subset of $\{\{i, j\} : |i - j| > 1, \pmod n\}$ and there are no crossing pairs in $\mathcal{F}^*$, then there exists a canonical partition Γ_a such that $\mathcal{F}^* = \mathcal{F}(\Gamma_a)$.

Proof of lemma 3.2. We prove this lemma by mathematical induction. First, if $\mathcal{F}(\Gamma_a) = \emptyset$, then the tree graph inside must be a star (for $n \geq 3$) as in Figure 5. From now on, we assume that the set in (3.1) is

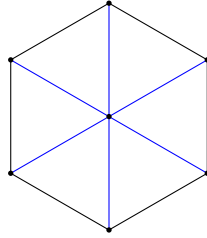
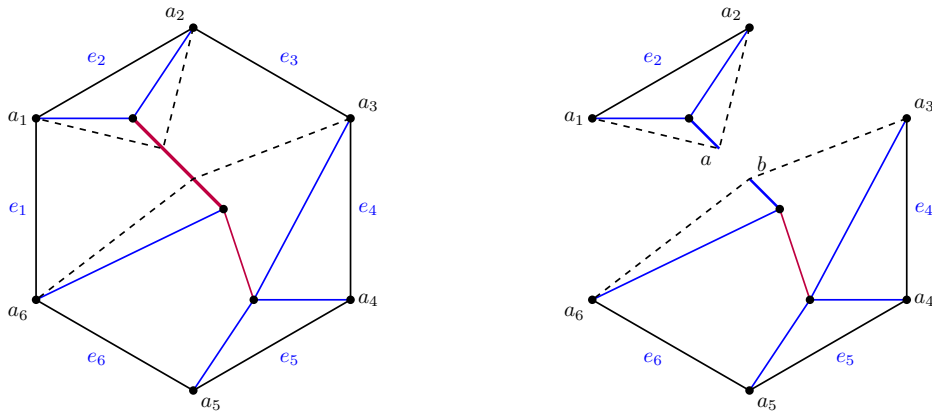


Figure 5: Star graph

nonempty. Assume for example that $(1, 3) \in \mathcal{F}$. By assumption, there exists an internal edge connecting R_1 and R_3 . Then the partition can be reduced to two partitions in the smaller polygons:

$$\mathcal{P}_{(a_1, a_2, a)}, \quad \mathcal{P}_{(a_3, a_4, a_5, a_6, b)}.$$

where a and b are two additional vertices to form two polygons. The rationale for this construction is that once $(1, 3) \in \mathcal{F}$ is given, the original polygon will be divided into two regions which will not “communicate”. The vertices a and b are added to get back to polygon language. Based on this observation, one can easily finish the induction proof.



□

Definition 3.3 (Representation of $\mathcal{K}_{t,\sigma,\mathbf{a}}$). Let $\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})$. To each edge e_i (equivalently, to the corresponding region R_i), associate a charge σ_i , and write $\sigma \in \{+, -\}^n$ for the collection of these charges. Let $b_1, \dots, b_m$ be the internal vertices of $\Gamma_{\mathbf{a}}$. Given σ , $\mathbf{a} = (a_1, \dots, a_n)$, $\mathbf{b} = (b_1, \dots, b_m)$, and $t \in [0, 1]$, define (with a slight abuse of notation, using the same symbols for vertices and their labels)

$$\Gamma_{\mathbf{a}}^{(b)}(t, \sigma) = \prod_{e \in \mathcal{E}(\Gamma)} (f_t(e))_{e_i, e_f}, \quad m_i = m(\sigma_i) \in \{m, \overline{m}\} \quad (3.2)$$

Here e_i and e_f denote the endpoints of the edge e , and $f_t(e)$ is defined by

$$f_t(e) = \Theta_{tm_k m_l}^{(B)} - \mathbf{1}(|k - l| \neq 1, \mod n), \quad R_k \cap R_l = e. \quad (3.3)$$

For $\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})$, define the value of the graph $(\Gamma_{\mathbf{a}}, \sigma)$ by

$$\Gamma_{t,\sigma,\mathbf{a}} := \Gamma_{\mathbf{a}}(t, \sigma) := \sum_{\mathbf{b} \in \mathbb{Z}_T^m} \Gamma_{\mathbf{a}}^{(b)}(t, \sigma). \quad (3.4)$$

Here we have followed the standard convention to use the same notation for the graph and its value.

We can unpack the concise definition of $f_t(e)$ by noting the following:

1. If $e = \{a_i, b_j\}$, then e is the boundary between R_i and R_{i+1} , and $f_t(e)$ is defined as:

$$f_t(e) = \Theta_{tm_i m_{i+1}}^{(B)}, \quad e = \{a_i, b_j\}$$

2. If $e = \{b_i, b_j\}$, and e is the boundary between R_k and R_l , then

$$f_t(e) = \Theta_{tm_k m_l}^{(B)} - 1, \quad e = \{b_i, b_j\}, \quad R_k \cap R_l = e.$$

For $n = 2$ the tree has no internal vertex: it is the single edge $e = \{a_1, a_2\}$, and (3.3) gives $f_t(e) = \Theta_{tm_1 m_2}^{(B)}$, so that (3.5) reduces to (2.57).

For example $\Gamma_{\mathbf{a}}^{(b)}$ in Fig. 4 is given by

$$\begin{aligned} \Gamma_{\mathbf{a}}^{(b)} &= (\Theta_{tm_1 m_2}^{(B)})_{a_1, b_1} \cdot (\Theta_{tm_2 m_3}^{(B)})_{a_2, b_1} \cdot (\Theta_{tm_3 m_4}^{(B)})_{a_3, b_1} \cdot (\Theta_{tm_4 m_5}^{(B)})_{a_4, b_3} \\ &\quad \times (\Theta_{tm_5 m_6}^{(B)})_{a_5, b_3} \cdot (\Theta_{tm_6 m_1}^{(B)})_{a_6, b_2} \cdot \left(\Theta_{tm_1 m_4}^{(B)} - 1 \right)_{b_1, b_2} \cdot \left(\Theta_{tm_4 m_6}^{(B)} - 1 \right)_{b_2, b_3} \end{aligned}$$

The key result in this subsection is the following representation of $\mathcal{K}_{t,\sigma,\mathbf{a}}$.

Lemma 3.4 (Tree Representation of $\mathcal{K}$). For $n \geq 2$, we have

$$\mathcal{K}_{t,\sigma,\mathbf{a}} = m_{\sigma} \cdot W^{-n+1} \sum_{\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})} \Gamma_{\mathbf{a}}(t, \sigma), \quad m_{\sigma} := \prod_{i=1}^n m(\sigma_i); \quad (3.5)$$

As an example, we give the tree graph representation of $\mathcal{K}$ for $n = 4$. There are three graphs for the case $n = 4$, as in Figure 6. The blue edges are boundary edges and equal to $\Theta^{(B)}$. The purple edges are internal edges equal to $\Theta^{(B)} - 1$. The r.h.s. of (3.5) equals

$$\begin{aligned} \sum_{\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})} \Gamma_{\mathbf{a}}(t, \sigma) &= \sum_{b_1, b_2, b_3, b_4} \left(\prod_{i=1}^4 \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i b_i} \right) \times \\ &\quad \times \left(\delta_{b_1 b_2 b_3 b_4} + \delta_{b_1 b_2} \delta_{b_3 b_4} \left(\Theta_{tm_1 m_3}^{(B)} - 1 \right)_{b_1 b_3} + \delta_{b_1 b_4} \delta_{b_2 b_3} \left(\Theta_{tm_2 m_4}^{(B)} - 1 \right)_{b_1 b_2} \right) \end{aligned}$$

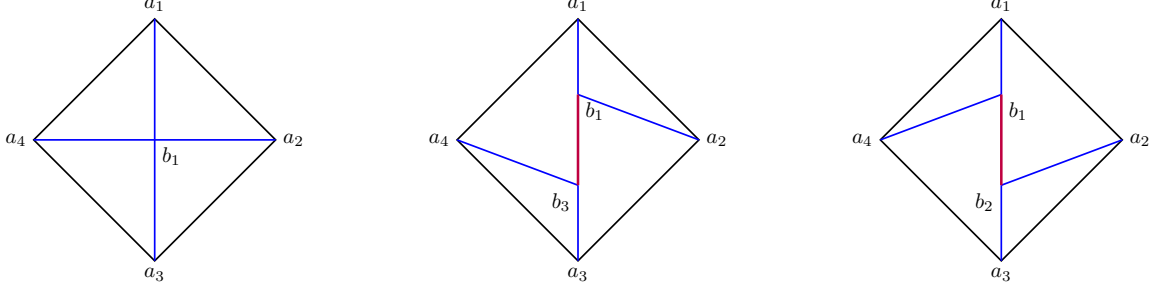


Figure 6: Graphs for $n = 4$

Corollary 3.5 (Pure loop $\mathcal{K}$). *In the special case $\sigma = (+, + \dots +)$, we have*

$$|\mathcal{K}_{t,\sigma,\mathbf{a}}| \leq C_n \exp \left(-c_n \max_{ij} \|a_i - a_j\| \right), \quad \|a_i - a_j\| := \min_{q \in \mathbb{Z}} |a_i - a_j + qL| \quad (3.6)$$

Proof of Corollary 3.5. Since $\sigma_i = +$ for all i , $f_t(e)$ in (3.3) is either $\Theta_{tm^2}^{(B)}$ or $\Theta_{tm^2}^{(B)} - 1$. Applying (2.52) with $\xi = tm^2$, we obtain that $\|\Theta_{tm^2}^{(B)}\|_{\max} = O(1)$ and $\Theta_{tm^2}^{(B)}$ decays exponentially. Thus

$$\left| \Gamma_{\mathbf{a}}^{(\mathbf{b})}(t, \sigma) \right| \leq C_n \exp \left(-c_n \max_{ij} \|b_i - b_j\| - c_n \max_{ij} \|a_i - b_j\| \right).$$

Together with the (3.4) and (3.5), we obtain the desired result (3.6). $\square$

Proof of Lemma 3.4. In this proof, we temporarily denote

$$\tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} := m_{\sigma} \cdot W^{-n+1} \sum_{\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})} \Gamma_{t,\sigma,\mathbf{a}}, \quad m_{\sigma} := \prod_{i=1}^n m(\sigma_i);$$

We will show that

$$\tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} = \mathcal{K}_{t,\sigma,\mathbf{a}}, \quad \text{for } t = 0 \quad (3.7)$$

and $\tilde{\mathcal{K}}$ satisfies the dynamics equation for $\mathcal{K}$ in (2.48), i.e.,

$$\frac{d}{dt} \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} = W \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} \left(\mathcal{G}_{k,l}^{(a),L} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right) S_{ab}^{(B)} \left(\mathcal{G}_{k,l}^{(b),R} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right) \quad (3.8)$$

here

$$\mathcal{G}_{k,l}^{(a),L} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} = \tilde{\mathcal{K}}_{t,\mathcal{G}_{k,l}^{(a),L}(\sigma, \mathbf{a})}, \quad \mathcal{G}_{k,l}^{(b),R} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} = \tilde{\mathcal{K}}_{t,\mathcal{G}_{k,l}^{(b),R}(\sigma, \mathbf{a})}.$$

For $t = 0$, we have by definition that

$$t = 0 \implies \Theta_{t\xi}^{(B)} = I$$

If the tree graph Γ has an internal edge, then $\Gamma_{t,\sigma,\mathbf{a}} = 0$ when $t = 0$. Hence for $t = 0$, the only non-trivial graph is the star-shaped graph (Figure 5), which has only one internal vertex. Hence (3.7) can be explicitly verified, i.e.,

$$\tilde{\mathcal{K}}_{0,\sigma,\mathbf{a}} = m_{\sigma} W^{-n+1} \cdot \delta_{a_1, a_2, \dots, a_n} = W^{-n+1} \Gamma_{0,\sigma,\mathbf{a}}^{(\text{star})} = \mathcal{K}_{0,\sigma,\mathbf{a}}.$$

Now we prove (3.8). By definition,

$$\frac{d}{dt} \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} = m_{\sigma} W^{-n+1} \cdot \sum_{\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})} \frac{d}{dt} \Gamma_{t,\sigma,\mathbf{a}} = m_{\sigma} W^{-n+1} \cdot \sum_{\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})} \sum_{\mathbf{b}} \frac{d}{dt} \Gamma_{t,\sigma,\mathbf{a}}^{(\mathbf{b})}$$

Recall $\Gamma_{t,\sigma,\mathbf{a}}^{(\mathbf{b})}$ as defined in (3.2). Due to $dm_i/dt = 0$, the derivative d/dt acts only on the $f_t(e)$. By definition,

$$\frac{d}{dt}\Theta_{t\xi}^{(B)} = \Theta_{t\xi}^{(B)} \cdot \xi S^{(B)} \cdot \Theta_{t\xi}^{(B)}$$

Therefore,

$$\frac{d}{dt}\Theta_{tm_im_j}^{(B)} = m_im_j \left(\Theta_{tm_im_j}^{(B)} \cdot S^{(B)} \cdot \Theta_{tm_im_j}^{(B)} \right)$$

In other words, the derivative of a blue edge $\Theta^{(B)}$ or a purple edge $\Theta^{(B)} - 1$ equals m_im_j times two blue edges with $S^{(B)}$ in the middle.

$$\begin{array}{c} \frac{d}{dt} \text{ (blue line)} \\ \frac{d}{dt} \text{ (purple line)} \end{array} = \text{ (blue line)} S^{(B)} \text{ (blue line)} \cdot m_im_j$$

Figure 7: Derivatives of edges

On the other hand, we know that for fixed $\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})$, $\sigma, \mathbf{a}, i, j$, there is at most one edge $e \in \mathcal{E}(\Gamma_{\mathbf{a}})$ such that $e = R_i \cap R_j$. Then we can write the derivative of $\Gamma_{t,\sigma,\mathbf{a}}$ as follows

$$\frac{d}{dt}\Gamma_{t,\sigma,\mathbf{a}}^{(\mathbf{b})} = \sum_{1 \leq i < j \leq n} \sum_{e \in \mathcal{E}(\Gamma_{\mathbf{a}})} \mathbf{1}(e = R_i \cap R_j) \cdot \frac{\Gamma_{t,\sigma,\mathbf{a}}^{(\mathbf{b})}}{(f_t(e))_{e_i,e_f}} \cdot m_im_j \left(\Theta_{tm_im_j}^{(B)} \cdot S^{(B)} \cdot \Theta_{tm_im_j}^{(B)} \right)_{e_i,e_f} \quad (3.9)$$

where $f_t(e)$ is defined in (3.3). The quotient notation in (3.9) is symbolic and does not denote numerical division. Indeed, for $e \in \mathcal{E}(\Gamma_{\mathbf{a}})$ with $e = R_i \cap R_j$, the definition (3.2) ensures that $(f_t(e))_{e_i,e_f}$ is one of the factors in $\Gamma_{t,\sigma,\mathbf{a}}^{(\mathbf{b})}$. Thus the quotient denotes the product defining $\Gamma_{t,\sigma,\mathbf{a}}^{(\mathbf{b})}$ with this factor omitted.

Suppose that there exists $e \in \mathcal{E}(\Gamma_{\mathbf{a}})$ such that $e = R_i \cap R_j$. Then

$$\frac{\Gamma_{t,\sigma,\mathbf{a}}^{(\mathbf{b})}}{(f_t(e))_{e_i,e_f}} \cdot m_im_j \left(\Theta_{tm_im_j}^{(B)} \cdot S^{(B)} \cdot \Theta_{tm_im_j}^{(B)} \right)_{e_i,e_f}, \quad e = R_i \cap R_j$$

is equal to removing the edge e in $\Gamma_{\mathbf{a}}$ and adding two edges $\{e_i, a\}$, $\{e_f, b\}$, and a $S_{ab}^{(B)}$ in the middle. Here is an example with $i = 1$ and $j = 3$ in Figure 8. Note: this statement also holds for the case $|i - j| = 1$. For example, if $i = 1, j = 2$, the triangle in the r.h.s. of Figure 8 will become an $n = 2$ polygon $\mathcal{P}_{(a_1,a)}$ with charge vector (σ_1, σ_2) and a blue edge (a_1, a) inside.

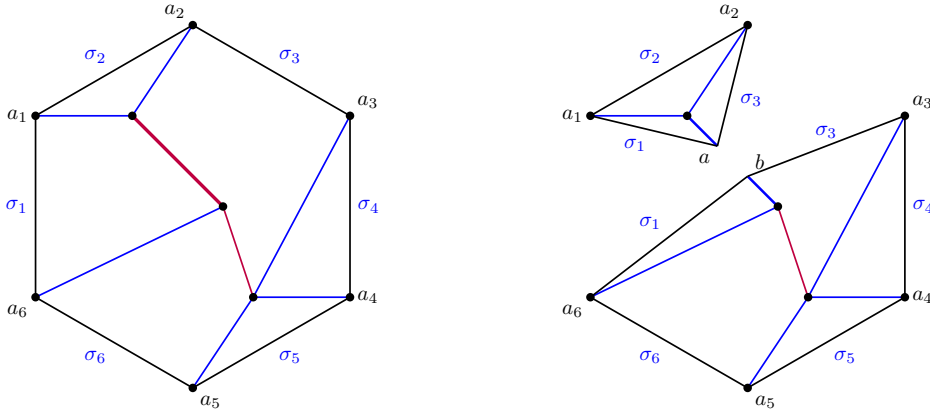


Figure 8: Derivative of tree

The new edges created by the partition will inherit the original charges. We denote them by

$$(\sigma', \mathbf{a}') = \mathcal{G}_{i,j}^{(a),L}(\sigma, \mathbf{a}) \quad (\sigma'', \mathbf{a}'') = \mathcal{G}_{i,j}^{(b),R}(\sigma, \mathbf{a}). \quad (3.10)$$

Therefore, there exist $\Gamma'_{\mathbf{a}'} \in TSP(\mathcal{P}_{\mathbf{a}'})$ and $\Gamma''_{\mathbf{a}''} \in TSP(\mathcal{P}_{\mathbf{a}''})$ such that (as in Figure 8)

$$m_{\boldsymbol{\sigma}} \cdot \sum_{\mathbf{b}} \frac{\Gamma_{t,\boldsymbol{\sigma},\mathbf{a}}^{(\mathbf{b})}}{(f_t(e))_{e_i,e_f}} \cdot m_i m_j \left(\Theta_{tm_i m_j}^{(B)} \cdot S_{ab}^{(B)} \cdot \Theta_{tm_i m_j}^{(B)} \right)_{e_i,e_f} = \sum_{a,b} m_{\boldsymbol{\sigma}'} \cdot \Gamma'_{t,\boldsymbol{\sigma}',\mathbf{a}'} \cdot S_{ab}^{(B)} \cdot m_{\boldsymbol{\sigma}''} \Gamma''_{t,\boldsymbol{\sigma}'',\mathbf{a}''}.$$

Thus,

$$m_{\boldsymbol{\sigma}} \cdot \frac{d}{dt} \sum_{\Gamma} \Gamma_{t,\boldsymbol{\sigma},\mathbf{a}} = \sum_{i < j} \sum_{a,b} \left(m_{\boldsymbol{\sigma}'} \sum_{\Gamma'} \Gamma'_{t,\boldsymbol{\sigma}',\mathbf{a}'} \right) \cdot S_{ab}^{(B)} \cdot \left(m_{\boldsymbol{\sigma}''} \sum_{\Gamma''} \Gamma''_{t,\boldsymbol{\sigma}'',\mathbf{a}''} \right)$$

where $i, j, a, b, \boldsymbol{\sigma}, \mathbf{a}, \boldsymbol{\sigma}', \mathbf{a}', \boldsymbol{\sigma}'',$ and $\mathbf{a}''$ satisfy relation (3.10). Clearly it is equivalent to (3.8) and this completes the proof of Lemma 3.4. $\square$

3.2 Ward's identity on $\mathcal{K}$

Recall Ward's identity for resolvents, which states that

$$G \cdot G^\dagger = G(z) \cdot G^\dagger(z) = G(z) \cdot G(\bar{z}) = \frac{G(z) - G(\bar{z})}{z - \bar{z}} = \frac{G - G^\dagger}{2i\eta}, \quad \eta = \text{Im } z.$$

In our setting, $G_t(+) = (H_t - z_t)^{-1} = (G_t(-))^\dagger$ and thus

$$G_t(+) \cdot G_t(-) = \frac{G_t(+) - G_t(-)}{2i\eta_t} = \frac{G_t(+) - G_t(-)}{2i(1-t)\text{Im } m}, \quad \eta_t = \text{Im } z_t. \quad (3.11)$$

From Ward's identity, we have the following identity for a 2- G -loop:

$$\sum_a \mathcal{L}_{t,(+,-),(a,b)} = \frac{1}{2W i \eta_t} (\mathcal{L}_{t,(+),(b)} - \mathcal{L}_{t,(-),(b)}).$$

We extend this identity to all $n - G$ loops $\mathcal{L}$ in the following lemma. The main purpose of this subsection is to show that the same identity holds for $\mathcal{K}$ loop as well.

Lemma 3.6. *For an n - G -loop $\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}$ with $\sigma_1 = +$ and $\sigma_n = -$, we have*

$$\sum_{a_n} \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}} = \frac{1}{2W i \eta_t} (\mathcal{L}_{t,\boldsymbol{\sigma}+, \mathbf{a}/a_n} - \mathcal{L}_{t,\boldsymbol{\sigma}-, \mathbf{a}/a_n}) \quad (3.12)$$

and

$$\sum_{a_n} \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}} = \frac{1}{2W i \eta_t} (\mathcal{K}_{t,\boldsymbol{\sigma}+, \mathbf{a}/a_n} - \mathcal{K}_{t,\boldsymbol{\sigma}-, \mathbf{a}/a_n}) \quad (3.13)$$

where

- $\boldsymbol{\sigma}\pm$ is obtained by removing σ_n from $\boldsymbol{\sigma}$ and replacing σ_1 with $\pm$, i.e.,

$$\boldsymbol{\sigma}\pm = (\pm, \sigma_2, \sigma_3, \dots, \sigma_{n-1}).$$

Notice that the length of $\boldsymbol{\sigma}\pm$ is $n - 1$.

- $\mathbf{a}/a_n$ is obtained by removing a_n from $\mathbf{a}$:

$$\mathbf{a}/a_n = (a_1, a_2, a_3, \dots, a_{n-1})$$

Corollary 3.7. *Under the assumption of Lemma 3.6, we have*

$$\sum_{a_2} \cdots \sum_{a_{n-1}} \sum_{a_n} \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}} = O((W\eta_t)^{-n+1}) \quad (3.14)$$

Proof of Corollary 3.7. By definition of $\mathcal{K}$, we know that $\mathcal{K}$ is translation invariant. Then the left side of equation (3.14) is independent of a_1 . Hence it is equivalent to

$$\frac{1}{L} \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}_{t, \sigma, \mathbf{a}} = O((W\eta_t)^{-n+1}).$$

Applying (3.13) repeatedly, we can reduce it to shorter pure loops to get

$$\left| L^{-1} \cdot \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}_{t, \sigma, \mathbf{a}} \right| \leq C_n \sum_{1 \leq m \leq n} (W\eta_t)^{-n+m} \cdot L^{-1} \cdot \max_{\sigma' \in \{+\}^m \cup \{-\}^m} \left| \sum_{\mathbf{a}' \in \mathbb{Z}_L^m} \mathcal{K}_{t, \sigma', \mathbf{a}'} \right|, \quad (3.15)$$

Here $\sigma' \in \{+\}^m \cup \{-\}^m$ means that $\sigma' = (+, +, \dots, +)$ or $(-, -, \dots, -)$. By the estimate of pure loop $\mathcal{K}$ in Lemma 3.5, we have

$$\sum_{\mathbf{a}' \in \mathbb{Z}_L^m} \mathcal{K}_{t, \sigma', \mathbf{a}'} = O(1).$$

Together with (3.15), this completes the proof of the corollary. $\square$

Proof of Lemma 3.6. We divide the proof into five steps. The main idea is to show that the difference between the two sides of (3.13) satisfies a homogeneous linear differential equation with zero initial value.

Step 1: The identity for $\mathcal{L}$ and the cases $n = 2, 3$ for $\mathcal{K}$.

Since $\sum_{a_n} E_{a_n} = W^{-1}I$, cyclicity of the trace and the two-resolvent Ward identity give

$$\begin{aligned} \sum_{a_n} \mathcal{L}_{t, \sigma, \mathbf{a}} &= \frac{1}{W} \left\langle G_t(-) G_t(+) E_{a_1} \prod_{j=2}^{n-1} G_t(\sigma_j) E_{a_j} \right\rangle \\ &= \frac{1}{2W i \eta_t} (\mathcal{L}_{t, \sigma+, \mathbf{a}/a_n} - \mathcal{L}_{t, \sigma-, \mathbf{a}/a_n}). \end{aligned}$$

This proves (3.12).

We next prove (3.13). For $n = 2$, the explicit two-loop formula and the stochasticity of $S^{(B)}$ yield

$$\sum_{a_2} \mathcal{K}_{t, (+, -), (a_1, a_2)} = \frac{1}{W} \sum_{a_2} \left(\frac{|m|^2}{1 - t|m|^2 S^{(B)}} \right)_{a_1 a_2} = \frac{1}{W(1-t)}. \quad (3.16)$$

On the other hand,

$$\frac{\mathcal{K}_{t, +, a_1} - \mathcal{K}_{t, -, a_1}}{2W i \eta_t} = \frac{m - \bar{m}}{2W i \eta_t} = \frac{1}{W(1-t)},$$

where we used $|m| = 1$, $m - \bar{m} = 2i \operatorname{Im} m$, and $\eta_t = (1-t) \operatorname{Im} m$.

For completeness, we give the calculation for $n = 3$. This also provides a representative example of the algebra used below. Put

$$q := m(\sigma_2), \quad A := \Theta_{tmq}^{(B)}, \quad B := \Theta_{t\bar{m}q}^{(B)}.$$

Since $\sigma_1 = +$ and $\sigma_3 = -$, the explicit three-loop formula gives

$$\sum_{a_3} \mathcal{K}_{t, (+, \sigma_2, -), (a_1, a_2, a_3)} = \frac{q}{W^2(1-t)} (AB)_{a_1 a_2}.$$

Here we used

$$\sum_{a_3} (\Theta_t^{(B)})_{a_3 x} = \frac{1}{1-t}$$

and the fact that all propagators are functions of $S^{(B)}$ and hence commute. The two loops on the right-hand side of (3.13) are

$$\mathcal{K}_{t,(+,\sigma_2),(a_1,a_2)} = \frac{mq}{W} A_{a_1 a_2}, \quad \mathcal{K}_{t,(-,\sigma_2),(a_1,a_2)} = \frac{\bar{m}q}{W} B_{a_1 a_2}.$$

The required matrix identity follows directly from the definitions of A and B :

$$\begin{aligned} mA - \bar{m}B &= AB(mB^{-1} - \bar{m}A^{-1}) \\ &= AB \left[m(1 - t\bar{m}qS^{(B)}) - \bar{m}(1 - tmqS^{(B)}) \right] \\ &= (m - \bar{m})AB. \end{aligned}$$

Consequently,

$$\frac{\mathcal{K}_{t,(+,\sigma_2),(a_1,a_2)} - \mathcal{K}_{t,(-,\sigma_2),(a_1,a_2)}}{2Wi\eta_t} = \frac{q}{W^2(1-t)} (AB)_{a_1 a_2},$$

which proves (3.13) for $n = 3$.

Step 2: Induction and the difference equation.

Let $n \geq 4$ and assume that (3.13) holds for all loops of length strictly smaller than n . Define

$$\mathcal{K}_{t,\sigma,\mathbf{a}}^{(*)} := \sum_{a_n} \mathcal{K}_{t,\sigma,\mathbf{a}}, \quad \mathcal{K}_{t,\sigma,\mathbf{a}}^{(\pm)} := \mathcal{K}_{t,\sigma \pm \mathbf{a}/a_n}, \quad (3.17)$$

and

$$\mathcal{K}_{t,\sigma,\mathbf{a}}^{(**)} := \frac{\mathcal{K}_{t,\sigma,\mathbf{a}}^{(+)} - \mathcal{K}_{t,\sigma,\mathbf{a}}^{(-)}}{2Wi\eta_t}, \quad D_{t,\sigma,\mathbf{a}} := \mathcal{K}_{t,\sigma,\mathbf{a}}^{(*)} - \mathcal{K}_{t,\sigma,\mathbf{a}}^{(**)}.$$

We shall prove that D vanishes identically. More precisely, the induction will be closed once we establish

$$\begin{aligned} \partial_t D_{t,\sigma,\mathbf{a}} &= W \sum_{k=1}^{n-1} \sum_{a,b} D_{t,\sigma,\mathbf{a}} \big|_{a_k \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_k, \sigma_{k+1}), (a_k, b)} \\ &\quad + \frac{1}{1-t} D_{t,\sigma,\mathbf{a}}. \end{aligned} \quad (3.18)$$

At $t = 0$, the initial condition for the primitive equation gives

$$\begin{aligned} \mathcal{K}_{0,\sigma,\mathbf{a}}^{(*)} &= W^{-n+1} \left(\prod_{j=2}^{n-1} m(\sigma_j) \right) \mathbf{1}(a_1 = \dots = a_{n-1}), \\ \mathcal{K}_{0,\sigma,\mathbf{a}}^{(**)} &= \frac{W^{-n+2}(m - \bar{m})}{2Wi\eta_0} \left(\prod_{j=2}^{n-1} m(\sigma_j) \right) \mathbf{1}(a_1 = \dots = a_{n-1}). \end{aligned}$$

Since $\eta_0 = \text{Im } m$, these expressions are equal. Thus $D_0 = 0$.

For $1 \leq k < l \leq n$, abbreviate

$$\mathcal{J}_{k,l,a}^L := \mathcal{G}_{k,l}^{(a),L} \circ \mathcal{K}_{t,\sigma,\mathbf{a}}, \quad \mathcal{J}_{k,l,b}^R := \mathcal{G}_{k,l}^{(b),R} \circ \mathcal{K}_{t,\sigma,\mathbf{a}},$$

and define $\mathcal{J}^{L,(\pm)}$ and $\mathcal{J}^{R,(\pm)}$ analogously with $\mathcal{K}^{(\pm)}$ in place of $\mathcal{K}$. The primitive equation gives

$$\partial_t \mathcal{K}^{(*)} = W \sum_{1 \leq k < l \leq n} \sum_{a,b} \left(\sum_{a_n} \mathcal{J}_{k,l,a}^L \right) S_{ab}^{(B)} \mathcal{J}_{k,l,b}^R. \quad (3.19)$$

Moreover, since $\partial_t \eta_t^{-1} = (1-t)^{-1} \eta_t^{-1}$,

$$\begin{aligned} \partial_t \mathcal{K}^{(**)} &= \frac{1}{1-t} \mathcal{K}^{(**)} \\ &\quad + \frac{W}{2Wi\eta_t} \sum_{1 \leq k < l \leq n-1} \sum_{a,b} \left(\mathcal{J}_{k,l,a}^{L,(+)} \mathcal{J}_{k,l,b}^{R,(+)} - \mathcal{J}_{k,l,a}^{L,(-)} \mathcal{J}_{k,l,b}^{R,(-)} \right) S_{ab}^{(B)}. \end{aligned} \quad (3.20)$$

Road map for Steps 3–4. To compare these two derivatives, we partition the pairs (k, l) into four classes. The following are the four claims that will be verified below.

1. For $2 \leq k < l \leq n-1$, the non-adjacent terms $l \geq k+2$ cancel by the induction hypothesis, while the adjacent terms give

$$W \sum_{k=2}^{n-2} \sum_{a,b} D_{t,\sigma,\mathbf{a}}|_{a_k \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_k, \sigma_{k+1}), (a_k, b)}.$$

2. For each $3 \leq m \leq n-2$, the two boundary cuts $(1, m)$ and (m, n) cancel after applying the induction hypothesis and using the symmetry of $S^{(B)}$.
3. The cut $(1, n)$ contributes $(1-t)^{-1} \mathcal{K}^{(*)}$. Together with the derivative of η_t^{-1} in $\partial_t \mathcal{K}^{(**)}$, this produces $(1-t)^{-1} D$.
4. The four remaining cuts $(1, 2)$, $(1, n-1)$, $(2, n)$ and $(n-1, n)$ give the two endpoint terms

$$W \sum_{k \in \{1, n-1\}} \sum_{a,b} D_{t,\sigma,\mathbf{a}}|_{a_k \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_k, \sigma_{k+1}), (a_k, b)}.$$

Step 3 proves the first claim, and Step 4 proves the remaining three.

Step 3: Internal cuts.

Suppose that $2 \leq k < l \leq n-1$. The right loop produced by the cut-and-glue operation is explicitly

$$\mathcal{J}_{k,l,b}^R = \mathcal{K}_{t,(\sigma_k, \dots, \sigma_l), (a_k, \dots, a_{l-1}, b)}.$$

Changing the first charge of the original loop or removing its last charge does not affect this segment. Hence

$$\mathcal{J}_{k,l,b}^{R,(+)} = \mathcal{J}_{k,l,b}^{R,(-)} = \mathcal{J}_{k,l,b}^R. \quad (3.21)$$

This is a representative calculation behind the corresponding cut-and-glue identities used in the proof.

If $l \geq k+2$, the left loop has length at most $n-1$. The induction hypothesis therefore gives

$$\sum_{a_n} \mathcal{J}_{k,l,a}^L = \frac{\mathcal{J}_{k,l,a}^{L,(+)} - \mathcal{J}_{k,l,a}^{L,(-)}}{2W i \eta_t}. \quad (3.22)$$

Thus all non-adjacent internal cuts cancel between $\partial_t \mathcal{K}^{(*)}$ and $\partial_t \mathcal{K}^{(**)}$.

If $l = k+1$, then

$$\mathcal{J}_{k,k+1,a}^L = \mathcal{K}_{t,\sigma,\mathbf{a}}|_{a_k \rightarrow a}, \quad \mathcal{J}_{k,k+1,b}^R = \mathcal{K}_{t,(\sigma_k, \sigma_{k+1}), (a_k, b)}. \quad (3.23)$$

Therefore, the contribution of all adjacent internal cuts is

$$W \sum_{k=2}^{n-2} \sum_{a,b} D_{t,\sigma,\mathbf{a}}|_{a_k \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_k, \sigma_{k+1}), (a_k, b)}. \quad (3.24)$$

Step 4: Boundary cuts.

We first consider $(k, l) = (1, n)$. In this case

$$\mathcal{J}_{1,n,a}^L = \mathcal{K}_{t,(+,-), (a, a_n)}, \quad \mathcal{J}_{1,n,b}^R = \mathcal{K}_{t,\sigma,\mathbf{a}}|_{a_n \rightarrow b}.$$

Using the already proved two-loop identity and $\sum_a S_{ab}^{(B)} = 1$, we obtain

$$W \sum_{a,b,a_n} \mathcal{J}_{1,n,a}^L S_{ab}^{(B)} \mathcal{J}_{1,n,b}^R = \frac{1}{1-t} \mathcal{K}_{t,\sigma,\mathbf{a}}^{(*)}. \quad (3.25)$$

Next fix $3 \leq m \leq n-2$. The two boundary cuts under consideration are

$$(k, l) = (1, m) \quad \text{and} \quad (k, l) = (m, n), \quad 3 \leq m \leq n-2. \quad (3.26)$$

Denote their respective contributions to $\partial_t D$ by $C_{1,m}$ and $C_{m,n}$. Applying the induction hypothesis to the shorter left loop shows that

$$C_{1,m} = \frac{W}{2W i \eta_t} \sum_{a,b} \mathcal{J}_{1,m,a}^{L,(-)} \left(\mathcal{J}_{1,m,b}^{R,(-)} - \mathcal{J}_{1,m,b}^{R,(+)} \right) S_{ab}^{(B)}. \quad (3.27)$$

After interchanging a and b , the cut (m, n) gives

$$C_{m,n} = -\frac{W}{2W i \eta_t} \sum_{a,b} \mathcal{J}_{1,m,a}^{L,(-)} \left(\mathcal{J}_{1,m,b}^{R,(-)} - \mathcal{J}_{1,m,b}^{R,(+)} \right) S_{ab}^{(B)}. \quad (3.28)$$

Here we used the symmetry of $S^{(B)}$. Hence

$$C_{1,m} + C_{m,n} = 0. \quad (3.29)$$

It remains to consider

$$(1, 2), \quad (1, n-1), \quad (2, n), \quad (n-1, n). \quad (3.30)$$

To keep track of the four terms, introduce the notation

$$U_a^\pm := \mathcal{J}_{1,2,a}^{L,(\pm)}, \quad u_b^\pm := \mathcal{J}_{1,2,b}^{R,(\pm)}, \quad v_a^\pm := \mathcal{J}_{1,n-1,a}^{L,(\pm)}, \quad V_b^\pm := \mathcal{J}_{1,n-1,b}^{R,(\pm)}. \quad (3.31)$$

Thus $U^\pm, V^\pm$ are loops of length $n-1$, whereas $u^\pm, v^\pm$ are two-loops. In particular,

$$\mathcal{K}^{(**)}|_{a_1 \rightarrow a} = \frac{U_a^+ - U_a^-}{2W i \eta_t}, \quad \mathcal{K}^{(**)}|_{a_{n-1} \rightarrow a} = \frac{V_a^+ - V_a^-}{2W i \eta_t}. \quad (3.32)$$

We now list the four full-loop contributions explicitly. For the two adjacent cuts, the left loop is the original loop with one index replaced:

$$\begin{aligned} \sum_{a_n} \mathcal{J}_{1,2,a}^L &= \mathcal{K}^{(*)}|_{a_1 \rightarrow a}, & \mathcal{J}_{1,2,b}^R &= u_b^+, \\ \sum_{a_n} \mathcal{J}_{n-1,n,a}^L &= \mathcal{K}^{(*)}|_{a_{n-1} \rightarrow a}, & \mathcal{J}_{n-1,n,b}^R &= v_b^-. \end{aligned}$$

In the second identity we used cyclicity to identify $v_b^- = \mathcal{K}_{t,(\sigma_{n-1},-),(a_{n-1},b)}$.

For the remaining cuts $(1, n-1)$ and $(2, n)$, the left loops have length three. Applying the three-loop Ward identity from Step 1 gives

$$\begin{aligned} \sum_{a_n} \mathcal{J}_{1,n-1,a}^L &= \frac{v_a^+ - v_a^-}{2W i \eta_t}, & \mathcal{J}_{1,n-1,b}^R &= V_b^+, \\ \sum_{a_n} \mathcal{J}_{2,n,a}^L &= \frac{u_a^+ - u_a^-}{2W i \eta_t}, & \mathcal{J}_{2,n,b}^R &= U_b^-. \end{aligned} \quad (3.33)$$

For example, the second line follows from

$$\sum_{a_n} \mathcal{K}_{t,(+,\sigma_2,-),(a_1,a,a_n)} = \frac{1}{2W i \eta_t} [\mathcal{K}_{t,(+,\sigma_2),(a_1,a)} - \mathcal{K}_{t,(\sigma_2,-),(a,a_1)}] = \frac{u_a^+ - u_a^-}{2W i \eta_t},$$

where the last equality again uses cyclicity of the two-loop.

We next compare these four terms with the $(1, 2)$ and $(1, n-1)$ terms in $\partial_t \mathcal{K}^{(**)}$. The latter are

$$\frac{W}{2W i \eta_t} \sum_{a,b} (U_a^+ u_b^+ - U_a^- u_b^- + v_a^+ V_b^+ - v_a^- V_b^-) S_{ab}^{(B)}.$$

For the left endpoint, the cuts $(1, 2)$ and $(2, n)$, after subtracting the $(1, 2)$ term above, give

$$\begin{aligned}
& W \sum_{a,b} \mathcal{K}^{(*)}|_{a_1 \rightarrow a} S_{ab}^{(B)} u_b^+ + \frac{W}{2Wi\eta_t} \sum_{a,b} (u_a^+ - u_a^-) S_{ab}^{(B)} U_b^- - \frac{W}{2Wi\eta_t} \sum_{a,b} (U_a^+ u_b^+ - U_a^- u_b^-) S_{ab}^{(B)} \\
&= W \sum_{a,b} \left(\mathcal{K}^{(*)}|_{a_1 \rightarrow a} - \frac{U_a^+ - U_a^-}{2Wi\eta_t} \right) S_{ab}^{(B)} u_b^+ \\
&= W \sum_{a,b} D|_{a_1 \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_1, \sigma_2), (a_1, b)}.
\end{aligned}$$

Here the symmetry of $S^{(B)}$ was used in the first equality to interchange a and b in the middle term.

Similarly, the cuts $(1, n-1)$ and $(n-1, n)$, after subtracting the $(1, n-1)$ term in $\partial_t \mathcal{K}^{(**)}$, give

$$\begin{aligned}
& W \sum_{a,b} \mathcal{K}^{(*)}|_{a_{n-1} \rightarrow a} S_{ab}^{(B)} v_b^- + \frac{W}{2Wi\eta_t} \sum_{a,b} (v_a^+ - v_a^-) S_{ab}^{(B)} V_b^+ - \frac{W}{2Wi\eta_t} \sum_{a,b} (v_a^+ V_b^+ - v_a^- V_b^-) S_{ab}^{(B)} \\
&= W \sum_{a,b} \left(\mathcal{K}^{(*)}|_{a_{n-1} \rightarrow a} - \frac{V_a^+ - V_a^-}{2Wi\eta_t} \right) S_{ab}^{(B)} v_b^- \\
&= W \sum_{a,b} D|_{a_{n-1} \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_{n-1}, \sigma_n), (a_{n-1}, b)}.
\end{aligned}$$

Therefore the total contribution of the four remaining cuts is

$$W \sum_{k \in \{1, n-1\}} \sum_{a,b} D_{t, \sigma, \mathbf{a}}|_{a_k \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_k, \sigma_{k+1}), (a_k, b)}. \quad (3.34)$$

Step 5: Closing the induction.

Combining Steps 2–4, we obtain

$$\partial_t D_{t, \sigma, \mathbf{a}} = W \sum_{k=1}^{n-1} \sum_{a,b} D_{t, \sigma, \mathbf{a}}|_{a_k \rightarrow a} S_{ab}^{(B)} \mathcal{K}_{t,(\sigma_k, \sigma_{k+1}), (a_k, b)} + \frac{1}{1-t} D_{t, \sigma, \mathbf{a}}.$$

For every $T < 1$, this is a finite-dimensional homogeneous linear differential equation on $[0, T]$. Since $D_0 = 0$, uniqueness gives $D_t = 0$ on $[0, T]$. As $T < 1$ is arbitrary,

$$\mathcal{K}_{t, \sigma, \mathbf{a}}^{(*)} = \mathcal{K}_{t, \sigma, \mathbf{a}}^{(**)}, \quad 0 \leq t < 1.$$

This is precisely (3.13), and the proof is complete. $\square$

3.3 Sum zero property of $\mathcal{K}$

In this section, we will state the sum-zero property of self-energies formally. This property will serve as a key tool in estimating the correct scaling of the primitive loops. Recall the tree representation of $\mathcal{K}$ in Lemma 3.4. The key quantity $\Gamma_{\mathbf{a}}(t, \sigma)$ in this representation contains only three types of edges, namely,

$$S^{(B)}, \quad \Theta_{tm^2}^{(B)}, \quad \Theta_{t|m|^2}^{(B)}$$

which commute with one another. By explicit computations, we have

$$\left\| \Theta_{tm^2}^{(B)} \right\|_{\max} \prec 1, \quad \left\| \Theta_{t|m|^2}^{(B)} \right\|_{\max} \prec \frac{1}{\ell_t \eta_t}; \quad (3.35)$$

$$\left\| \Theta_{tm^2}^{(B)} \right\|_1 \prec 1, \quad \left\| \Theta_{t|m|^2}^{(B)} \right\|_1 \prec \frac{1}{\eta_t} \quad (3.36)$$

Due to the big difference in ranges between $\Theta_{tm^2}^{(B)}$ and $\Theta_{t|m|^2}^{(B)}$, we call them short and long edges respectively:

$$\Theta_{t\xi}^{(B)} \text{ or } \left(\Theta_{t\xi}^{(B)} - 1\right) = \begin{cases} \text{short } \Theta \text{ edge,} & \xi = m^2, \bar{m}^2 \\ \text{long } \Theta \text{ edge,} & \xi = |m|^2 \end{cases}$$

Definition 3.8. Fix $\sigma = (\sigma_1, \dots, \sigma_n)$ and $\mathbf{a} = (a_1, \dots, a_n)$. For a partition $\Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}})$, recall that $\mathcal{F}(\Gamma_{\mathbf{a}})$, defined in (3.1), is the collection of pairs of non-adjacent subregions that share an internal edge. Thus, the elements of $\mathcal{F}(\Gamma_{\mathbf{a}})$ are in one-to-one correspondence with the internal edges in $\mathcal{E}(\Gamma_{\mathbf{a}})$.

I. The set of long internal edges $\mathcal{F}_{\text{long}}(\Gamma_{\mathbf{a}}, \sigma)$. Given σ , define $\mathcal{F}_{\text{long}}(\Gamma_{\mathbf{a}}, \sigma)$ to be the collection of long internal edges:

$$\mathcal{F}_{\text{long}}(\Gamma_{\mathbf{a}}, \sigma) := \left\{ \{i, j\} \in \mathcal{F}(\Gamma_{\mathbf{a}}) : \{\sigma_i, \sigma_j\} = \{+, -\} \right\}.$$

In other words, for $\{i, j\} \in \mathcal{F}_{\text{long}}(\Gamma_{\mathbf{a}}, \sigma)$, there exists an internal (i.e., purple) edge in $\Gamma_{\mathbf{a}}$ separating R_i and R_j with $\sigma_i \neq \sigma_j$.

By definition,

$$\mathcal{F}_{\text{long}}(\Gamma_{\mathbf{a}}, \sigma) \subset \mathcal{F}(\Gamma_{\mathbf{a}}) \subset \mathbb{Z}_n^{\text{off}} := \left\{ \{i, j\} : 1 \leq i < j \leq n, \quad |i - j| \neq 1 \pmod{n} \right\}.$$

II. Partitions with a fixed set of long edges π . For a subset π satisfying $\pi \subset \mathbb{Z}_n^{\text{off}}$, we denote by

$$TSP(\mathcal{P}_{\mathbf{a}}, \sigma, \pi) := \left\{ \Gamma_{\mathbf{a}} \in TSP(\mathcal{P}_{\mathbf{a}}) : \mathcal{F}_{\text{long}}(\Gamma_{\mathbf{a}}, \sigma) = \pi \right\}, \quad \pi \subset \mathbb{Z}_n^{\text{off}},$$

the subset of $TSP(\mathcal{P}_{\mathbf{a}})$ with π as the collection of long internal edges (which may be empty, i.e., $\pi = \emptyset$).

Example: In the case $\pi = \emptyset$, $\Gamma_{\mathbf{a}}$ contains no internal long edges. We will ignore all short edges and use a big dot representing some tree structure consisting entirely of short edges. In previous band papers [71] and [72], we called this dot a molecule.

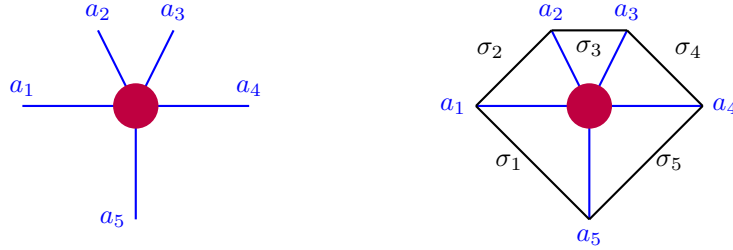


Figure 9: Single molecule partition

Example: In the following example in Figure 10, we have

$$\mathbf{a} = (a_1, a_2, \dots, a_{10}), \quad \pi = \{\{1, 5\}, \{5, 7\}, \{8, 1\}\} \quad (3.37)$$

Then $\Gamma_{\mathbf{a}}$ has the following structure. The big dots are connected via internal long edges. Locally, each sub-tree containing a red dot $\mathcal{M}$ and the edges incident to this $\mathcal{M}$ matches a tree structure of a single-molecule partition. More precisely, for these 4 local sub-trees, we have

$$\begin{aligned} \Gamma^{(top)} &\in TSP(\mathcal{P}_{(a_1, a_2, a_3, a_4, c_1)}, (\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5), \pi = \emptyset) \\ \Gamma^{(left)} &\in TSP(\mathcal{P}_{(a_9, a_{10}, c_2, a_8)}, (\sigma_9, \sigma_{10}, \sigma_1, \sigma_8), \pi = \emptyset) \\ \Gamma^{(middle)} &\in TSP(\mathcal{P}_{(a_7, c_2, c_1, c_3)}, (\sigma_7, \sigma_8, \sigma_1, \sigma_5), \pi = \emptyset) \\ \Gamma^{(right)} &\in TSP(\mathcal{P}_{(a_5, a_6, c_3)}, (\sigma_5, \sigma_6, \sigma_7), \pi = \emptyset) \end{aligned} \quad (3.38)$$

Here c_1 , c_2 and c_3 are not in the initial tree $\Gamma_{\mathbf{a}}$. We only use them to represent the local structure.

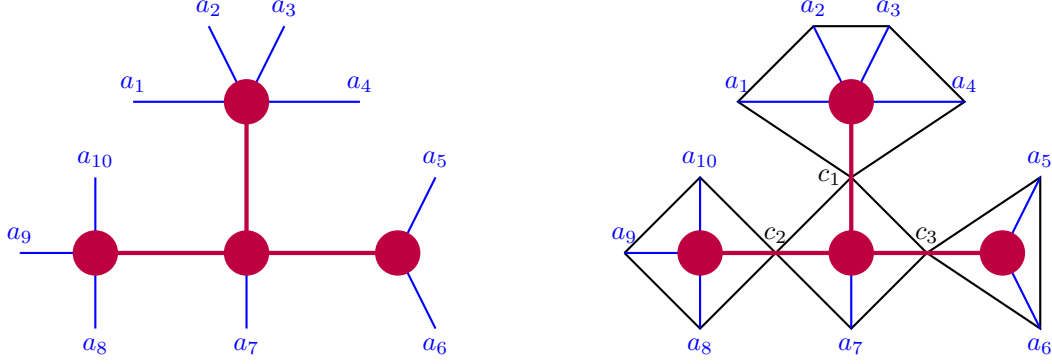


Figure 10: Multiple molecule Tree

Definition 3.9 (Definition of $\mathcal{K}^{(\pi)}$ and $\Sigma^{(\pi)}$). *I. Resummation of the kernel $\mathcal{K}$. Given a subset*

$$\pi \subset \mathbb{Z}_n^{\text{off}} \quad (3.39)$$

we define $\mathcal{K}^{(\pi)}$ to be the sum of all trees whose long edges are specified by π , namely

$$\mathcal{K}_{t,\sigma,a}^{(\pi)} = \sum_{\Gamma_a \in TSP(\mathcal{P}_a, \sigma, \pi)} \Gamma_a(t, \sigma). \quad (3.40)$$

Here $\Gamma_a(t, \sigma)$ is defined in Definition 3.3. Clearly, for $n \geq 2$,

$$\mathcal{K}_{t,\sigma,a} = W^{-n+1} \cdot m_\sigma \cdot \sum_{\pi} \mathcal{K}_{t,\sigma,a}^{(\pi)}, \quad m_\sigma = \prod_i m_i. \quad (3.41)$$

Notice that there is a factor $m_\sigma W^{-n+1}$ in the last equation due to our convention that $\mathcal{K}^{(\pi)}$ is independent of W .

II. Definition of the self-energy. We define the self-energy $\Sigma^{(\pi)}$ of $\mathcal{K}^{(\pi)}$ as follows. Relabel the vertices by letting d_i denote the endpoint of the boundary edge that starts at a_i . When two edges ending at d_i and d_j meet, we identify their endpoints by inserting a Kronecker delta. We label the remaining internal vertices by $s_1, s_2, \dots$. The self-energy is obtained by removing the boundary edges from $\mathcal{K}_{t,\sigma,a}^{(\pi)}$ and summing over all internal indices s . Clearly, we have

$$\Sigma^{(\pi)}(t, \sigma, \mathbf{d}) : \quad \mathcal{K}_{t,\sigma,a}^{(\pi)} = \sum_{\mathbf{d}} \left(\Sigma^{(\pi)}(t, \sigma, \mathbf{d}) \right) \cdot \prod_{i=1}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} \quad (3.42)$$

Example: For Figure 6 with $n = 4$ and $\sigma = (+, -, +, -)$,

$$\Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) = \delta_{d_1 d_2 d_3 d_4} + \delta_{d_1 d_2} \delta_{d_3 d_4} \left(\Theta_{tm^2}^{(B)} - 1 \right)_{d_1 d_3} + \delta_{d_1 d_4} \delta_{d_2 d_3} \left(\Theta_{tm^2}^{(B)} - 1 \right)_{d_1 d_2}.$$

On the other hand, if $\sigma = (+, -, +, -)$ then $\Sigma^{(\pi)}(t, \sigma, \mathbf{d}) = 0$ if $\pi \neq \emptyset$.

By (2.52) and explicit calculations, we have $\Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) = O(1)$ and that $\Sigma^{(\emptyset)}(t, \sigma, \mathbf{d})$ is short-ranged in the sense that

$$\left| \Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) \right| \leq C_n \exp \left(-c_n \max_{ij} \|d_i - d_j\| \right). \quad (3.43)$$

In addition, we have the following sum zero property :

$$\sum_{d_1, d_2, d_3} \Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) = O(1 - t) = O(\eta_t).$$

(The name sum zero comes from the fact that the above quantity equals 0 when $t = 1$) Due to the translation invariance, the last bound is equivalent to

$$L^{-1} \cdot \sum_{d_1, d_2, d_3, d_4} \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) = O(1 - t) = O(\eta_t).$$

It turns out that this property holds for all alternating $\boldsymbol{\sigma}$ with $\pi = \emptyset$.

Lemma 3.10 (Symmetry and sum zero property). *For fixed $4 \leq n \in 2\mathbb{Z}$ and an alternating loop $\boldsymbol{\sigma}^{(alt)}$ with*

$$1 \leq k \leq n, \quad \sigma_k^{(alt)} = \begin{cases} +, & k \in 2\mathbb{Z} - 1 \\ -, & k \in 2\mathbb{Z} \end{cases}, \quad t \in [0, 1],$$

the single molecule tree graph (i.e., $\pi = \emptyset$) satisfies the following two properties:

1. $\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(alt)}, \mathbf{d})$ is translationally invariant and symmetric in the sense that

$$\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(alt)}, \mathbf{d}) = \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(alt)}, -\mathbf{d})$$

2. $\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(alt)}, \mathbf{d})$ has the following sum zero property

$$L^{-1} \cdot \sum_{\mathbf{d} \in \mathbb{Z}_L^n} \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(alt)}, \mathbf{d}) = O(1 - t) = O(\eta_t) \quad (3.44)$$

The sum-zero property is the key input for the following estimates on $\mathcal{K}^{(\pi)}$ and $\mathcal{K}$.

Lemma 3.11 (Bound on $\mathcal{K}$). *For any $\mathcal{K}_{t, \boldsymbol{\sigma}, \mathbf{a}}^{(\pi)}$ defined in Definition 3.9, we have*

$$\mathcal{K}_{t, \boldsymbol{\sigma}, \mathbf{a}}^{(\pi)} = O_{\prec}((\ell_t \cdot \eta_t)^{-n+1}) \quad (3.45)$$

Together with (3.41), we have

$$\mathcal{K}_{t, \boldsymbol{\sigma}, \mathbf{a}} = O_{\prec}((W\ell_t \cdot \eta_t)^{-n+1}). \quad (3.46)$$

Notice that there is no W factor in $\mathcal{K}^{(\pi)}$ due to its definition.

3.4 Proof of Lemma 3.10

Proof of Lemma 3.10. We note that the symmetry and translational-invariance properties are simple consequences of the explicit expression for $\Sigma^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{d})$. We now turn to the sum-zero property. By definitions of $\mathcal{K}^{(\pi)}$ and $\Sigma^{(\pi)}$, for any $n \geq 3$ we have that

$$\sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \sum_{\mathbf{d} \in \mathbb{Z}_L^n} \prod_{i=1}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} \cdot \Sigma^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{d}) \quad (3.47)$$

Using $S^{(B)} \mathbf{1} = \mathbf{1}$, we obtain

$$\sum_{a_i} \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} = \frac{1}{1 - tm_i m_{i+1}}.$$

Therefore, we have

$$\sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) \Bigg/ \sum_{\mathbf{d} \in \mathbb{Z}_L^n} \Sigma^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{d}) \sim \eta_t^{-|\{i: \sigma_i \neq \sigma_{i+1}, i \in \mathbb{Z}_n\}|}. \quad (3.48)$$

By (3.41) and Corollary 3.7, we have

$$L^{-1} \cdot \sum_{\pi} \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = O(\eta_t^{-n+1}). \quad (3.49)$$

We now show that it holds without sum over π , i.e., for any fixed σ , $n \geq 3$ and π in (3.39),

$$L^{-1} \cdot \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a}) = O(\eta_t^{-n+1}) \quad (3.50)$$

Assuming that the last equation holds in the case that $\pi = \emptyset$ and $\sigma = \sigma^{(alt)}$, together with (3.48), we have

$$L^{-1} \cdot \sum_{\mathbf{d} \in \mathbb{Z}_L^n} \Sigma^{(\emptyset)}(t, \sigma^{(alt)}, \mathbf{d}) = O(\eta_t). \quad (3.51)$$

This implies the desired result (3.44).

We now prove (3.50) by induction. If $n = 3$, then π can only be $\emptyset$. Hence (3.49) implies (3.50) in the case $n = 3$. Next, we assume that (3.50) holds with n replaced by $m < n$. Under this assumption, we first prove that if $\pi \neq \emptyset$ then (3.50) holds, i.e.,

$$\pi \neq \emptyset \implies L^{-1} \cdot \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a}) = O(\eta_t^{-n+1}) \quad (3.52)$$

If $\pi \neq \emptyset$, we can always represent $\mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a})$ with the molecule structure and the self-energy $\Sigma^{(\emptyset)}$. For example, for $n = 10$ and π in (3.37), the molecule structure is the one in Figure 10. As shown in Figure 11, let d_i denote the vertex connected to a_i (these vertices are not marked in the figure), and let c_i denote the vertices of edges connecting molecules (i.e., big dots). Then

$$\mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a}) = \sum_{\mathbf{d}} \sum_{\mathbf{c}} \left(\prod_{i=1}^{10} \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^3 \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=1}^4 \Sigma^{(\emptyset)}(t, \sigma^{(k)}, \mathbf{d}^{(k)})$$

Here $\Sigma^{(\emptyset)}(t, \sigma^{(k)}, \mathbf{d}^{(k)})$, $1 \leq k \leq 4$, represent the four self-energies (i.e., four big dots, top, bottom, left,

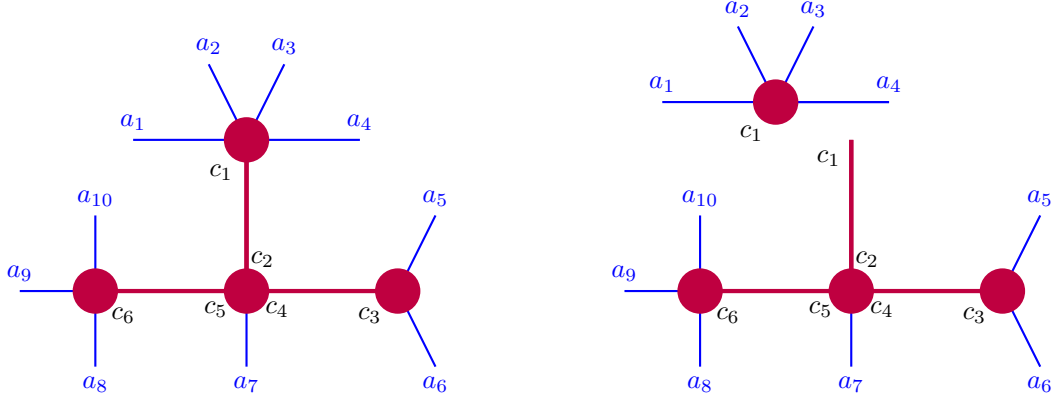


Figure 11: Decomposition of $\mathcal{K}^{(\pi)}$

right). More precisely,

$$\mathbf{d}^{(top)} = (c_1, d_1, d_2, d_3, d_4), \quad \mathbf{d}^{(left)} = (c_6, d_8, d_9, d_{10}), \quad \mathbf{d}^{(bottom)} = (c_2, c_4, d_7, c_5), \quad \mathbf{d}^{(right)} = (c_3, d_5, d_6),$$

and $\mathcal{K}^{(\pi)}$ is decomposed into three parts:

- Edges incident to external vertices, i.e., the (blue) boundary edges.
- Edges connecting two different molecules, each weighted by $(\Theta_{t|m|^2} - 1)$.
- The cores $\Sigma^{(\emptyset)}(t, \sigma^{(k)}, \mathbf{d}^{(k)})$ of the individual molecules.

One can easily extend it to the general cases. Given a set π (which can be the empty set), we label all vertices of long internal edges for π by c_i , ($1 \leq i \leq 2|\pi|$). Recall that the vertex connecting with the boundary vertex a_i is denoted by d_i . We now denote all internal vertices other than d_i, c_j by s_k . The index set for k has fewer than n elements, but we will not specify it. We now explain how to construct the molecule and their tree structure. Given $\{i, j\} \in \pi$, we draw a line in the polygon from the center of edge i to that of j . In this way, we have a partition of the polygon. The set π for which there is $\Gamma_{\mathbf{a}}$ such that $\mathcal{F}_{\text{long}}(\Gamma_{\mathbf{a}}, \boldsymbol{\sigma}) = \pi$ satisfies that these lines representing the pairing are non-crossing. From now on, we will call π a noncrossing pairing. Given a non-crossing pairing, we divide the polygon into several regions, say, M regions (note that $M = |\pi| + 1$). We represent each region by a big dot (molecule), and there is an edge connecting two dots if and only if these two regions are neighboring. Notice that each dot typically has many vertices connecting to it, as shown in Figure 12.

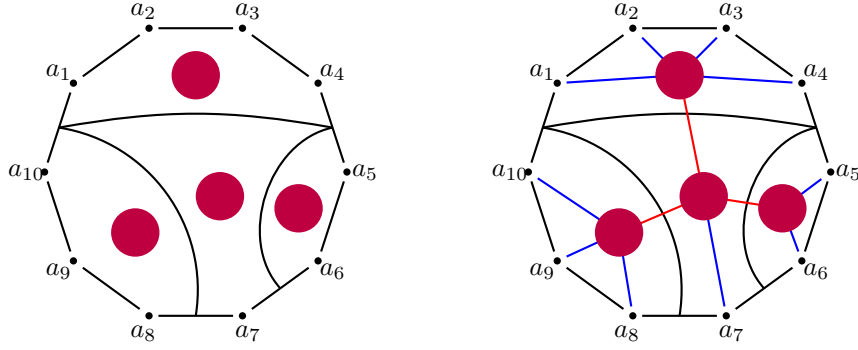


Figure 12: $\pi = \{\{1, 5\}, \{5, 7\}, \{8, 1\}\}$

In general, there are complicated structures inside these molecules; there are short edges and other vertices labeled by s_k . All vertices labeled by s_k are required to be summed. With this convention, for π with M molecules, we can write

$$\mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{\mathbf{d}} \sum_{\mathbf{c}} \left(\prod_{i=1}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^{M-1} \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=1}^M \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{d}^{(k)}) \quad (3.53)$$

Here the vertices labelled by s_k are summed and thus they no longer appear explicitly in the formula above. Summing over $\mathbf{a}$, we have

$$\sum_{\mathbf{a}} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{\mathbf{d}, \mathbf{c}} \left(\prod_{i=1}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^{M-1} \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=1}^M \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{d}^{(k)}) \quad (3.54)$$

Given a molecule structure (or equivalently a set π representing non-crossing pairings), there must be a molecule containing just one c vertex, i.e., the big dot for this molecule connects to only one $(\Theta_{t|m|^2} - 1)_{c_{2k-1} c_{2k}}$ edge. (For example, the top, left and right molecules in Figure 11). In a different language, this molecule represents a region with exactly one pairing line. For simplicity, we assume that it is the first molecule containing c_1 and connecting with $a_1, a_2, \dots, a_{m-1}$. With these notations, we have

$$\{1, m\} \in \pi, \quad \sigma_1 \neq \sigma_m \quad (3.55)$$

The expression in the formula of $\mathcal{K}^{(\pi)}$ related to this molecule is

$$\sum_{d_1, \dots, d_{m-1}} \Sigma^{(\emptyset)} \left(t, (\sigma_1, \dots, \sigma_m), (d_1, \dots, d_{m-1}, c_1) \right) \cdot \left(\prod_{i=1}^{m-1} \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right)$$

(For example: the top right part in Figure 11 is for the case $m = 5$.) Notice that $d_i, a_i, 1 \leq i \leq m-1$ do

not appear in other molecules. The following part is separated from other parts in (3.54), i.e.,

$$f^*(c_1) := \sum_{a_1, \dots, a_{m-1}} \sum_{d_1, \dots, d_{m-1}} \Sigma^{(\emptyset)} \left(t, (\sigma_1, \dots, \sigma_m), (d_1, \dots, d_{m-1}, c_1) \right) \cdot \left(\prod_{i=1}^{m-1} \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right)$$

Now we can write the $\mathcal{K}^{(\pi)}$ in (3.54) in the terms of $f^*(c_1)$ as follows:

$$\begin{aligned} \sum_{\mathbf{a}} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) &= \sum_{c_1} f^*(c_1) \cdot \sum_{a_m, \dots, a_n} \sum_{d_m, \dots, d_n} \sum_{c_2, \dots, c_{2M-2}} \\ &\quad \left(\prod_{i=m}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^{M-1} \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=2}^M \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{d}^{(k)}) \end{aligned} \quad (3.56)$$

In the Figure 11, the upper right part represents the $f^*(c_1)$ and the lower right part represents the 2nd line of the (3.56).

By translation invariance, $f^*(c_1)$ does not depend on c_1 , i.e., $f^*(c_1) = f^*(1) \in \mathbb{C}$. Inserting it back to (3.56), we obtain that

$$\begin{aligned} \sum_{\mathbf{a}} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) &= f^*(1) \cdot \sum_{a_m, \dots, a_n} \sum_{d_m, \dots, d_n} \sum_{\mathbf{c}} \\ &\quad \left(\prod_{i=m}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^{M-1} \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=2}^M \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{d}^{(k)}) \end{aligned} \quad (3.57)$$

Since c_1 appears only in the internal edge $(\Theta_{t|m|^2} - 1)_{c_1 c_2}$, we can sum over c_1 . By definition of $\Theta_{t|m|^2}$, $S^{(B)} \mathbf{1} = \mathbf{1}$ and $|m| = 1$, we have

$$\sum_a (\Theta_{t|m|^2} - 1)_{ab} = \frac{t}{1-t} = t \sum_a (\Theta_{t|m|^2})_{ab}$$

Thus we can replace $(\Theta_{t|m|^2}^{(B)} - 1)_{c_1 c_2}$ in (3.57) with $t(\Theta_{t|m|^2}^{(B)})_{c_1 c_2}$. This replacement shows that after summing over c_1 , the internal edge $c_1 - c_2$ becomes an external edge with a factor t . This will be crucial later on when we split the graph. We can now rewrite

$$\begin{aligned} \sum_{\mathbf{a}} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) &= t \cdot f^*(1) \cdot \sum_{a_m, \dots, a_n} \sum_{d_m, \dots, d_n} \sum_{\mathbf{c}} \\ &\quad \left(\prod_{i=m}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \left(\Theta_{t|m|^2}^{(B)} \right)_{c_1 c_2} \prod_{k=2}^{M-1} \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=2}^M \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{d}^{(k)}) \end{aligned} \quad (3.58)$$

Define π' to be π with the pair for edge $\{c_1, c_2\}$ removed, i.e.

$$\pi' = \pi \setminus \{\{1, m\}\}$$

Denote

$$\boldsymbol{\sigma}' = (\sigma_1, \sigma_m, \sigma_{m+1}, \dots, \sigma_n), \quad \mathbf{a}' = (c_1, a_m, a_{m+1}, \dots, a_n)$$

Then we have (see Figure 11 for an example)

$$\sum_{\mathbf{a}} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = t \cdot f^*(1) \cdot \sum_{\mathbf{a}'} \mathcal{K}^{(\pi')}(t, \boldsymbol{\sigma}', \mathbf{a}') \quad (3.59)$$

Notice that $\{c_1, c_2\}$ is a boundary edge in $\mathcal{K}^{(\pi')}(t, \boldsymbol{\sigma}', \mathbf{a}')$ and it needs to be of the form $\left(\Theta_{t|m|^2}^{(B)} \right)_{c_1 c_2}$. Since $\mathbf{a}' \in \mathbb{Z}_L^{n-m+2}$, we can apply the induction hypothesis (3.50) to $\mathcal{K}^{(\pi')}$. Thus

$$\sum_{\mathbf{a}} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = O(t \cdot f^*(1) \cdot L \cdot (\eta_t)^{-n+m-1}) \quad (3.60)$$

Multiplying $(\Theta_{t|m|^2}^{(B)})_{c_1,a} = (\Theta_{tm_1m_m}^{(B)})_{c_1,a}$ to $f^*(c_1)$ and summing up c_1 and a , we obtain

$$\begin{aligned} & \sum_a \sum_{c_1} (\Theta_{tm_1m_m}^{(B)})_{c_1,a} \cdot f^*(c_1) \\ &= \sum_{a, c_1} \sum_{a_1, \dots, a_{m-1}} \sum_{d_1, \dots, d_{m-1}} \Sigma^{(\emptyset)} \left(t, (\sigma_1, \dots, \sigma_m), (d_1, \dots, d_{m-1}, c_1) \right) \cdot \left(\prod_{i=1}^{m-1} (\Theta_{tm_i m_{i+1}}^{(B)})_{a_i d_i} \right) (\Theta_{tm_1 m_m}^{(B)})_{c_1, a} \end{aligned} \quad (3.61)$$

By definition, the right hand side can be written in terms of $\mathcal{K}^{(\emptyset)}(t, (\sigma_1, \dots, \sigma_m), (a_1, \dots, a_{m-1}, a))$, namely,

$$\sum_a \sum_{c_1} f^*(c_1) (\Theta_t^{(B)})_{c_1, a} = \sum_{a_1, \dots, a_{m-1}, a} \mathcal{K}^{(\emptyset)} \left(t, (\sigma_1, \dots, \sigma_m), (a_1, \dots, a_{m-1}, a) \right). \quad (3.62)$$

By the induction hypothesis (3.50), the right-hand side of (3.62) is $O(L \cdot \eta_t^{-m+1})$. Thus

$$\sum_a \sum_{c_1} f^*(c_1) (\Theta_t^{(B)})_{c_1, a} = O(L \cdot \eta_t^{-m+1}) \quad (3.63)$$

On the other hand, since $f^*(c_1) = f^*(1)$, we have the identity

$$\sum_a \sum_{c_1} f^*(c_1) (\Theta_t^{(B)})_{c_1, a} = L \cdot f^*(1) \cdot (1-t)^{-1}. \quad (3.64)$$

Hence

$$f^*(c_1) = f^*(1) = O(\eta_t^{-m+2})$$

Together with (3.60), we have proved (3.50) if $\pi \neq \emptyset$.

For $\pi = \emptyset$, we write $\mathcal{K}^{(\emptyset)}$ as

$$\sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{\pi} \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) - \sum_{\pi \neq \emptyset} \sum_{\mathbf{a} \in \mathbb{Z}_L^n} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = O(L \cdot \eta_t^{-n+1}) \quad (3.65)$$

Here we have used (3.49) to bound the first term on the right hand side and (3.50) for the second term. We have thus proved (3.50) and Lemma 3.10. $\square$

3.5 Proof of Lemma 3.11

Proof of Lemma 3.11. We first focus on the case $\pi = \emptyset$. By symmetry, without loss of generality, we can split it into three cases

1. Pure loop, i.e., $\sigma_k = +$ for all $1 \leq k \leq n$ or $\sigma_k = -$ for all $1 \leq k \leq n$. In this case, all edges are short edges, hence we can easily obtain (3.45).
2. $\sigma_1 = +$, $\sigma_2 = -$, and there exists j s.t. $\sigma_j = \sigma_{j+1} = +$.
3. $\boldsymbol{\sigma}$ is alternating as in (3.44), i.e. $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{(alt)}$.

Recall that for $\sigma_1 \neq \sigma_2$, and $\pi = \emptyset$, we have

$$\begin{aligned} \mathcal{K}_{t, \boldsymbol{\sigma}, \mathbf{a}}^{(\emptyset)} &= \sum_{\mathbf{d}} \left(\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) \right) \cdot \prod_{i=1}^n (\Theta_{tm_i m_{i+1}}^{(B)})_{a_i, d_i} \\ &= \sum_{d_1} (\Theta_{t|m|^2}^{(B)})_{a_1, d_1} \cdot \sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) \right) \cdot \prod_{i=2}^n (\Theta_{tm_i m_{i+1}}^{(B)})_{a_i, d_i} \end{aligned}$$

We are going to prove the following statement, which is slightly stronger than (3.45):

$$\sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) \right) \cdot \prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} = O((\ell_t \eta_t)^{-n+2}) \cdot \left(\min_{2 \leq k \leq n} \|a_k - d_1\| + 1 \right)^{-1} \quad (3.66)$$

First, if there exists $2 \leq j \leq n$ s.t. $\sigma_j = \sigma_{j+1} = +$, then (2.52) shows that

$$\left(\Theta_{tm_j m_{j+1}}^{(B)} \right)_{a_j, d_j} = \left(\Theta_{tm^2}^{(B)} \right)_{a_j, d_j} = O(1)$$

It implies that

$$\prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} = O((\ell_t \eta_t)^{-n+2})$$

On the other hand, due to the short range property of $\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d})$ (in (3.43)), we have

$$\left| \sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) \right) \cdot \prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} \right| \leq C \prod_{i=2}^n \left\| \Theta_{tm_i m_{i+1}}^{(B)} \right\|_{\max}$$

By combining these two bounds, we derive (3.66) for this case.

Next we prove (3.66) in the case that $\pi = \emptyset$ and $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{(alt)}$. In this case,

$$\prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} = \prod_{i=2}^n \left(\Theta_{t|m|^2}^{(B)} \right)_{a_i, d_i}$$

With short range property of $\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d})$ (in (3.43)), and the estimate of $(\Theta_{t|m|^2}^{(B)})_{a_i d_i}$ in (2.52), one can easily bound the left-hand side of (3.66) by $O((\ell_t \eta_t)^{-n+1})$. To obtain the missing factor for (3.66), we need to apply the sum zero property which we proved in Lemma 3.10, i.e.,

$$\sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) \right) = O(\eta_t) \quad (3.67)$$

Here we use the translation invariance of $\Sigma^{(\emptyset)}$. More precisely, the left-hand side of (3.67) is independent of d_1 . Hence the full sum over $d_1, \dots, d_n$ is L times this partial sum. Lemma 3.10, applied to the full sum, therefore yields (3.67). For simplicity, for fixed d_1 and $\mathbf{a}$, we temporarily denote s, f and g as follows

$$s_i := d_i - d_1, \quad f(a_i, s_i) := \left(\Theta_{t|m|^2}^{(B)} \right)_{a_i, (d_1 + s_i)}, \quad g(s_2, \dots, s_n) := \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d})$$

Then the l.h.s. of (3.66) can be written as

$$\sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) \right) \cdot \prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} = \sum_{s_2, \dots, s_n} \prod_{i=2}^n f(a_i, s_i) \cdot g(\mathbf{s}) \quad (3.68)$$

Note due the fast decay of $g(\mathbf{s})$, we can focus on the case that $\max_i |s_i| \prec 1$. By Lemma 3.10, we have

$$g(\mathbf{s}) = g(-\mathbf{s}), \quad \sum_{s_2, \dots, s_n} g(s_2, \dots, s_n) = O(\eta_t).$$

For each i , we write

$$f(a_i, s_i) = f_0(a_i, s_i) + f_1(a_i, s_i) + f_2(a_i, s_i)$$

where

$$f_0(a_i, s_i) = f(a_i, 0) \quad (3.69)$$

$$f_1(a_i, s_i) = \frac{1}{2} (f(a_i, s_i) - f(a_i, -s_i))$$

$$f_2(a_i, s_i) = f(a_i, s_i) - f(a_i, 0) - \frac{1}{2} (f(a_i, s_i) - f(a_i, -s_i))$$

By Lemma 2.14

$$\begin{aligned} f_0(a_i, s_i) &= O((\ell_t \eta_t)^{-1}), \\ f_1(a_i, s_i) &= O(\ell_t^{-1} \eta_t^{-1/2}), \\ f_2(a_i, s_i) &= O((\|a_i - d_1\| + 1)^{-1}). \end{aligned} \tag{3.70}$$

Inserting

$$\prod_i f(a_i, s_i) = \prod_i (f_0(a_i, s_i) + f_1(a_i, s_i) + f_2(a_i, s_i))$$

into (3.68), we obtain

$$\sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{d}) \right) \cdot \prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} = \sum_{s_2, \dots, s_n} \sum_{0 \leq \xi_2, \xi_3, \dots, \xi_n \leq 2} \prod_{i=2}^n f_{\xi_i}(a_i, s_i) \cdot g(\mathbf{s}) \tag{3.71}$$

We claim that for g and f satisfying the estimates just stated and any fixed ξ 's,

$$\sum_{s_2, \dots, s_n} \prod_{i=2}^n f_{\xi_i}(a_i, s_i) \cdot g(\mathbf{s}) = O((\ell_t \eta_t)^{-n+2}) \cdot \left(\min_{2 \leq k \leq n} \|a_k - d_1\| + 1 \right)^{-1}. \tag{3.72}$$

We estimate the l.h.s. in the following cases:

- If $\xi_k = 2$ for some k , assume without loss of generality that $\xi_n = 2$. Then

$$\prod_{i=2}^n f_{\xi_i}(a_i, s_i) = O((\ell_t \eta_t)^{-n+2}) \cdot \|a_n - d_1\|^{-1}$$

It implies that in this case (3.72) holds.

- If $0 \leq \xi_k \leq 1$ for all $2 \leq k \leq n$ and two of the ξ_k are equal to 1, then, similarly to the previous case,

$$\prod_{i=2}^n f_{\xi_i}(a_i, s_i) = O((\ell_t \eta_t)^{-n+1}) \cdot \eta_t = O((\ell_t \eta_t)^{-n+2}) \cdot \ell_t^{-1}$$

If $\|a_k - d_1\| = O(\ell_t)$ for all $2 \leq k \leq n$, this implies (3.72); if $\|a_k - d_1\| \gg \ell_t$ for some $2 \leq k \leq n$, then the l.h.s. of (3.72) is exponentially small.

- If $0 \leq \xi_k \leq 1$ for all $2 \leq k \leq n$ and exactly one of the ξ_k equals 1, assume without loss of generality that $\xi_n = 1$. Similarly to the previous case,

$$\prod_{i=2}^n f_{\xi_i}(a_i, s_i) g(\mathbf{s}) = \prod_{i=2}^{n-1} f(a_i, 0) f_1(a_n, s_n) g(\mathbf{s})$$

By definition, we have

$$f_1(a_n, s_n) = -f_1(a_n, -s_n).$$

By symmetry and the property $g(\mathbf{s}) = g(-\mathbf{s})$, the left-hand side of (3.72) equals zero in this case.

- At last, if $\xi_k = 0$ for all $2 \leq k \leq n$. Then

$$\prod_{i=2}^n f_{\xi_i}(a_i, s_i) g(\mathbf{s}) = \prod_{i=2}^n f(a_i, 0) g(\mathbf{s})$$

Applying (3.67), we have

$$\sum_{\mathbf{s}} g(\mathbf{s}) = O(\eta_t)$$

It implies that in this case (3.72) holds.

Therefore, we have proved that for any ξ 's, the (3.72) holds, which completes the proof of (3.66) for the case of $\pi = \emptyset$ and $\sigma_1 = +$, $\sigma_2 = -$.

We now prove (3.45) in the case $\pi \neq \emptyset$. As above, we use the decomposition method as in Figure 11. By (3.53) we write

$$\mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a}) = \sum_{\mathbf{d}} \sum_{\mathbf{c}} \left(\prod_{i=1}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^{M-1} \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=1}^M \Sigma^{(\emptyset)}(t, \sigma^{(k)}, \mathbf{d}^{(k)}) \quad (3.73)$$

Here, each molecule contains at least one c -vertex; equivalently, each molecule has degree at least one with respect to the internal $(\Theta - 1)$ edges. On the other hand, if every molecule had degree larger than one, then the molecule graph would contain an internal $(\Theta - 1)$ -edge cycle, contradicting its tree structure. Therefore at least one molecule has degree one, i.e. the big dot corresponding to this molecule connects to only one $(\Theta_{t|m|^2}^{(B)} - 1)_{c_{2k-1}, c_{2k}}$ edge. (For example, the top, left and right molecules in Figure 11). For simplicity, we assume that it is the first molecule, and it contains c_1 and connects with $a_1, a_2, \dots, a_{m-1}$. Note that in this case

$$\{1, m\} \in \pi, \quad \sigma_1 \neq \sigma_m \quad (3.74)$$

As in the r.h.s of Figure 11, we decompose $\mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a})$ into a product of two parts, $\mathcal{A}$ and $\mathcal{B}$, such that

$$\mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a}) = \sum_{c_1} (\mathcal{A}_{a_1, \dots, a_{m-1}, c_1}) \cdot (\mathcal{B}_{c_1, a_m, \dots, a_n}) \quad (3.75)$$

$$\mathcal{A}_{a_1, \dots, a_{m-1}, c_1} = \sum_{d_1, \dots, d_{m-1}} \left(\prod_{i=1}^{m-1} \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \Sigma^{(\emptyset)}(t, \sigma^{(1)}, \mathbf{d}^{(1)}), \quad \mathbf{d}^{(1)} = (d_1, d_2, \dots, d_{m-1}, c_1)$$

$$\mathcal{B}_{c_1, a_m, \dots, a_n} = \sum_{d_m, \dots, d_n} \sum_{c_2, \dots, c_{2M-2}} \left(\prod_{i=m}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^{M-1} \left(\Theta_{t|m|^2} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=2}^M \Sigma^{(\emptyset)}(t, \sigma^{(k)}, \mathbf{d}^{(k)}).$$

Using (3.66), we have

$$\mathcal{A}_{a_1, \dots, a_{m-1}, c_1} = O((\ell_t \eta_t)^{-m+2}) \cdot \left(\min_{1 \leq k \leq m} \|a_k - c_1\| + 1 \right)^{-1}$$

On the other hand, with the induction hypothesis and $(\Theta_t - 1)_{c_1 c_2} = t (S^{(B)} \cdot \Theta_t)_{c_1 c_2}$, we have

$$\mathcal{B}_{c_1, a_m, \dots, a_n} = O((\ell_t \eta_t)^{-n+m-1})$$

Combining these two bounds and inserting them back to (3.75), we obtain (3.45) for the case that $\pi \neq \emptyset$, and complete the proof of Lemma 3.11 $\square$

4 G -entry estimates

In this section, we bound G -entries and 1-loops in terms of 2-loops. The proof of Lemma 4.1 uses a time-independent method. It relies on a standard decomposition widely employed for Wigner matrices (e.g., [39]) and band matrices (e.g., [34]).

Lemma 4.1 (Resolvent entry estimate). *Recall z_t and G_t (which depend on N) from Definitions 2.7 and 2.9. Suppose that $|E| < 2 - \kappa$, $0 \leq t < 1$. For a fixed constant $c > 0$, define the event*

$$\Omega(t, c) := \{ \|G_t - m\|_{\max} \leq W^{-c} \}. \quad (4.1)$$

Then the entries of G_t can be bounded in terms of 2- G -loops as follows:

$$\mathbf{1}_{\Omega(t, c)} \cdot \max_{i \in \mathcal{I}_a} \max_{j \in \mathcal{I}_b} |(G_t)_{ij}|^2 \prec \sum_{a'=a-1}^{a+1} \sum_{b'=b-1}^{b+1} \mathcal{L}_{t, (+, -), (a', b')} + W^{-1} \cdot \mathbf{1}(|a - b| \leq 1), \quad (4.2)$$

$$\mathbf{1}_{\Omega(t,c)} \cdot \max_i |(G_t)_{ii} - m|^2 \prec \max_{a,b} \mathcal{L}_{t,(+,-),(a,b)}. \quad (4.3)$$

In particular, if for some $c > 0$,

$$\|G_t - m\|_{\max} \prec W^{-c}, \quad (4.4)$$

then (4.2) and (4.3) hold without the indicator $\mathbf{1}_{\Omega(t,c)}$. Furthermore, under the same assumption, the 1-loop estimate (interpreted as an average local law) holds:

$$\max_a |\langle (G_t - m) E_a \rangle| \prec \max_{a,b} \mathcal{L}_{t,(+,-),(a,b)}. \quad (4.5)$$

We first recall the following perturbation formulas from Lemma 4.2 of [38]:

Lemma 4.2. *Let H be a Hermitian matrix and let $H^{(i)}$ denote the $(N-1) \times (N-1)$ submatrix of H obtained by removing the i -th row and column. Define*

$$G_{kl}^{(i)} := \left[H^{(i)} - z \right]^{-1}(k, l). \quad (4.6)$$

Then we have

$$G_{ii} = \left(H_{ii} - z - \sum_{kl} H_{ik} G_{kl}^{(i)} H_{li} \right)^{-1}, \quad (4.7)$$

$$G_{ij} = -G_{ii} \sum_k H_{ik} G_{kj}^{(i)}, \quad (4.8)$$

$$G_{jk}^{(i)} = G_{jk} - \frac{G_{ji} G_{ik}}{G_{ii}}. \quad (4.9)$$

Proof of Lemma 4.1. For simplicity, we ignore the subscript t in our proof and write $\Omega = \Omega(t, c)$. For $i \neq j$, using (4.8), we have

$$|G_{ij}| = |G_{ii}| \left| \sum_k H_{ik} G_{kj}^{(i)} \right|.$$

Here $\mathbb{E}|H_{ij}|^2 = t \cdot S_{ij} \leq S_{ij}$. By definition in (4.4), we have $\mathbf{1}_{\Omega} |G_{ii}| = O(1)$. Since $G^{(i)}$ is independent of the i -th row of H , we can apply Lemma 3.3 in [39] to get

$$\left| \sum_k H_{ik} G_{kj}^{(i)} \right| \prec \left(\sum_k S_{ik} |G_{kj}^{(i)}|^2 \right)^{1/2}.$$

Using (4.9), we further deduce that

$$\mathbf{1}_{\Omega} \cdot |G_{kj}^{(i)}| \leq |G_{kj}| + O(W^{-c} |G_{ij}|).$$

Combining these estimates, we obtain

$$\mathbf{1}_{\Omega} \cdot |G_{ij}|^2 \prec \sum_k S_{ik} |G_{kj}|^2 + O(W^{-c} |G_{ij}|^2) \prec \sum_k S_{ik} |G_{kj}|^2. \quad (4.10)$$

Applying this bound on G_{kj} and iterating the process (for the special case $k = j$, we can bound $G_{kk} \prec 1$), we find

$$\mathbf{1}_{\Omega} \cdot |G_{ij}|^2 \prec \sum_{kl} S_{ik} |G_{kl}|^2 S_{lj} + W^{-1} \mathbf{1}(|[j] - [i]| \leq 1). \quad (4.11)$$

This implies (4.2).

Now we prove (4.3). Using (4.7) and Lemma 3.3 in [39], we get

$$\sum_{kl} H_{ik} G_{kl}^{(i)} H_{li} = \sum_k t S_{ik} G_{kk}^{(i)} + O \prec \left(\left(\sum_{kl} S_{ik} |G_{kl}^{(i)}|^2 S_{li} \right)^{1/2} \right).$$

As above, using (4.9) to remove the (i) superscript and applying (4.2), we find

$$\mathbf{1}_\Omega \cdot \sum_{kl} H_{ik} G_{kl}^{(i)} H_{li} = \mathbf{1}_\Omega \cdot \sum_k t S_{ik} G_{kk} + O_{\prec}(\mathbf{1}_\Omega \cdot \mathcal{E}),$$

where

$$\mathcal{E}^2 = \mathbf{1}_\Omega \cdot \max_{a,b} \mathcal{L}_{t,(+,-),(a,b)} + \left(\max_{a,b} \mathcal{L}_{t,(+,-),(a,b)} \right)^2 + W^{-1}.$$

On the other hand, it is easy to verify that in $\Omega(t, c)$,

$$cW^{-1} \leq \max_{a,b} \mathcal{L}_{t,(+,-),(a,b)} \leq CW^{-1} + W^{-c}.$$

Substituting back into (4.7), we get

$$\mathbf{1}_\Omega \cdot G_{ii} = \mathbf{1}_\Omega \left(-z - \sum_k t S_{ik} G_{kk} - O_{\prec} \left(\left(\max_{a,b} \mathcal{L}_{t,(+,-),(a,b)} \right)^{1/2} \right) \right)^{-1}.$$

By definition, one can easily check that $m = -(tm+z)^{-1}$. Expanding the right-hand side around $(-z-tm)^{-1}$, we have

$$\mathbf{1}_\Omega \cdot (G_{ii} - m) = \sum_j [(1 - tm^2 S)^{-1}]_{ij} \cdot \mathcal{E}_j, \quad \mathcal{E}_j \prec \left(\max_{a,b} \mathcal{L}_{t,(+,-),(a,b)} \right)^{1/2} + W^{-c} \cdot \max_j |G_{jj} - m|.$$

Together with the fact that $\|(1 - tm^2 S)^{-1}\|_{\max \rightarrow \max} = O(1)$, we conclude (4.3).

Next, we prove (4.5), which is a type of estimate commonly referred to as fluctuation averaging. A brief historical context is provided in Section 10.3.1 of [37]. In particular, very similar results are established in equation (3.7) of [34] and equation (4.11) of [40].

Recall the definition of $H^{(i)}$ introduced above (4.6). Denote $\mathbb{E}_i[X] = \mathbb{E}[X \mid H^{(i)}]$, i.e., the conditional expectation with respect to the i -th row and column of H . Previously, we showed that

$$\max_{i,j} |G_{ij} - m \delta_{ij}|^2 \prec \Psi^2 := \max_{\mathbf{a}} \mathcal{L}_{t,(+,-),\mathbf{a}},$$

where Ψ aligns with the notation in [40]. It is established in equation (4.11) of [40] that, for any $\{t_k\}_{k \in \mathbb{Z}_N}$ satisfying

$$0 \leq |t_k| \leq W^{-1}, \quad \sum_k |t_k| \leq 1,$$

we have

$$\sum_k t_k (1 - \mathbb{E}_k)(G_{kk} - m) \prec \Psi^2 = \max_{\mathbf{a}} \mathcal{L}_{t,(+,-),\mathbf{a}}. \quad (4.12)$$

On the other hand, using the identity $G - m = m(-H - tm)G$ and applying Gaussian integration by parts, we obtain

$$\mathbb{E}_i(G_{ii} - m) = \mathbb{E}_i[m(-H - tm)G]_{ii} = \sum_k \mathbb{E}_i[m(G_{kk} - m)t S_{ki} G_{ii}] = tm^2 \sum_k S_{ik} \cdot \mathbb{E}_i(G_{kk} - m) + O_{\prec}(\Psi^2).$$

Using (4.9), we deduce that

$$\mathbb{E}_i(G_{kk} - m) = \mathbb{E}_i(G_{kk}^{(i)} - m) + O_{\prec}(\Psi^2) = (G_{kk}^{(i)} - m) + O_{\prec}(\Psi^2) = G_{kk} - m + O_{\prec}(\Psi^2).$$

Substituting this into the earlier equation, we find

$$\mathbb{E}_i(G_{ii} - m) = tm^2 \sum_k S_{ik} \cdot (G_{kk} - m) + O_{\prec}(\Psi^2) = tm^2 \sum_k S_{ik} \cdot \mathbb{E}_k(G_{kk} - m) + O_{\prec}(\Psi^2),$$

where the last estimate follows from (4.12). Solving this equation, we conclude that

$$\mathbb{E}_i(G_{ii} - m) = O_{\prec}(\Psi^2).$$

Combining this with (4.12), we derive (4.5) with $t_k = W^{-1} \cdot \mathbf{1}(k \in \mathcal{I}_a)$.

□

5 Analysis of loop hierarchy

In this section, we prove Theorem 2.21. Except for Step 1, the proof primarily relies on analyzing the G-loop hierarchy.

5.1 Proof of Theorem 2.21, step 1

Proof. Our goal is to establish (2.73) and (2.74). Using the assumption (2.72) and the definitions $\ell_t = \ell(z_t)$ and $\eta_t = \text{Im } z_t$, we have

$$1 - u \gg N^{-1}, \quad \eta_u \gg N^{-1}, \quad s \leq u \leq t.$$

Given that $\partial_z(H - z)^{-1} = (H - z)^{-2}$ and the entries of H_t follow a Gaussian distribution, it follows that for any $C > 0$, there exists a constant C' such that

$$\max_{u \geq N^{-1}} \max_{|u - u'| \leq N^{-C'}} \|G_u - G_{u'}\|_{\max} \leq N^{-C}, \quad (5.1)$$

holds up to events that are exponentially small (negligible). Hence, through a standard N^{-C} net argument, we can reduce the proof of (2.73) and (2.74) for all u to the case $u = t$. This standard procedure, which we will refer to as a continuity argument, will be used repeatedly in this paper. We now focus on the proof for the case $u = t$.

Case 1: $s < t \leq 1/2$. Under this assumption on t ,

$$\|G_t\|_{op} = 1/\eta_t = O(1).$$

Combined with the fact that $\|E_a\|_{op} \leq W^{-1}$ for all $a \in \mathbb{Z}_L$, we have

$$\mathcal{L}_{t,\sigma,\mathbf{a}} = \left\langle \prod_{i=1}^n (G_t(\sigma_i) E_{a_i}) \right\rangle = O(W^{-n+1}), \quad t \leq 1/2, \quad (5.2)$$

which implies (2.73) for $t \leq 1/2$. Applying (4.2) and (4.3) from Lemma 4.1 to $\|G_t - m\|_{\max}$, and using the bound $\mathcal{L}$ in (5.2), we find that for $t \leq 1/2$,

$$\mathbf{1} \left(\|G_t - m\|_{\max} \leq W^{-1/10} \right) \cdot \|G_t - m\|_{\max} \prec W^{-1/2}, \quad t \leq 1/2,$$

where we have used the fact that ℓ_t and η_t are of order one. On the other hand, using the assumption (2.70) for s , we have $\|G_s - m\|_{\max} \prec W^{-1/2}$. Then, applying a standard continuity argument (with an N^{-C} net between s and t) and (5.1), we obtain that for any $u \in [s, t]$,

$$\|G_u - m\|_{\max} \prec W^{-1/2}, \quad u \leq 1/2. \quad (5.3)$$

Hence, we have proved (2.73) and (2.74) for the case $t \leq 1/2$.

Case 2: $t > s \geq 1/2$. Combining the assumption (2.68) and (2.59) from Lemma 2.17, i.e.,

$$\mathcal{K}_{s,\sigma,\mathbf{a}} \prec (W\ell_s\eta_s)^{-n+1},$$

we obtain that for any fixed n and s ,

$$\mathcal{L}_{s,\sigma,\mathbf{a}} \prec (W\ell_s\eta_s)^{-n+1}. \quad (5.4)$$

The following lemma will be proved in Section 6.

Lemma 5.1 (Continuity estimate on loops). *Suppose that $c < t_1 \leq t_2 \leq 1$ for some constant $c > 0$. Assume that for any fixed $n \in \mathbb{N}$, the following bounds hold at time t_1 :*

$$\max_{\sigma, \mathbf{a}} \mathcal{L}_{t_1,\sigma,\mathbf{a}} \prec (W\ell_1\eta_1)^{-n+1}, \quad \eta_i = \eta_{t_i}, \quad \ell_i = \ell_{t_i}. \quad (5.5)$$

Define Ω as

$$\Omega := \{\|G_{t_2}\|_{\max} \leq 2\}. \quad (5.6)$$

Then, for any fixed $n \geq 1$, we have

$$\mathbf{1}_\Omega \cdot \max_{\sigma, a} \mathcal{L}_{t_2, \sigma, a} \prec (W\ell_1\eta_2)^{-n+1} = \left(\frac{\ell_2}{\ell_1}\right)^{n-1} \cdot (W\ell_2\eta_2)^{-n+1}, \quad \eta_i = \eta_{t_i}, \quad \ell_i = \ell_{t_i}. \quad (5.7)$$

Using (5.2) and (5.4), we know that the estimate (5.5) for $\mathcal{L}_s$ holds. Applying Lemma 5.1, we obtain

$$\mathbf{1}(\|G_u\|_{\max} \leq 2) \cdot \max_{\sigma, a} \mathcal{L}_{u, \sigma, a} \prec \left(\frac{\ell_u}{\ell_s}\right)^{n-1} \cdot (W\ell_u\eta_u)^{-n+1}, \quad s \leq u \leq t. \quad (5.8)$$

Combining the special case $n = 2$ of (5.8) with Lemma 4.1, we have

$$\mathbf{1}\left(\|G_u - m\|_{\max} \leq (W\ell_u\eta_u)^{-1/6}\right) \cdot \|G_u - m\|_{\max} \prec \left(\frac{\ell_u}{\ell_s}\right)^{1/2} \cdot (W\ell_u\eta_u)^{-1/2}, \quad s \leq u \leq t.$$

Using the assumption (2.72), we find that for any fixed $D > 0$,

$$\mathbb{P}\left((W\ell_u\eta_u)^{-1/4} \leq \|G_u - m\|_{\max} \leq (W\ell_u\eta_u)^{-1/6}\right) \leq N^{-D}, \quad s \leq u \leq t. \quad (5.9)$$

This shows that the interval $((W\ell_u\eta_u)^{-1/4}, (W\ell_u\eta_u)^{-1/6})$ is a forbidden region for $\|G_u - m\|_{\max}$ for any u between s and t . Since the complement of this event holds with very high probability, the standard continuity argument implies that the forbidden region is avoided for all times between s and t simultaneously.

On the other hand, by assumption and (5.3), we have the initial bound at time s :

$$\|G_s - m\|_{\max} \prec (W\ell_s\eta_s)^{-1/2}.$$

Using a standard continuity argument between s and t , we conclude

$$\|G_t - m\|_{\max} \prec (W\ell_t\eta_t)^{-1/4}.$$

Combining this with (5.8), we have proved (2.73) in this case.

Case 3: $t > 1/2 > s$. From Case 1, we know that (5.2) and (5.3) hold for $t = 1/2$. Note that the proof of Case 2 uses only these two conditions. Therefore, the current case follows as a consequence of Case 2 with $s = 1/2$. □

5.2 Dynamics of $\mathcal{L} - \mathcal{K}$

Recall the loop hierarchy from Lemma 2.11. Using the notations

$$\mathcal{L}_{t, \mathcal{G}_{k,l}^{(a), L}(\sigma, \mathbf{a})} := \left(\mathcal{G}_{k,l}^{(a), L} \circ \mathcal{L}_{t, \sigma, \mathbf{a}}\right), \quad \mathcal{L}_{t, \mathcal{G}_{k,l}^{(b), R}(\sigma, \mathbf{a})} := \left(\mathcal{G}_{k,l}^{(b), R} \circ \mathcal{L}_{t, \sigma, \mathbf{a}}\right),$$

the hierarchy takes the compact form:

$$d\mathcal{L}_{t, \sigma, \mathbf{a}} = \mathcal{E}_{t, \sigma, \mathbf{a}}^{(M)} + \mathcal{E}_{t, \sigma, \mathbf{a}}^{(\tilde{G})} dt + W \cdot \sum_{1 \leq k < l \leq n} \sum_{a, b} \mathcal{L}_{t, \mathcal{G}_{k,l}^{(a), L}(\sigma, \mathbf{a})} \cdot S_{ab}^{(B)} \cdot \mathcal{L}_{t, \mathcal{G}_{k,l}^{(b), R}(\sigma, \mathbf{a})} dt. \quad (5.10)$$

The primitive hierarchy governing the evolution of $\mathcal{K}$ is given by:

$$d\mathcal{K}_{t, \sigma, \mathbf{a}} = W \cdot \sum_{1 \leq k < l \leq n} \sum_{a, b} \mathcal{K}_{t, \mathcal{G}_{k,l}^{(a), L}(\sigma, \mathbf{a})} \cdot S_{ab}^{(B)} \cdot \mathcal{K}_{t, \mathcal{G}_{k,l}^{(b), R}(\sigma, \mathbf{a})} dt. \quad (5.11)$$

Combining the two equations, we obtain:

$$\begin{aligned} d(\mathcal{L} - \mathcal{K})_{t, \sigma, \mathbf{a}} = & W \cdot \sum_{1 \leq k < l \leq n} \sum_{a, b} \left((\mathcal{L} - \mathcal{K})_{t, \mathcal{G}_{k, l}^{(a), L}(\sigma, \mathbf{a})} \cdot S_{ab}^{(B)} \cdot \mathcal{K}_{t, \mathcal{G}_{k, l}^{(b), R}(\sigma, \mathbf{a})} + (\mathcal{K} \iff \mathcal{L} - \mathcal{K}) \right) dt \\ & + \mathcal{E}_{t, \sigma, \mathbf{a}}^{((\mathcal{L} - \mathcal{K}) \times (\mathcal{L} - \mathcal{K}))} dt + \mathcal{E}_{t, \sigma, \mathbf{a}}^{(\tilde{G})} dt + \mathcal{E}_{t, \sigma, \mathbf{a}}^{(M)}, \end{aligned} \quad (5.12)$$

where

$$\mathcal{E}_{t, \sigma, \mathbf{a}}^{((\mathcal{L} - \mathcal{K}) \times (\mathcal{L} - \mathcal{K}))} := W \cdot \sum_{1 \leq k < l \leq n} \sum_{a, b} (\mathcal{L} - \mathcal{K})_{t, \mathcal{G}_{k, l}^{(a), L}(\sigma, \mathbf{a})} \cdot S_{ab}^{(B)} \cdot (\mathcal{L} - \mathcal{K})_{t, \mathcal{G}_{k, l}^{(b), R}(\sigma, \mathbf{a})}. \quad (5.13)$$

Here $\mathcal{K} \iff \mathcal{L} - \mathcal{K}$ represents the terms obtained by swapping $\mathcal{K}$ and $\mathcal{L} - \mathcal{K}$. The terms on the first line of the right-hand side correspond to $(\mathcal{L} - \mathcal{K})$ "loops" connected with a $\mathcal{K}$ "loop" via $S_{ab}^{(B)}$. These terms can be rearranged by the rank of $\mathcal{K}$, i.e., the length of the corresponding G -loop. For $2 \leq l_{\mathcal{K}} \leq n$, define

$$\begin{aligned} [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}} := & W \cdot \sum_{1 \leq k < l \leq n} \sum_{a, b} \left(\right. \\ & (\mathcal{L} - \mathcal{K})_{t, \mathcal{G}_{k, l}^{(a), L}(\sigma, \mathbf{a})} \cdot S_{ab}^{(B)} \\ & \cdot \mathcal{K}_{t, \mathcal{G}_{k, l}^{(b), R}(\sigma, \mathbf{a})} \cdot \mathbf{1}(\text{length}(\mathcal{K}) = l_{\mathcal{K}}) \\ & \left. + (\mathcal{K} \iff \mathcal{L} - \mathcal{K}) \right). \end{aligned} \quad (5.14)$$

The first line on the right-hand side of (5.12) is therefore $\sum_{l_{\mathcal{K}}=2}^n [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}}$. Separating the special case $l_{\mathcal{K}} = 2$, we have the $\mathcal{L} - \mathcal{K}$ hierarchy

$$d(\mathcal{L} - \mathcal{K})_{t, \sigma, \mathbf{a}} = [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}=2} dt + \sum_{l_{\mathcal{K}} > 2} [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}} dt + \mathcal{E}_{t, \sigma, \mathbf{a}}^{((\mathcal{L} - \mathcal{K}) \times (\mathcal{L} - \mathcal{K}))} dt + \mathcal{E}_{t, \sigma, \mathbf{a}}^{(\tilde{G})} dt + \mathcal{E}_{t, \sigma, \mathbf{a}}^{(M)}. \quad (5.15)$$

Clearly, we can view $[\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}=2}$ as a linear transform of the tensor $(\mathcal{L} - \mathcal{K})_{t, \sigma, \mathbf{a}}$.

Definition 5.2 (Definition of $\Theta_{t, \sigma}$ and $\mathcal{U}_{s, t, \sigma}$). Define the linear operator $\Theta_{t, \sigma}$ on a tensor $\mathcal{A} : \mathbb{Z}_L^n \rightarrow \mathbb{C}$ by

$$\begin{aligned} (\Theta_{t, \sigma} \circ \mathcal{A})_{\mathbf{a}} = & \sum_{i=1}^n \sum_{b_i} \left(\frac{m_i m_{i+1} S^{(B)}}{1 - t m_i m_{i+1} S^{(B)}} \right)_{a_i b_i} \cdot \mathcal{A}_{\mathbf{a}^{(i)}}, \quad m_i = m(\sigma_i) \\ \mathbf{a}^{(i)} = & (a_1, \dots, a_{i-1}, b_i, a_{i+1}, \dots, a_n). \end{aligned} \quad (5.16)$$

We also define the evolution kernel

$$(\mathcal{U}_{s, t, \sigma} \circ \mathcal{A})_{\mathbf{a}} = \sum_{b_1, \dots, b_n} \prod_{i=1}^n \left(\frac{1 - s \cdot m_i m_{i+1} S^{(B)}}{1 - t \cdot m_i m_{i+1} S^{(B)}} \right)_{a_i b_i} \cdot \mathcal{A}_{\mathbf{b}}, \quad \mathbf{b} = (b_1, \dots, b_n), \quad m_i = m(\sigma_i) \quad (5.17)$$

For any $\xi \in \mathbb{C}$, we have

$$\left(\frac{1 - s \xi S^{(B)}}{1 - t \xi S^{(B)}} \right) = I - (s - t) \xi \cdot \Theta_{t \xi}^{(B)} \cdot S^{(B)}. \quad (5.18)$$

Using these notations and the rank-2 $\mathcal{K}$ tensor from (2.57), we derive the following identity for the first term on the right-hand side of (5.15), for $n \geq 2$.

$$(\Theta_{t, \sigma} \circ (\mathcal{L} - \mathcal{K}))_{\mathbf{a}} = [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}=2}. \quad (5.19)$$

The following lemma is just a form of Duhamel formula.

Lemma 5.3 (Integrated loop hierarchy). *Let $\mathcal{A}_t$ be a tensor satisfying the stochastic equation:*

$$d\mathcal{A}_t = \Theta_{t,\sigma} \circ \mathcal{A}_t dt + \mathcal{D}_t dt + \mathcal{C}_t \cdot d\mathcal{B}_t.$$

Then for $s < t$ the $\mathcal{A}_t$ satisfies

$$\mathcal{A}_t = \mathcal{U}_{s,t,\sigma} \circ \mathcal{A}_s + \int_s^t \mathcal{U}_{u,t,\sigma} \circ \mathcal{D}_u du + \int_s^t \mathcal{U}_{u,t,\sigma} \circ (\mathcal{C}_u \cdot d\mathcal{B}_u).$$

Inserting this solution into (5.15), we have the following integrated loop hierarchy

$$\begin{aligned} (\mathcal{L} - \mathcal{K})_{t,\sigma,a} &= (\mathcal{U}_{s,t,\sigma} \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_a \\ &\quad + \sum_{l_{\mathcal{K}} > 2} \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{u,\sigma}^{l_{\mathcal{K}}} \right)_a du \\ &\quad + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \right)_a du \\ &\quad + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_a du + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_a du \end{aligned} \quad (5.20)$$

Furthermore, let T be a stopping time with respect to the matrix Brownian motion H_t and denote $\tau := T \wedge t$. Then we have the stopped integrated loop hierarchy

$$\begin{aligned} (\mathcal{L} - \mathcal{K})_{\tau,\sigma,a} &= (\mathcal{U}_{s,\tau,\sigma} \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_a \\ &\quad + \sum_{l_{\mathcal{K}} > 2} \int_s^{\tau} \left(\mathcal{U}_{u,\tau,\sigma} \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{u,\sigma}^{l_{\mathcal{K}}} \right)_a du \\ &\quad + \int_s^{\tau} \left(\mathcal{U}_{u,\tau,\sigma} \circ \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \right)_a du \\ &\quad + \int_s^{\tau} \left(\mathcal{U}_{u,\tau,\sigma} \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_a du + \mathcal{U}_{t,\tau} \circ \int_s^{\tau} \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_a du. \end{aligned} \quad (5.21)$$

Here and below, $\mathcal{E}^{(M)}$ denotes an Itô differential containing $d\mathcal{B}_u$. Its stochastic integral is therefore written without an additional factor du . All other $\mathcal{E}$ -terms in (5.20) are drift terms and are integrated with respect to du . The equation (5.20) will serve as our fundamental equation for estimating $\mathcal{L} - \mathcal{K}$. To analyze this equation, we first introduce the following notation, which is necessary to compute the quadratic variation of the martingale term.

Definition 5.4 (Definition of $\mathcal{E} \otimes \mathcal{E}$). *Denote*

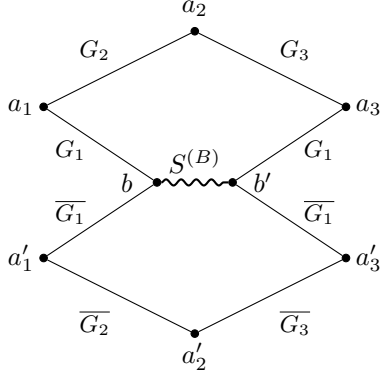
$$\begin{aligned} (\mathcal{E} \otimes \mathcal{E})_{t,\sigma,a,a'} &:= \sum_{k=1}^n (\mathcal{E} \otimes \mathcal{E})_{t,\sigma,a,a'}^{(k)} \\ (\mathcal{E} \otimes \mathcal{E})_{t,\sigma,a,a'}^{(k)} &:= W \sum_{b,b'} S_{b,b'}^{(B)} \mathcal{L}_{t,\sigma^{(k)},a^{(k)}}, \quad \sigma^{(k)} \in \{+, -\}^{2n+2}, \end{aligned} \quad (5.22)$$

where the loop $\mathcal{L}_{t,\sigma^{(k)},a^{(k)}}$ is obtained by cutting the k -th edge of $\mathcal{L}_{t,\sigma,a}$ and attaching it to its complex-conjugate loop (with indices a') to form a larger loop, with new indices b and b' . Hence, there are n indices between b and b' so that

$$\begin{aligned} a^{(k)} &= (a_k, a_{k+1}, \dots, a_n, a_1, \dots, a_{k-1}, b', a'_{k-1}, \dots, a'_1, a'_n, \dots, a'_k, b) \\ \sigma^{(k)} &= (\sigma_k, \sigma_{k+1}, \dots, \sigma_n, \sigma_1, \dots, \sigma_k, \bar{\sigma}_k, \dots, \bar{\sigma}_1, \bar{\sigma}_n, \dots, \bar{\sigma}_k) \end{aligned} \quad (5.23)$$

The symbol $\otimes$ in the notation $\mathcal{E} \otimes \mathcal{E}$ was used to emphasize the symmetric structure (as illustrated in Figure 13); it does not denote a tensor product.

Example: For a 3-G loop $\mathcal{L}_{\sigma,a}$, Figure 13 represents one loop that appears in $\mathcal{E} \otimes \mathcal{E}$. Although $S^{(B)}$ is not part of the loop, it does appear within $\mathcal{E} \otimes \mathcal{E}$.



$$k = 1$$

$$\mathbf{a}^{(1)} = (a_1, a_2, a_3, b', a'_3, a'_2, a'_1, b);$$

$$\boldsymbol{\sigma}^{(1)} = (\sigma_1, \sigma_2, \sigma_3, \sigma_1, \overline{\sigma_1}, \overline{\sigma_3}, \overline{\sigma_2}, \overline{\sigma_1}).$$

Figure 13: Example of the 8 – G loop in $\mathcal{E} \otimes \mathcal{E}$: for $\sigma \in \{+, -\}^3$

Lemma 5.5 (The martingale term). *For any stopping time T with respect to H_t and $\tau := t \wedge T$, we have*

$$\mathbb{E} \left| \int_s^\tau (\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)})_{\mathbf{a}} \right|^{2p} \leq C_{n,p} \mathbb{E} \left(\int_s^\tau \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\overline{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} du \right)^p \quad (5.24)$$

where $\overline{\sigma}$ is the conjugate sign vector of σ . More precisely, as in (5.17),

$$[(\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\overline{\sigma}}) \circ \mathcal{A}]_{\mathbf{a},\mathbf{a}'} = \sum_{b,b'} \prod_{i=1}^n \left(\frac{1-u \cdot m_i m_{i+1} S^{(B)}}{1-t \cdot m_i m_{i+1} S^{(B)}} \right)_{a_i b_i} \cdot \prod_{i=1}^n \left(\frac{1-u \cdot \overline{m_i} \overline{m_{i+1}} S^{(B)}}{1-t \cdot \overline{m_i} \overline{m_{i+1}} S^{(B)}} \right)_{a'_i b'_i} \cdot \mathcal{A}_{b,b'}.$$

Proof of Lemma 5.5. We will prove the case $\tau = t$; the general case is identical. By Definition 2.46, $\mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)} = \sum_{\alpha=(i,j)} \mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)}(\alpha) \cdot d\mathcal{B}_\alpha(t)$ and

$$\left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} = \sum_{\alpha=(i,j)} \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)}(\alpha) \right)_{\mathbf{a}} \cdot d\mathcal{B}_\alpha(u)$$

Here with $\alpha = (i, j)$ we have

$$\mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)}(\alpha) = (S_{ij})^{1/2} \cdot \partial_{(H_t)_{ij}} \mathcal{L}_{t,\sigma,\mathbf{a}}.$$

Using the chain rule and the structure of $\mathcal{L}$, we can write

$$\mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)}(\alpha) = \sum_{k=1}^n \mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)}(\alpha, k), \quad \mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)}(\alpha, k) := (S_{ij})^{1/2} \cdot \mathcal{L}_{t,\sigma,\mathbf{a}} \Big| (G_k \rightarrow \partial_{(H_t)_{ij}} G_k)$$

where (α, k) denotes the term in which the derivative acts on the k -th G -edge of $\mathcal{L}$. By definition (5.22), we have

$$\sum_{\alpha} \mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)}(\alpha, k) \cdot \overline{\mathcal{E}_{t,\sigma,\mathbf{a}'}^{(M)}(\alpha, k)} = (\mathcal{E} \otimes \mathcal{E})_{t,\sigma,\mathbf{a},\mathbf{a}'}^{(k)}$$

Since $\mathcal{U}$ is a deterministic linear operator, the quadratic variation of the martingale term in (5.20) can be bounded by

$$\begin{aligned} \left[\int (\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)})_{\mathbf{a}} \right]_t &= \int_s^t \sum_{\alpha} \left| (\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)}(\alpha))_{\mathbf{a}} \right|^2 du = \int_s^t \sum_{\alpha} \left| \left(\sum_{k=1}^n \mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)}(\alpha, k) \right)_{\mathbf{a}} \right|^2 du \\ &\leq C_n \int_s^t \sum_{\alpha} \sum_{k=1}^n \left| (\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)}(\alpha, k))_{\mathbf{a}} \right|^2 du \\ &= C_n \int_s^t \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\overline{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} du. \end{aligned} \quad (5.25)$$

Here we have used the Schwarz inequality in expanding the square. Our desired result, (5.24), follows from the BDG inequality. $\square$

5.3 Proof of Theorem 2.21, step 2

In this section, we focus on the $(+, -)$ 2- G -loop, i.e. $\sigma = (+, -)$. The subscript σ will be dropped in this subsection. We will prove (2.76) first, and (2.75) will be proved at the end of this section. Define the tail functions $\mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell)$ and $\mathcal{T}_{u,D}(\ell)$

$$\mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell) := \max_{a,b: |a-b| \geq \ell} |(\mathcal{L}-\mathcal{K})_{u,\sigma,(a,b)}|, \quad \sigma = (+, -), \quad (5.26)$$

$$\mathcal{T}_{u,D}(\ell) := (W\ell_u\eta_u)^{-2} \exp\left(-(\ell/\ell_u)_+^{1/2}\right) + W^{-D}. \quad (5.27)$$

Both $\mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}$ and $\mathcal{T}_{u,D}$ are non-increasing functions:

$$0 \leq \ell_1 \leq \ell_2, \quad \mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell_1) \geq \mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell_2), \quad \mathcal{T}_{u,D}(\ell_1) \geq \mathcal{T}_{u,D}(\ell_2)$$

Denote the ratio between them by

$$\mathcal{J}_{u,D}(\ell) := \left(\mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell) / \mathcal{T}_{u,D}(\ell)\right) + 1 \quad (5.28)$$

We aim to bound $\mathcal{J}_{u,D}(\ell)$ for any $u \in [s, t]$ and large $D > 0$, i.e.,

$$\mathcal{J}_{u,D}^* := \max_{\ell} \mathcal{J}_{u,D}(\ell) \leq (\eta_s/\eta_u)^4 \quad (5.29)$$

Define a scale parameter

$$\ell_t^* := (\log W)^{3/2} \cdot \ell_t.$$

In this scale ℓ_t^* , Θ_t is exponentially small, whereas $\mathcal{T}_t$ is not. More precisely, we have the following lemma.

Lemma 5.6. *For any fixed large D and small $\delta > 0$,*

$$|b-a| \geq \delta \cdot \ell_t^* \implies (\Theta_t)_{ab} \leq W^{-D}, \quad (\Theta_s^{-1}\Theta_t)_{ab} \leq W^{-D} \quad (5.30)$$

$$|b-a| \geq \delta \cdot \ell_t^* \implies \mathcal{L}_{t,(-,+),(a,b)} \prec \mathcal{J}_{t,D}^* \cdot \mathcal{T}_{t,D}(|a-b|). \quad (5.31)$$

For any constant $C > 0$,

$$\mathcal{T}_{u,D}(\ell - C \cdot \ell_u^*) \prec \mathcal{T}_{u,D}(\ell). \quad (5.32)$$

Proof. We will only prove (5.31). By definition,

$$\mathcal{L}_{t,(-,+),(a,b)} = (\mathcal{L}-\mathcal{K})_{t,(-,+),(a,b)} + \mathcal{K}_{t,(-,+),(a,b)}.$$

Using the Θ representation of the 2- $\mathcal{K}$ in (2.57) and the decay property of Θ from (2.52), we have for any $D > 0$ that

$$\mathcal{K}_{t,(-,+),(a,b)} \prec N^{-D} \leq \mathcal{T}_{t,D}(|a-b|).$$

Together with the definition (5.29) of $\mathcal{J}^*$, (5.28), and (5.26), we have proved (5.31). We remark that (5.31) is significant because $\mathcal{T}_{t,D}$ is smaller by a factor of $(W\eta_t\ell_t)^{-1}$ than the typical size of a 2- $\mathcal{L}$ loop. The extra small factor comes from the assumption $|b-a| \geq \delta\ell_t^*$. $\square$

Our proof of (2.76) relies on the loop hierarchy (5.21) in the special case $n = 2$. We begin the analysis of the hierarchy by bounding terms in (5.21).

Lemma 5.7. *Suppose the assumptions of Theorem 2.21 and the conclusions of Step 1, i.e., (2.73) and (2.74), hold. Assume that*

$$s \leq u \leq t, \quad D \geq 10, \quad \mathbf{a} = (a_1, a_2), \quad \mathbf{a}' = (a'_1, a'_2), \quad \sigma = (+, -), \quad \max_i |a_i - a'_i| \leq \ell_t^*. \quad (5.33)$$

We have

$$\mathcal{E}_{u,\sigma,\mathbf{a}}^{((L-K) \times (L-K))} / \mathcal{T}_{t,D}(|a_1 - a_2|) \prec (\eta_u)^{-1} \cdot (W\eta_u\ell_u)^{-1} \cdot (\mathcal{J}_{u,D}^*)^2 \quad (5.34)$$

$$\begin{aligned} \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} / \mathcal{T}_{t,D}(|a_1 - a_2|) &\prec (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^2 \cdot \mathbf{1}(|a_1 - a_2| \leq \ell_u^*) \\ &+ (\eta_u)^{-1} \cdot (W\eta_u\ell_u)^{-3/7} \cdot (\mathcal{J}_{u,D}^*)^3 \end{aligned} \quad (5.35)$$

$$\begin{aligned} (\mathcal{E} \otimes \mathcal{E})_{u,\sigma,\mathbf{a},\mathbf{a}'} / (\mathcal{T}_{t,D}(|a_1 - a_2|))^2 &\prec (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^5 \cdot \mathbf{1}(|a_1 - a_2| \leq 4\ell_t^*) \\ &+ (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^{3/2} \cdot (W\eta_u\ell_u)^{-1/2} \cdot (\mathcal{J}_{u,D}^*)^3 \end{aligned} \quad (5.36)$$

By assumption (2.72), s and t are near each other. So the exact exponents on the right side of (5.34)-(5.36) are not important for our purpose. We only need the errors to be of the form

$$(\eta_s/\eta_t)^\alpha \cdot (W\eta_u\ell_u)^{-\beta} \cdot (\mathcal{J}_{u,D}^*)^\gamma$$

for some positive constants α , β and γ .

We now provide a power counting to guess the sizes of terms in the previous lemma. Since $\mathcal{E}^{((L-K) \times (L-K))}$ is a higher order term, we will ignore it in the following heuristic. Denote by $A_u \sim W\ell_u\eta_u$. By definition, $\mathcal{E}_u^{(\tilde{G})}$ is a product of a $1\text{-}\mathcal{L} - \mathcal{K}$ loop and a $3\text{-}\mathcal{L}$ loop. We know that $1\text{-}\mathcal{L} - \mathcal{K}$ loop is of order A_u^{-1} and $3\text{-}\mathcal{L}$ loop is of order A_u^{-2} . In addition, the summation index in $\mathcal{E}^{(\tilde{G})}$ yields a factor ℓ_u if we assume the correct decay property. Since there is an additional W factor in $\mathcal{E}_u^{(\tilde{G})}$,

$$\mathcal{E}_u^{(\tilde{G})} \sim A_u^{-3} W\ell_u = \eta_u^{-1} A_u^{-2}. \quad (5.37)$$

The factor A_u^{-2} is exactly the prefactor in the definition of $\mathcal{T}_u$ in (5.35). For $\mathcal{E} \otimes \mathcal{E}_u$, it is a $6\text{-}\mathcal{L}$ loop of order A_u^{-5} . Hence

$$(\mathcal{E} \otimes \mathcal{E})_u \sim A_u^{-5} W\ell_u = \eta_u^{-1} A_u^{-4}, \quad (5.38)$$

which explains the order in (5.36). Notice that in both (5.35) and (5.36), we used $\mathcal{T}_t$ on the left sides of the equations while both $\mathcal{E}_u^{(\tilde{G})}$ and $(\mathcal{E} \otimes \mathcal{E})_u$ are at the time u .

So far we only used the loop bounds which are consequences of Step 1. It remains to understand the last terms in (5.35) and (5.36). Assuming Lemma 5.7, we now prove (2.76). We will use extensively the kernel estimates concerning the operator $\mathcal{U}$ in Section 7.1.

Proof of (2.76). By assumption (2.69) on $(\mathcal{L} - \mathcal{K})_s$, the operator norm bound on $\mathcal{U}$ in Lemma 7.1, and the tail estimate (7.2), we can bound the first term on the right side of (5.21) by

$$(\mathcal{U}_{s,t,\sigma} \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_{\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|) \prec (\ell_t/\ell_s)^2 \cdot \mathbf{1}(|a_1 - a_2| \leq \ell_t^*) + 1, \quad (5.39)$$

where the last term of order one comes from applying (7.2). Notice that the expansion factor $(\eta_u/\eta_t)^2$ from applying Lemma 7.1 has become $(\ell_t/\ell_s)^2$ due to the prefactors in $\mathcal{T}_{s,D}$ and $\mathcal{T}_{t,D}$. For any $\mathbf{a}$ fixed and any function f , we decompose $f = f_1 + f_2$ where $f_1(\mathbf{b}) = f(\mathbf{b})\mathbf{1}(\|\mathbf{b} - \mathbf{a}\| \leq \ell_t^*)$. From the decay of $\mathcal{U}_{u,t}$, $(\mathcal{U}_{u,t,\sigma} \circ f_2)_{\mathbf{a}}$ is exponentially small. Hence we only have to bound $\mathcal{U}_{u,t,\sigma} \circ f_1$, for which we apply Lemma 7.1. Therefore, we can bound the second term of (5.21) by

$$\begin{aligned} \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{((L-K) \times (L-K))} \right)_{\mathbf{a}} &\prec (\eta_u/\eta_t)^2 \max_{\|\mathbf{b} - \mathbf{a}\| \leq \ell_t^*} \mathcal{E}_{u,\sigma,\mathbf{b}}^{((L-K) \times (L-K))} + W^{-D}, \\ \|\mathbf{b} - \mathbf{a}\| &= \max_i |b_i - a_i| \leq \ell_t^*. \end{aligned}$$

Under the last condition, (5.32) implies that

$$\mathcal{T}_{t,D}(|b_1 - b_2|) \prec \mathcal{T}_{t,D}(|a_1 - a_2|).$$

Using the estimate (5.34) for $\mathcal{E}^{((L-K) \times (L-K))}$ and the (s, t) -condition (2.72), we have

$$\left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{((L-K) \times (L-K))} \right)_{\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|) \prec \frac{1}{\eta_u} (\eta_u/\eta_t)^2 \cdot (W\eta_u\ell_u)^{-1} \cdot (\mathcal{J}_{u,D}^*)^2 \quad (5.40)$$

Similarly, the estimate (5.35) on $\mathcal{E}^{(\tilde{G})}$ implies

$$\begin{aligned} \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_{\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|) &\prec \mathbf{1}(|a_1 - a_2| \leq 3\ell_t^*) \cdot \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^2 \cdot (\ell_u/\ell_s)^2 \\ &+ \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^2 \cdot (W\eta_u\ell_u)^{-3/7} \cdot (\mathcal{J}_{u,D}^*)^3, \end{aligned} \quad (5.41)$$

and the estimate (5.36) on $\mathcal{E} \otimes \mathcal{E}$ implies

$$\begin{aligned} \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} / (\mathcal{T}_{t,D}(|a_1 - a_2|))^2 &\prec \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) \cdot \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^4 \cdot (\ell_u/\ell_s)^5 \\ &+ \frac{1}{\eta_u} \cdot (W\eta_u\ell_u)^{-1/3} \cdot (\mathcal{J}_{u,D}^*)^3 \end{aligned} \quad (5.42)$$

In the last inequality, we have absorbed the expansion factor $(\eta_u/\eta_t)^4$ by the change of the exponent in $(W\eta_u\ell_u)$ from $-1/2$ to $-1/3$.

We now insert these bounds into the 2- G -loop equation (5.21) and bound the martingale term by (5.24). Denote by T the stopping time

$$T := \min\{u : \mathcal{J}_{u,D}^* \geq N^\delta (\eta_s/\eta_t)^4\} \quad (5.43)$$

where $\delta > 0$ is fixed and arbitrarily small, and set $\tau = T \wedge t$. Clearly, the stopped versions of Lemma 5.7 and the previous bounds in this proof are valid by similar arguments. The quadratic variation of the stopped martingale term is then bounded by

$$\begin{aligned} &\int_s^\tau \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} du / \mathcal{T}_{t,D}(|a_1 - a_2|)^2 \\ &\prec \int_s^\tau du \left\{ \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) \cdot \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^4 \cdot (\ell_u/\ell_s)^5 + \frac{1}{\eta_u} \cdot (W\eta_u\ell_u)^{-1/3} \cdot (\mathcal{J}_{u,D}^*)^3 \right\} \\ &\prec [(\eta_s/\eta_t)^4 (\ell_t/\ell_s)^5 \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) + (\eta_s/\eta_t)^3]. \end{aligned} \quad (5.44)$$

Applying (5.24), we have

$$\int_s^\tau \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|) \prec [(\eta_s/\eta_t)^2 (\ell_t/\ell_s)^{5/2} \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) + (\eta_s/\eta_t)^{3/2}].$$

This statement holds for any $t \geq s$. Furthermore, due to the monotonicity of ℓ and η , for any $s \leq t' \leq t$ we have

$$\begin{aligned} \int_s^{T \wedge t'} \left(\mathcal{U}_{u,t',\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} / \mathcal{T}_{t',D}(|a_1 - a_2|) &\prec [(\eta_s/\eta_{t'})^2 (\ell_{t'}/\ell_s)^{5/2} \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_{t'}^*) + (\eta_s/\eta_{t'})^{3/2}] \\ &\prec [(\eta_s/\eta_t)^2 (\ell_t/\ell_s)^{5/2} \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) + (\eta_s/\eta_t)^{3/2}] \end{aligned} \quad (5.45)$$

We extend this bound to a N^{-C} net between s and t , i.e., $t^{(k)} = s + (t - s) \cdot k \cdot N^{-C}$ for a fixed large C .

$$\max_k \int_s^{T \wedge t^{(k)}} \left(\mathcal{U}_{u,t^{(k)},\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} / \mathcal{T}_{t^{(k)},D}(|a_1 - a_2|) \prec [(\eta_s/\eta_t)^2 (\ell_t/\ell_s)^{5/2} \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) + (\eta_s/\eta_t)^{3/2}] \quad (5.46)$$

For any k , we can rewrite the last term in (5.21) as follows:

$$\mathcal{U}_{t,\tau} \circ \int_s^\tau \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} = \mathcal{U}_{t^{(k)},\tau} \circ \int_s^\tau \left(\mathcal{U}_{u,t^{(k)},\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}}$$

Choose a k such that $t^{(k-1)} \leq \tau < t^{(k)}$. Then $T \wedge t^{(k)} = \tau$. Since C can be chosen arbitrarily large, $\mathcal{U}_{t^{(k)}, \tau}$ is nearly an identity operator. Then

$$\mathcal{U}_{t^{(k)}, \tau} \int_s^\tau \left(\mathcal{U}_{u, t^{(k)}, \sigma} \circ \mathcal{E}_{u, \sigma}^{(M)} \right)_{\mathbf{a}} = \int_s^\tau \left(\mathcal{U}_{u, t^{(k)}, \sigma} \circ \mathcal{E}_{u, \sigma}^{(M)} \right)_{\mathbf{a}} + O(N^{-C/2})$$

and

$$\mathcal{T}_{t^{(k)}, D}(|a_1 - a_2|) / \mathcal{T}_{\tau, D}(|a_1 - a_2|) = 1 + O(N^{-C/2})$$

Combining these two bounds, we have

$$\mathcal{U}_{t, \tau} \circ \int_s^\tau \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{E}_{u, \sigma}^{(M)} \right)_{\mathbf{a}} / \mathcal{T}_{\tau, D}(|a_1 - a_2|) \prec [(\eta_s/\eta_t)^2 (\ell_t/\ell_s)^{5/2} \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) + (\eta_s/\eta_t)^{3/2}]$$

Combining this bound with (5.39), (5.40), (5.41) and (5.21), we have

$$(\mathcal{L} - \mathcal{K})_{\tau, \mathbf{a}} / \mathcal{T}_{t, D}(|a_1 - a_2|) \prec [(\eta_s/\eta_t)^2 (\ell_t/\ell_s)^{5/2} \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) + (\eta_s/\eta_t)^{3/2}],$$

where we have used the initial condition $\mathcal{J}_{s, D}^* \prec 1$ from (2.68) and (2.69).

This implies that

$$\mathcal{J}_{\tau, D}^* \prec (\eta_s/\eta_t)^2 (\ell_t/\ell_s)^{5/2} + (\eta_s/\eta_t)^{3/2} \quad (5.47)$$

Since $\ell_t/\ell_s \leq (\eta_s/\eta_t)^{1/2}$ and $\eta_s/\eta_t \geq 1$, the right side of (5.47) is at most $2(\eta_s/\eta_t)^{13/4} < N^\delta (\eta_s/\eta_t)^4$. Hence $\mathbb{P}(T \leq t) \leq N^{-D}$ and we have completed the proof of (2.76). We remark that integrating the near-field drift jointly in time yields the stronger bound $\mathcal{J}_{\tau, D}^* \prec (\eta_s/\eta_t)^2$. Notice that we have also proved

$$(\mathcal{L} - \mathcal{K})_{t, \mathbf{a}} / \mathcal{T}_{t, D}(|a_1 - a_2|) \prec [(\eta_s/\eta_t)^2 (\ell_t/\ell_s)^{5/2} \cdot \mathbf{1}(|a_1 - a_2| \leq 6\ell_t^*) + (\eta_s/\eta_t)^{3/2}]. \quad (5.48)$$

The region $|a_1 - a_2| > 6\ell_t^$.* The far-field prefactor $(\eta_s/\eta_t)^{3/2}$ in (5.48) is not bounded; we remove it by a second pass. Applying (5.47) with t replaced by any $u \in [s, t]$ and using $\ell_u/\ell_s \leq (\eta_s/\eta_u)^{1/2}$, we have $\mathcal{J}_{u, D}^* \prec (\eta_s/\eta_u)^{13/4}$ uniformly in $u \in [s, t]$. Fix $v \in [s, t]$ and write $r := \eta_s/\eta_v$ and $A := W\ell_v\eta_v$, so that $r^{30} \leq A$ by (2.72). We insert this bound once more into (5.39)–(5.42), with t replaced by v , using $\eta_s/\eta_u \leq r$ and $W\ell_u\eta_u \geq A$ for $u \leq v$, and we localize the martingale term at the stopping time $T' := \min\{u : \mathcal{J}_{u, D}^* \geq N^{\delta'} r^{13/4}\}$ for a fixed small $\delta' > 0$; $\mathbb{P}(T' \leq v)$ is negligible by the first pass. For $|a_1 - a_2| > 6\ell_v^*$ all terms containing an indicator function vanish. Relative to $\mathcal{T}_{v, D}(|a_1 - a_2|)$, and up to the factor $\int_s^v \eta_u^{-1} du = O(\log N)$, the remaining terms are bounded by $r^{17/2} A^{-1} \leq A^{-43/60}$ (from (5.40)), $r^{47/4} A^{-3/7} \leq A^{-31/840}$ (from (5.41)) and, for the quadratic variation, $r^{39/4} A^{-1/3} \leq A^{-1/120}$ (from (5.42)), so that the martingale term is $\prec A^{-1/240}$; the far part of (5.39) is $\prec 1$. Since $A \geq W\ell_t\eta_t \geq N^c$ for some $c > 0$ (by $t \leq 1 - N^{-1+\tau}$), these negative powers of A absorb the factors N^ε . Hence

$$(\mathcal{L} - \mathcal{K})_{v, \mathbf{a}} / \mathcal{T}_{v, D}(|a_1 - a_2|) \prec 1, \quad |a_1 - a_2| > 6\ell_v^*,$$

uniformly in $v \in [s, t]$. □

Proof of Lemma 5.7. Proof of (5.34). We first note the monotonicity properties

$$u \leq t \implies \ell_u \leq \ell_t, \quad \mathcal{T}_{u, D} \leq \mathcal{T}_{t, D}.$$

By definition,

$$\mathcal{E}_{u, \sigma, \mathbf{a}}^{((\mathcal{L} - \mathcal{K}) \times (\mathcal{L} - \mathcal{K}))} = W \sum_{b_1, b_2} (\mathcal{L} - \mathcal{K})_{u, \sigma, (a_1, b_1)} S_{b_1, b_2}^{(B)} (\mathcal{L} - \mathcal{K})_{u, \sigma, (b_2, a_2)} \quad (5.49)$$

By definition of $S^{(B)}$, we have $|b_1 - b_2| \leq 1$. Using

$$\int_0^a \exp\left(-\sqrt{(a-x)} - \sqrt{x} + \sqrt{a}\right) dx \leq C \approx 6.12, \quad \text{and} \quad \int_0^\infty \exp(-\sqrt{x}) dx = 2$$

we have

$$\begin{aligned} \mathcal{E}_{u,\sigma,\mathbf{a}}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} &\prec W(\mathcal{J}_{u,D}^*)^2 \cdot \sum_x \mathcal{T}_{u,D}(|a_1 - x|) \cdot \mathcal{T}_{u,D}(|a_2 - x|) \\ &\prec (\mathcal{J}_{u,D}^*)^2 \cdot \eta_u^{-1} \cdot (W\ell_u\eta_u)^{-1} \cdot \mathcal{T}_{u,D}(|a_1 - a_2|). \end{aligned} \quad (5.50)$$

We have thus proved (5.34).

Proof of (5.35). By definition, we write

$$\max_{\mathbf{a}} \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} \prec W \sum_{b_1, b_2} \langle \tilde{G}_u E_{b_1} \rangle \cdot S_{b_1, b_2}^{(B)} \cdot \mathcal{L}_{u,(-,+,+), (a_2, b_2, a_1)} + c.c., \quad \tilde{G} = G - m \quad (5.51)$$

By (4.5) and (2.74), we have

$$\langle \tilde{G}_u E_{b_1} \rangle \prec \Xi_{u,2}^{(\mathcal{L})} \cdot (W\ell_u\eta_u)^{-1} \prec (W\ell_u\eta_u)^{-1} + \mathcal{J}_{u,D}^* \cdot (W\ell_u\eta_u)^{-2}.$$

Therefore

$$\max_{\mathbf{a}} \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} \prec \frac{1}{\ell_u\eta_u} \cdot \sum_b |\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| (1 + \mathcal{J}_{u,D}^* (W\ell_u\eta_u)^{-1}) \quad (5.52)$$

Consider first the case $|a_1 - a_2| \leq \ell_u^*$. Denote

$$\ell_u^{**} := \ell_u \cdot (\log W)^3 = \ell_u^* \cdot (\log W)^{3/2}$$

Clearly, $\mathcal{T}_{u,D}(\ell_u^{**}) \prec W^{-D}$. For $|b - a_1| \leq \ell_u^{**}$, the loop bound (2.73) proved in the Step 1 implies that

$$\sum_b \mathbf{1}(|b - a_1| \leq \ell_u^{**}) \cdot |\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| \prec (\ell_u/\ell_s)^2 \cdot (W\ell_u\eta_u)^{-2} \cdot \ell_u \quad (5.53)$$

where the summation over b provides a factor $\ell_u^{**} \prec \ell_u$. If $|b - a_1| \geq \ell_u^{**}$, we can easily bound

$$|\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| \prec \max_{x_1 \in \mathcal{I}_{a_1}, y \in \mathcal{I}_b} |G_{x_1, y}|.$$

Using (4.2), we have that

$$\max_{x_1 \in \mathcal{I}_{a_1}, y \in \mathcal{I}_b} |G_{x_1, y}| \prec \sum_{a'_1, b'} (\mathcal{L}_{u,(+,-), (a'_1, b')})^{1/2} \mathbf{1}(|a'_1 - a_1| \leq 1, |b - b'| \leq 1).$$

Since $|b - a_1| \geq \ell_u^{**}$, by (5.31) and $\mathcal{T}_{u,D}(|a_1 - b|) \prec W^{-D}$, we obtain that

$$\mathcal{L}_{u,(+,-), (a'_1, b')} \prec \mathcal{J}_{u,D}^* \cdot W^{-D}.$$

Therefore, the contribution from $|b - a_1| \geq \ell_u^{**}$ part is negligible in the sense

$$\sum_b \mathbf{1}(|b - a_1| \geq \ell_u^{**}) \cdot |\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| \prec \mathcal{J}_{u,D}^* \cdot W^{-10}. \quad (5.54)$$

It is easy to check that the contribution of the last term does not affect the argument given below. Furthermore, the contributions from both (5.53) and (5.52) are bounded by

$$|a_1 - a_2| \leq \ell_u^* \implies \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} / \mathcal{T}_{u,D}(|a_1 - a_2|) \prec (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^2 \cdot \left(1 + (\mathcal{J}_{u,D}^*)^2 (W\ell_u\eta_u)^{-1}\right). \quad (5.55)$$

This implies that (5.35) holds for $|a_1 - a_2| \leq \ell_u^*$. Here the term $(\eta_u)^{-1}(\ell_u/\ell_s)^2(\mathcal{J}_{u,D}^*)^2(W\ell_u\eta_u)^{-1}$ of (5.55) is bounded by the second term of (5.35), since $\mathcal{J}_{u,D}^* \geq 1$ and $(\ell_u/\ell_s)^2 \leq \eta_s/\eta_u \leq (W\ell_u\eta_u)^{1/30}$ by (2.72).

For $|a_1 - a_2| \geq \ell_u^*$, we split it into two cases

$$(1) : \min_i |a_i - b| \leq \ell_u^*/2, \quad (2) : \min_i |a_i - b| \geq \ell_u^*/2.$$

In the first case, we assume without loss of generality that $|a_1 - b| \leq \ell_u^*/2$. Applying the Schwarz inequality to the two G edges in $\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)}$ connecting the block a_2 , we have

$$G_{x_1x_2}G_{x_2y}G_{yx_1} \prec |G_{x_1x_2}|^2|G_{yx_1}| + |G_{x_2y}|^2|G_{yx_1}|.$$

Therefore, we have

$$\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec W^{-1} \left(\mathcal{L}_{u,(-,+),(a_1,a_2)} \max_{x_1 \in \mathcal{I}_{a_1}} \sum_{y \in \mathcal{I}_b} |G_{x_1y}| + \mathcal{L}_{u,(-,+),(a_2,b)} \max_{y \in \mathcal{I}_b} \sum_{x_1 \in \mathcal{I}_{a_1}} |G_{x_1y}| \right). \quad (5.56)$$

Using (4.2), the $\mathcal{K}$ bound and the trivial fact $\sqrt{1+c} \leq 1+c$, we can bound $|G_{x_1x_2}|$ by

$$\mathbf{1}(x_1 \neq x_2)|G_{x_1x_2}| \prec (W\ell_u\eta_u)^{-1/2}(1 + \mathcal{J}_{u,D}^*(W\ell_u\eta_u)^{-1}).$$

It implies that for any a and b ,

$$W^{-1} \max_{y \in \mathcal{I}_b} \sum_{x \in \mathcal{I}_a} |G_{xy}| \prec (W\ell_u\eta_u)^{-1/2}(1 + \mathcal{J}_{u,D}^*(W\ell_u\eta_u)^{-1}). \quad (5.57)$$

Inserting this bound into (5.56), we have

$$\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec (\mathcal{L}_{u,(-,+),(a_1,a_2)} + \mathcal{L}_{u,(-,+),(a_2,b)}) \cdot (W\ell_u\eta_u)^{-1/2}(1 + \mathcal{J}_{u,D}^*(W\ell_u\eta_u)^{-1}) \quad (5.58)$$

Since $|a_2 - b| \geq \ell_u^*/2$ and $|a_1 - a_2| \geq \ell_u^*$, we can estimate $\mathcal{L}_{u,(-,+),(a_1,a_2)} + \mathcal{L}_{u,(-,+),(a_2,b)}$ with (5.31) to have

$$\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec \mathcal{J}_{u,D}^* \cdot (\mathcal{T}_{u,D}(|a_1 - a_2|) + \mathcal{T}_{u,D}(|b - a_2|)) \cdot (W\ell_u\eta_u)^{-1/2}(1 + \mathcal{J}_{u,D}^*(W\ell_u\eta_u)^{-1}).$$

Furthermore since $|b - a_2| \geq |a_1 - a_2| - |a_1 - b| \geq |a_1 - a_2| - \ell_u^*/2$, then

$$\mathcal{T}_{u,D}(|b - a_2|) \prec \mathcal{T}_{u,D}(|a_1 - a_2|).$$

Combining these bounds, we obtain, for $|a_1 - a_2| \geq \ell_u^*$, that

$$\sum_b^{\text{Case 1}} |\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)}| \prec (\mathcal{J}_{u,D}^*)^2 \cdot \mathcal{T}_{u,D}(|a_1 - a_2|) \cdot \ell_u(W\ell_u\eta_u)^{-1/2}. \quad (5.59)$$

For case (2), we bound $\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)}$ as follows

$$\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec \max_{x_1, x_2, y} |G_{x_1y}| |G_{yx_2}| |G_{x_1x_2}| \mathbf{1}(x_1 \in \mathcal{I}_{a_1}) \mathbf{1}(x_2 \in \mathcal{I}_{a_2}) \mathbf{1}(y \in \mathcal{I}_b). \quad (5.60)$$

In this case, since a_1, a_2 and b are all different, x_1, x_2 and y must be different.

Next we use (4.2) to bound a single G with the 2- G -loops. Because a_1, a_2 and b are away from each other by $\ell_u^*/2$, we have, by (5.31),

$$|G_{yx_2}| \cdot |G_{x_1x_2}| \cdot |G_{x_1y}| \prec (\mathcal{J}_{u,D}^*)^{3/2} \left(\mathcal{T}_{u,D}(|a_1 - a_2|) \mathcal{T}_{u,D}(|a_1 - b|) \mathcal{T}_{u,D}(|a_2 - b|) \right)^{1/2} \quad (5.61)$$

Then we can bound

$$\sum_b^{\text{Case 2}} \mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec (\mathcal{J}_{u,D}^*)^{3/2} \left(\mathcal{T}_{u,D}(|a_1 - a_2|) \right)^{1/2} \sum_b \left(\mathcal{T}_{u,D}(|a_1 - b|) \mathcal{T}_{u,D}(|a_2 - b|) \right)^{1/2}$$

$$\prec (\mathcal{J}_{u,D}^*)^{3/2} \left(\mathcal{T}_{u,D}(|a_1 - a_2|) \right) \cdot \ell_u \cdot (W\ell_u\eta_u)^{-1} \quad (5.62)$$

where we have used

$$\int_0^a \exp \left(-\sqrt{(a-x)/2} - \sqrt{x/2} + \sqrt{a/2} \right) dx \leq C.$$

Combining this estimate with (5.59), we obtain that if $|a_1 - a_2| \geq \ell_u^*$ then

$$\sum_b \mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec (\mathcal{J}_{u,D}^*)^2 \left(\mathcal{T}_{u,D}(|a_1 - a_2|) \right) \cdot \ell_u \cdot (W\ell_u\eta_u)^{-1/2} \quad (5.63)$$

Inserting it into (5.52) and using (5.55), we have completed the proof of (5.35).

Proof of (5.36). By definition (5.22), $(\mathcal{E} \otimes \mathcal{E})$ is the sum of $(\mathcal{E} \otimes \mathcal{E})^{(k)}$, and the latter ones can be written in terms of the following loops $\mathcal{L}^{(k)} := \mathcal{L}_{u,\sigma^{(k)},\mathbf{a}^{(k)}}$. Here $\sigma^{(1)} = (+, -, +, -, +, -)$, $\mathbf{a}^{(1)} = (a_1, a_2, b', a'_2, a'_1, b)$ and $\sigma^{(2)} = (-, +, -, +, -, +)$, $\mathbf{a}^{(2)} = (a_2, a_1, b', a'_1, a'_2, b)$. Hence by symmetry, we only need to prove (5.36) for $(\mathcal{E} \otimes \mathcal{E})^{(1)}$, i.e., $k = 1$ case. Notice that $|b - b'| = 1$ in this case and we can treat $b = b'$ for all practical purposes in the following proof.

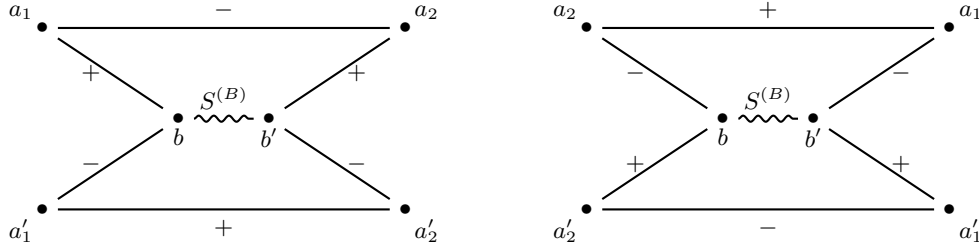


Figure 14: Left: $k = 1$;

right: $k = 2$.

Case 1: $|a_1 - a_2| \leq 4\ell_t^*$ We split the sum $\sum_{b,b'}$ into two parts

$$|b - a_1| \leq \ell_u^{**} := (\log W)^3 \ell_u, \quad |b - a_1| \geq \ell_u^{**}$$

Using $\mathcal{T}_{u,D}(\ell_u^{**})$ is very small, one can easily bound

$$\sum_{b,b'} \mathbf{1}(|b - a_1| \geq \ell_u^{**}) \mathcal{L}^{(1)} \prec \mathcal{J}_{u,D}^* \cdot W^{-3},$$

by arguments similar to those used in (5.54). For $|b - a_1| \leq \ell_u^{**}$, we use the loop bound (2.73) proved in Step 1 to have

$$\sum_{b,b'} \mathbf{1}(|b - a_1| \leq \ell_u^{**}) \mathcal{L}^{(1)} \prec (\ell_u/\ell_s)^5 \cdot (W\ell_u\eta_u)^{-5} \cdot \ell_u. \quad (5.64)$$

Combining these two bounds, we obtain that (5.36) holds for $|a_1 - a_2| \leq 4\ell_t^*$. Notice that the application of the loop bound of length 6 yields a very strong bound $(W\ell_u\eta_u)^{-5}$ which is not easy to see without the loop estimate.

Case 2: $|a_1 - a_2| \geq 4\ell_t^*$. Recall the assumption $|a'_i - a_i| \leq \ell_t^*$ (5.33) and the fact that we can treat $b = b'$ in the following proof. We split the sum over b (and b') into two parts

$$(1) : |b - a_1| \leq |b - a_2|, \quad (2) : |b - a_1| \geq |b - a_2|.$$

By symmetry (exchanging $a_1 \leftrightarrow a_2$ and cutting the loop at b' instead of b), we only consider the first case. Similar to (5.60) (see also figure 14), we bound $\mathcal{L}^{(1)}$ by the product of four G 's and $G^\dagger E_b G$ as follows:

$$\mathcal{L}^{(1)} \leq W^{-2} \max_{x_2 \in \mathcal{I}_{a_2}} \max_{y' \in \mathcal{I}_{b'}} \max_{x'_2 \in \mathcal{I}_{a'_2}} \sum_{x_1 \in \mathcal{I}_{a_1}} \sum_{x'_1 \in \mathcal{I}_{a'_1}} |G_{x_1 x_2}| |G_{x_2 y'}| |G_{y' x'_2}| |G_{x'_2 x'_1}| \left| (G^\dagger E_b G)_{x_1 x'_1} \right|. \quad (5.65)$$

Using the Cauchy–Schwarz inequality (on $\sum_{x_1, x'_1}$) and the definition $\mathcal{L}$, we have

$$\mathcal{L}^{(1)} \leq \max_{x_1, x_2, y', x'_1, x'_2}^* |G_{x_1 x_2}| |G_{x_2 y'}| |G_{y' x'_2}| |G_{x'_2 x'_1}| \cdot \left(\max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^4} \mathcal{L}_{u, \sigma, \mathbf{a}} \right)^{1/2}, \quad (5.66)$$

where the max is over the condition

$$\mathbf{1}(x_1 \in \mathcal{I}_{a_1}) \mathbf{1}(x_2 \in \mathcal{I}_{a_2}) \mathbf{1}(y' \in \mathcal{I}_{b'}) \mathbf{1}(x'_1 \in \mathcal{I}_{a'_1}) \mathbf{1}(x'_2 \in \mathcal{I}_{a'_2}).$$

We claim that

$$|G_{y' x_2}| \cdot |G_{y' x'_2}| \cdot |G_{x_1 x_2}| \cdot |G_{x'_1 x'_2}| \prec (\mathcal{J}_{u, D}^*)^2 \cdot \mathcal{T}_{t, D}(|a_1 - a_2|) \cdot \mathcal{T}_{t, D}(|b - a_2|). \quad (5.67)$$

To prove this bound, we split it into two cases:

$$\begin{aligned} (1a) : & |a_1 - b| \leq \ell_u^*, \quad \text{or} \quad |a'_1 - b| \leq \ell_u^* \\ (1b) : & |a_1 - b| \geq \ell_u^*, \quad \text{and} \quad |a'_1 - b| \geq \ell_u^*. \end{aligned} \quad (5.68)$$

Since $|a_1 - a_2| \geq 4\ell_t^*$, we have, similar to (5.61), that

$$|G_{x_1 x_2}|^2 \prec \mathcal{J}_{u, D}^* \cdot \mathcal{T}_{u, D}(|a_1 - a_2|), \quad (5.69)$$

$$|G_{x'_1 x'_2}|^2 \prec \mathcal{J}_{u, D}^* \cdot \mathcal{T}_{u, D}(|a'_1 - a'_2|) \prec \mathcal{J}_{u, D}^* \cdot \mathcal{T}_{t, D}(|a_1 - a_2|), \quad (5.70)$$

where we have used $\mathcal{T}_u \leq \mathcal{T}_t$, the assumption $|a'_i - a_i| \leq \ell_t^*$ and (5.32) in the second inequality. Thus we have

$$|G_{x_1 x_2}| \cdot |G_{x'_1 x'_2}| \prec \mathcal{J}_{u, D}^* \cdot \mathcal{T}_{t, D}(|a_1 - a_2|).$$

For edges connecting with b' , we have

$$|G_{y' x_2}| \cdot |G_{y' x'_2}| \prec \mathcal{J}_{u, D}^* \cdot \mathcal{T}_{u, D}^{1/2}(|b - a_2|) \cdot \mathcal{T}_{u, D}^{1/2}(|b - a'_2|),$$

where we have used that both $|b - a_2|$ and $|b - a'_2|$ are larger than $\ell_t^* \geq \ell_u^*$. Using $\mathcal{T}_u \leq \mathcal{T}_t$ and $|a_2 - a'_2| \leq \ell_t^*$, we have

$$|G_{y' x_2}| \cdot |G_{y' x'_2}| \prec \mathcal{J}_{u, D}^* \cdot \mathcal{T}_{t, D}(|b - a_2|).$$

Combining these bounds, we have proved (5.67).

Next, applying (2.73) on $\mathcal{L}$ term in (5.66), together with (5.67) we obtain that

$$\mathcal{L}^{(1)} \prec (\ell_u / \ell_s)^{3/2} (W \ell_u \eta_u)^{-3/2} \cdot (\mathcal{J}_{u, D}^*)^2 \cdot \mathcal{T}_{t, D}(|a_1 - a_2|) \cdot \mathcal{T}_{t, D}(|b - a_2|).$$

Consider the case (1a) so that $|a_2 - b| \geq |a_1 - a_2| - \ell_t^*$. Therefore, $\mathcal{T}_{t, D}(|b - a_2|) \prec \mathcal{T}_{t, D}(|a_1 - a_2|)$. Combining these bounds with (5.67), we have

$$\sum_{b, b'}^{(1a)} \mathcal{L}^{(1)} \prec (\ell_u / \ell_s)^{3/2} \cdot \ell_u \cdot (W \ell_u \eta_u)^{-3/2} \cdot (\mathcal{J}_{u, D}^*)^2 \cdot (\mathcal{T}_{t, D}(|a_1 - a_2|))^2. \quad (5.71)$$

For (1b), we bound $(G^\dagger E_b G)_{x_1 x'_1}$ by

$$(G^\dagger E_b G)_{x_1 x'_1} \leq \max_{y \in \mathcal{I}_b} |G_{y x_1}| |G_{y x'_1}|.$$

Since $|a_1 - b|, |a'_1 - b| \geq \ell_u^*$, we have, similar to (5.61),

$$(G^\dagger E_b G)_{x_1 x'_1} \prec \mathcal{J}_{u, D}^* \cdot \mathcal{T}_{t, D}(|b - a_1|).$$

Together with (5.67), we have

$$\begin{aligned} \sum_{b, b'}^{(1b)} \mathcal{L}^{(1)} & \prec (\mathcal{J}_{u, D}^*)^3 \cdot \mathcal{T}_{t, D}(|a_1 - a_2|) \cdot \sum_b \mathcal{T}_{t, D}(|b - a_1|) \mathcal{T}_{t, D}(|b - a_2|) \\ & \prec (\mathcal{J}_{u, D}^*)^3 \cdot \ell_t \cdot (W \ell_t \eta_t)^{-2} \cdot (\mathcal{T}_{t, D}(|a_1 - a_2|))^2. \end{aligned} \quad (5.72)$$

Here the last line was bounded as in the proof of (5.50). Putting the bounds for (1a) and (1b) together, we obtain (5.36) and complete the proof of Lemma 5.7. $\square$

Proof of (2.75). Combining the $\mathcal{L} - \mathcal{K}$ estimate (2.76) and the $\mathcal{K}$ bound in Lemma 2.17, we obtain that

$$\max_{a,b} \mathcal{L}_{u,(+,-),(a,b)} \prec (\eta_s/\eta_u)^4 \cdot (W\ell_u\eta_u)^{-2} + (W\ell_u\eta_u)^{-1} \prec (W\ell_u\eta_u)^{-1}, \quad u \in [s, t] \quad (5.73)$$

where we have used the induction hypothesis (2.72). Then we can apply the resolvent entry estimate in Lemma 4.1. The weak local law in (2.74) implies the assumptions of Lemma 4.1. From (4.3), (4.2) and (5.73), we have

$$\|G_u - m\|_{\max}^2 \prec \max_{a,b} \mathcal{L}_{u,(+,-),(a,b)} \prec (W\ell_u\eta_u)^{-1}.$$

This completes the proof of (2.75). $\square$

5.4 Fast decay property

In the second step of the proof for Theorem 2.21, we established the decay property of the 2- G loop in (2.76). In this subsection, we extend this decay property to general loops $\mathcal{L}$ and $\mathcal{L} - \mathcal{K}$. This decay property is highly useful for bounding the $\mathcal{E}$ terms in equation (5.20). We begin by defining the ℓ_u -decay property.

Definition 5.8 (The decay property). *Let $\mathcal{A}$ be a tensor $\mathbb{Z}_L^n \rightarrow \mathbb{R}$. We say $\mathcal{A}$ has (u, τ, D) decay at the time u if for some fixed small $\tau > 0$ and large $D > 0$, we have*

$$\max_{1 \leq i, j \leq n} \|a_i - a_j\| \geq \ell_u W^\tau \implies \mathcal{A}_a = O(W^{-D}), \quad a = (a_1, a_2, \dots, a_n). \quad (5.74)$$

Lemma 5.9 (The decay property of $\mathcal{L}_u$). *Assume that (2.75) and (2.76) hold. Then for any $n \geq 2$, any small τ , large D and $D' > 0$, the $\mathcal{L}_u$ has (u, τ, D) decay with probability $1 - O(W^{-D'})$. More precisely,*

$$\mathbb{P} \left(\max_{\sigma, a} \left(|\mathcal{L}_{u, \sigma, a}| + |(\mathcal{L} - \mathcal{K})_{u, \sigma, a}| \right) \cdot \mathbf{1} \left(\max_{i,j} \|a_i - a_j\| \geq \ell_u W^\tau \right) \geq W^{-D} \right) \leq W^{-D'}. \quad (5.75)$$

Proof of Lemma 5.9. Combining the decay properties of $\mathcal{K}_{u,(+,-)}$ and $\mathcal{L} - \mathcal{K}$ (2.76), we obtain that $\mathcal{L}_{u,(+,-)}$ has (u, τ, D) decay property for any fixed (τ, D) with high probability. Applying the G_{ij} estimate in (4.2) and the new $\mathcal{L}_u$ decay property, we have

$$\forall \tau, D, D', \quad \mathbb{P} \left(\max_{i \in \mathcal{I}_{a_1}} \max_{j \in \mathcal{I}_{a_2}} |G_{ij}| \cdot \mathbf{1}(|a_1 - a_2| \geq \ell_u W^\tau) \geq W^{-D} \right) \leq W^{-D'}.$$

By definition $\mathcal{L} = \langle \prod_{i=1}^n (G_i E_{a_i}) \rangle$ has the (u, τ, D) decay property. On the other hand, by the decay of $\Theta_t^{(B)}$ and the tree representation of $\mathcal{K}$ in Lemma 3.5, $\mathcal{K}$ also has (u, τ, D) decay property. Therefore $(\mathcal{L} - \mathcal{K})_u$ has (u, τ, D) decay property. $\square$

The evolution kernel satisfies a straightforward $L_\infty \rightarrow L_\infty$ bound, stated as a bound on $\|\mathcal{U}_{u,t}\|_{\max \rightarrow \max}$ in Lemma 7.1. When the tensor acted on by $\mathcal{U}_{u,t}$ is short-ranged and satisfies a sum-zero property, Lemma 7.3 gives a substantial improvement. These two estimates yield the following bounds on the $\mathcal{E}$ terms.

Lemma 5.10 (Bounds on $\mathcal{E}$ terms). *Assume that (2.75) and (2.76) hold. Define $\Xi^{(\mathcal{L})}$ and $\Xi^{(\mathcal{L}-\mathcal{K})}$ as follows*

$$\begin{aligned} \Xi_{t,m}^{(\mathcal{L})} &:= \max_{\sigma, a} |\mathcal{L}_{t, \sigma, a}| \cdot (W\ell_t\eta_t)^{m-1} \cdot \mathbf{1}(\sigma \in \{+, -\}^m), \\ \Xi_{t,m}^{(\mathcal{L}-\mathcal{K})} &:= \max_{\sigma, a} |(\mathcal{L} - \mathcal{K})_{t, \sigma, a}| \cdot (W\ell_t\eta_t)^m \cdot \mathbf{1}(\sigma \in \{+, -\}^m). \end{aligned} \quad (5.76)$$

Then

$$\begin{aligned} [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{u, \sigma}^{l_{\mathcal{K}}} &\prec \left(\max_{k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \right) \cdot (W\ell_u\eta_u)^{-n} \cdot \frac{1}{\eta_u}, \\ \mathcal{E}_{u, \sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} &\prec \left(\max_{k: 2 \leq k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u, n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-1} \right) \cdot (W\ell_u\eta_u)^{-n} \cdot \frac{1}{\eta_u}, \end{aligned}$$

$$\begin{aligned}\mathcal{E}_{u,\sigma}^{(\tilde{G})} &\prec \Xi_{u,n+1}^{(\mathcal{L})} \cdot (W\ell_u\eta_u)^{-n} \cdot \frac{1}{\eta_u}, \\ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} &\prec \Xi_{u,2n+2}^{(\mathcal{L})} \cdot (W\ell_u\eta_u)^{-2n} \cdot \frac{1}{\eta_u}.\end{aligned}\tag{5.77}$$

Furthermore, all these $\mathcal{E}$ terms have the (u, τ, D) decay property for $u \in [s, t]$.

Proof of Lemma 5.10. The proof follows directly from the definitions of these terms and the decay properties in Lemma 5.9.

- By definition (5.14), $\mathcal{K} \sim (\mathcal{L} - \mathcal{K})$ can be written as the product of $\mathcal{K}$ and $\mathcal{L} - \mathcal{K}$ loops. Bounding $\mathcal{K}$ with (2.59), we have

$$[\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{u,\sigma}^{l_{\mathcal{K}}} \prec W \cdot (W\ell_u\eta_u)^{-l_{\mathcal{K}}+1} \cdot \ell_u \cdot \Xi_{u,(n-l_{\mathcal{K}}+2)}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-(n-l_{\mathcal{K}}+2)} \tag{5.78}$$

where we used $l_{\mathcal{K}} \geq 3$ and the ℓ_u factor comes from the sum of the index a , which was restricted by the decay property of $\mathcal{K}$. As a side remark, the other index b was restricted by $S_{ab}^{(B)}$.

- By definition (5.13), $\mathcal{E}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))}$ can be written as the product of two loops, whose total length is $n+2$.

$$\mathcal{E}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \prec W \cdot \sum_{k=2}^n \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-k} \cdot \ell_u \cdot \Xi_{u,(n-k+2)}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-(n-k+2)} \tag{5.79}$$

- By definition (2.47), $\mathcal{E}^{(\tilde{G})}$ can be written as the product of a $(\mathcal{L} - \mathcal{K})$ loop of length one and an $(n+1) - G$ loop. With (4.5), we have

$$\mathcal{E}^{(\tilde{G})} \prec W \cdot \max_{a,b} \mathcal{L}_{u,(+,-),(a,b)} \cdot \ell_u \cdot \Xi_{u,(n+1)}^{\mathcal{L}} \cdot (W\ell_u\eta_u)^{-n} \tag{5.80}$$

Furthermore, by (2.76), Lemma 2.17 and (2.72) (see (5.73)), we have $\max_{a,b} \mathcal{L}_{u,(+,-),(a,b)} \prec (W\ell_u\eta_u)^{-1}$.

- By definition (5.23), $(\mathcal{E} \otimes \mathcal{E})$ can be written in terms of $(2n+2)$ - G -loops. Thus

$$(\mathcal{E} \otimes \mathcal{E}) \prec \Xi_{u,2n+2}^{(\mathcal{L})} \cdot W \cdot \ell_u \cdot (W\ell_u\eta_u)^{-2n-1}. \tag{5.81}$$

Here all ℓ_u factors come from summing an index restricted to a range ℓ_u . $\square$

In the remainder of this subsection, we will use Lemma 5.10 to improve estimates on $\mathcal{L} - \mathcal{K}$.

Lemma 5.11. Assume that σ is not an alternating sign vector, i.e.,

$$\exists k, \text{ s.t. } \sigma_k = \sigma_{k+1}. \tag{5.82}$$

Suppose that the assumptions of Theorem 2.21, the local law (2.75) and the decay property for $\mathcal{L} - \mathcal{K}$ (2.76) hold. If

$$\max_{u \in [s,t]} \Xi_{u,2n+2}^{(\mathcal{L})} \prec \Lambda$$

for some deterministic quantity $\Lambda \geq 1$, then

$$\Xi_{u,n}^{(\mathcal{L}-\mathcal{K})} \prec \Lambda^{1/2} + \max_{u \in [s,t]} \left(\max_{k < n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-1} + \Xi_{u,n+1}^{(\mathcal{L})} \right). \tag{5.83}$$

Proof of Lemma 5.11. By case 1 of (7.16),

$$\|\mathcal{U}_{u,t,\sigma}\|_{\max \rightarrow \max} \left| \text{(fast decay tensors satisfying (5.82))} \right| \leq C_n W^{C_n \tau} \cdot \left(\frac{\ell_u \eta_u}{\ell_t \eta_t} \right)^n.$$

By assumptions of Theorem 2.21 on $(\mathcal{L} - \mathcal{K})_s$, the $\mathcal{E}$ estimates in Lemma 5.10, and the previous bound on $\max \rightarrow \max$ norm of $\mathcal{U}_{u,t}$, we have

$$\begin{aligned} (W\ell_t\eta_t)^n \cdot (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &\prec 1 + \int_s^t \left(\max_{k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \right) \cdot \frac{1}{\eta_u} du \\ &+ \int_s^t \left(\max_{k: 2 \leq k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-1} \right) \cdot \frac{1}{\eta_u} du \\ &+ \int_s^t \Xi_{u,n+1}^{(\mathcal{L})} \cdot \frac{1}{\eta_u} du + (W\ell_t\eta_t)^n \cdot \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} du. \end{aligned} \quad (5.84)$$

By Lemma 5.5, we know that

$$\mathbb{E} \left[\int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} \right]^{2p} \leq C_{n,p} \mathbb{E} \left(\int_s^t \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} du \right)^p. \quad (5.85)$$

Using (5.77) to bound $\mathcal{E} \otimes \mathcal{E}$, we have

$$(W\ell_t\eta_t)^{2n} \cdot \int_s^t \left(\mathcal{U}_{u,t,(\sigma,\bar{\sigma})} \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} du \prec \int_s^t \Xi_{u,2n+2}^{(\mathcal{L})} \cdot \frac{1}{\eta_u} du. \quad (5.86)$$

This completes the proof of Lemma 5.11. □

5.5 Sum zero operator $\mathcal{Q}_t$

In this subsection, we introduce another key tool, $\mathcal{Q}_t$, for estimating the loop hierarchy.

Definition 5.12 (Definition of $\mathcal{Q}_t$). *Let $\mathcal{A}_a$ be a tensor with $\mathbf{a} \in \mathbb{Z}_L^n$, $n \geq 2$. Define $\mathcal{P}$ by*

$$(\mathcal{P} \circ \mathcal{A})_{a_1} = \sum_{a_2, a_3, \dots, a_n} \mathcal{A}_a, \quad \mathbf{a} = (a_1, \dots, a_n),$$

where the index a_1 was fixed. A tensor $\mathcal{A}$ has a sum zero property if and only if $\mathcal{P} \circ \mathcal{A} = 0$. Define

$$(\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}} = \mathcal{A}_{\mathbf{a}} - (\mathcal{P} \circ \mathcal{A})_{a_1} \cdot \vartheta_{t,\mathbf{a}}, \quad \vartheta_{t,\mathbf{a}} := (1-t)^{n-1} \prod_{i=2}^n \left(\Theta_t^{(B)} \right)_{a_1 a_i}.$$

Since $\sum_b \left(\Theta_t^{(B)} \right)_{ab} = (1-t)^{-1}$, it is easy to check that $\mathcal{P} \circ \vartheta_{t,\mathbf{a}} = 1$ and $\mathcal{P} \circ \mathcal{Q}_t = 0$.

The reason for using $\vartheta_{t,\mathbf{a}}$ in the definition of $\mathcal{Q}_t$ is to preserve both the max norm and the rapid decay properties of $\mathcal{A}$, provided that $\mathcal{A}$ possesses such properties.

Lemma 5.13 (Properties of $\mathcal{Q}_t$). *Let $\mathcal{A}$ be a tensor $\mathbb{Z}_L^n \rightarrow \mathbb{R}$. Assume that $\mathcal{A}$ has (t, τ, D) decay property. Then*

$$\max_{\mathbf{a}} |(\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}}| \leq W^{C_n \tau} \cdot \max_{\mathbf{a}} |\mathcal{A}_{\mathbf{a}}| + W^{-D+C_n}. \quad (5.87)$$

Furthermore, $(\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}}$ has the (t, τ, D) decay property.

We now use $\mathcal{Q}_t$ to improve our estimates on $\mathcal{L} - \mathcal{K}$ integral representation (5.20). Denote by $\dot{\vartheta}_{t,\mathbf{a}} = \frac{d}{dt} \vartheta_{t,\mathbf{a}}$. From the hierarchy of $\mathcal{L} - \mathcal{K}$ (5.15), we have

$$\begin{aligned} d\mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &= \mathcal{Q}_t \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{t,\sigma,\mathbf{a}}^{l_{\mathcal{K}}=2} dt + \sum_{l_{\mathcal{K}} > 2} \mathcal{Q}_t \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{t,\sigma,\mathbf{a}}^{l_{\mathcal{K}}} dt \\ &+ \mathcal{Q}_t \circ \mathcal{E}_{t,\sigma,\mathbf{a}}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} dt + \mathcal{Q}_t \circ \mathcal{E}_{t,\sigma,\mathbf{a}}^{(M)} + \mathcal{Q}_t \circ \mathcal{E}_{t,\sigma,\mathbf{a}}^{(\tilde{G})} dt - \left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma} \cdot \dot{\vartheta}_t \right]_{\mathbf{a}} dt. \end{aligned} \quad (5.88)$$

To match the form appearing in Lemma 5.3, we write the first term on the right-hand side as

$$\begin{aligned}\mathcal{Q}_t \circ [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t,\sigma,\mathbf{a}}^{l_{\mathcal{K}}=2} &= \mathcal{Q}_t \circ \Theta_{t,\sigma} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} \\ &= \Theta_{t,\sigma} \circ \mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} + [\mathcal{Q}_t, \Theta_{t,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}},\end{aligned}\tag{5.89}$$

where $[\mathcal{Q}_t, \Theta_{t,\sigma}]$ is the commutator of $\mathcal{Q}_t$ and $\Theta_{t,\sigma}$. Since $\mathcal{P} \circ \mathcal{Q}_t = 0$, the left-hand side of (5.89) satisfies the sum-zero property. Furthermore, recall the definition of $\Theta_{t,\sigma}$ in Definition 5.2. We use the identity

$$\sum_{a_i} \left(\frac{m_i m_{i+1}}{1 - t m_i m_{i+1} S^{(B)}} \right)_{a_i b_i} = \frac{m_i m_{i+1}}{1 - t m_i m_{i+1}},$$

whose right-hand side is independent of b_i . It follows directly that the sum-zero property is preserved under multiplication by $\Theta_{t,\sigma}$. More precisely, if $\mathcal{A}$ has the sum-zero property, then so does $\Theta_{t,\sigma} \circ \mathcal{A}$; equivalently,

$$\mathcal{P} \circ \mathcal{A} = 0 \implies \mathcal{P} \circ \Theta_{t,\sigma} \circ \mathcal{A} = 0.$$

It implies the first term in the right hand side of (5.89) has the sum-zero property. Since the left hand side of (5.89) also has this property, we obtain that the last term in (5.89) has sum zero property too, i.e.,

$$\mathcal{P} \circ [\mathcal{Q}_t, \Theta_{t,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} = 0.\tag{5.90}$$

Then combining (5.88) and (5.89), and applying Lemma 5.3, we obtain the following equation, analogous to (5.20),

$$\begin{aligned}\mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &= (\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_{\mathbf{a}} \\ &+ \sum_{l_{\mathcal{K}} > 2} \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{u,\sigma}^{l_{\mathcal{K}}} \right)_{\mathbf{a}} du \\ &+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \right)_{\mathbf{a}} du \\ &+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_{\mathbf{a}} du + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} du \\ &+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ ([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}) \right)_{\mathbf{a}} du \\ &- \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{v}_u] \right)_{\mathbf{a}} du.\end{aligned}\tag{5.91}$$

In the next lemma, we will utilize the fact that all terms on the right-hand side of (5.91) exhibit both fast decay and sum-zero properties.

Lemma 5.14. *Suppose that the assumptions of Theorem 2.21, the local law (2.75) and the decay property of $\mathcal{L} - \mathcal{K}$ (2.76) hold. Let $n \geq 2$. If*

$$\max_{u \in [s,t]} \Xi_{u,2n+2}^{(\mathcal{L})} \prec \Lambda$$

for some deterministic quantity $\Lambda \geq 1$, then

$$\Xi_{u,n}^{(\mathcal{L}-\mathcal{K})} \prec \Lambda^{1/2} + \max_{u \in [s,t]} \left(\max_{k < n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W \ell_u \eta_u)^{-1} + \Xi_{u,n+1}^{(\mathcal{L})} \right).\tag{5.92}$$

Notice that there is no prefactor depending on (η_s/η_t) on the right hand side of the last inequality. For $n = 2$, the sum over $l_{\mathcal{K}} > 2$ in (5.91) is empty.

Proof of Lemma 5.14. In the previous subsection, we have proved (5.92) for the non-alternating case, i.e., (5.82). Hence we only need to focus on the alternating case $\forall k \quad \sigma_k = \bar{\sigma}_{k+1}$. By Lemma 5.13,

$$\mathcal{Q}_s \circ (\mathcal{L} - \mathcal{K})_s, \quad \mathcal{Q} \circ (\mathcal{K} \sim \mathcal{L} - \mathcal{K}), \quad \mathcal{Q} \circ \mathcal{E}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))}, \quad \mathcal{Q} \circ \mathcal{E}^{(\tilde{G})}$$

have the sum zero and fast decay properties. Furthermore, their max norms can be bounded with (5.87), which amounts to $\|\mathcal{Q}\|_{\max \rightarrow \max} \prec 1$. By (7.16), we have the following bound for sum zero fast decay tensors:

$$\|\mathcal{U}_{u,t,\sigma}\|_{\max \rightarrow \max} \left| \text{(fast decay and sum zero tensor)} \right| \leq C_n W^{C_n \tau} \cdot \left(\frac{\ell_u \eta_u}{\ell_t \eta_t} \right)^n. \quad (5.93)$$

Following the same argument as in the proof of Lemma 5.11, we have

$$\begin{aligned} (W \ell_t \eta_t)^n \cdot \mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &\prec 1 + \max_{u \in [s,t]} \left(\max_{k < n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{2 \leq k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W \ell_u \eta_u)^{-1} + \Xi_{u,n+1}^{(\mathcal{L})} \right) \\ &+ (W \ell_t \eta_t)^n \cdot \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} du \\ &+ (W \ell_t \eta_t)^n \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ ([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}) \right)_{\mathbf{a}} du \\ &- (W \ell_t \eta_t)^n \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u] \right)_{\mathbf{a}} du. \end{aligned} \quad (5.94)$$

Next, we estimate the last line in (5.94). Since $\mathcal{P} \circ \vartheta_{t,\mathbf{a}} = 1$, we have $\mathcal{P} \circ \dot{\vartheta}_{t,\mathbf{a}} = 0$ and

$$\mathcal{P} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u]_{\mathbf{a}} = \left(\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \right)_{a_1} \mathcal{P} \circ \dot{\vartheta}_{u,\mathbf{a}} = 0.$$

Therefore, this term also has the sum zero property. From its definition, ϑ has a fast decay property. Then with (7.16) (case 2), we have

$$\int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u] \right)_{\mathbf{a}} du \prec \int_s^t \left(\frac{\ell_u \eta_u}{\ell_t \eta_t} \right)^n \cdot \max_{\mathbf{a}} \left| [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}]_{\mathbf{a}} \right| \max_{\mathbf{a}} |\dot{\vartheta}_{u,\mathbf{a}}| du. \quad (5.95)$$

We claim that for non-constant σ , i.e., $\{\sigma_1, \dots, \sigma_n\} = \{+, -\}$, the following holds:

$$[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}]_{\mathbf{a}} \prec (W \eta_u)^{-n} \ell_u^{-1} \cdot \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})} \quad (5.96)$$

for any loop length n . To see this, for a fixed non-constant σ , there exists $1 < k \leq n$ such that $\sigma_k = \bar{\sigma}_{k+1}$ (i.e., a pair of opposite charges). By Ward's identity (Lemma 3.6), we can sum $\mathcal{L} - \mathcal{K}$ of rank n over the index a_k and express the resulting sum in terms of $\mathcal{L} - \mathcal{K}$ of rank $n-1$ with a multiplicative factor $(W \eta_u)^{-1}$. The summation over the remaining indices contributes an additional factor of ℓ_u^{n-2} due to the fast decay property.

On the other hand, with the definition of ϑ and $\Theta^{(B)}$, we have

$$\max_{\mathbf{a}} |\dot{\vartheta}_{u,\mathbf{a}}| \prec \ell_u^{-n+1} \cdot \eta_u^{-1}.$$

Inserting them back to (5.95), we obtain that

$$\int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u] \right)_{\mathbf{a}} du \prec (W \ell_t \eta_t)^{-n} \max_{u \in [s,t]} \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}. \quad (5.97)$$

Similarly, we can estimate the second line of (5.94). By definition of Θ and Lemma 5.13, $\Theta_u \circ \mathcal{A}$ and $\mathcal{Q}_u \circ \mathcal{A}$ are rapidly decaying if $\mathcal{A}_u$ is rapidly decaying. Since $(\mathcal{L} - \mathcal{K})_u$ has the fast-decay property at time u , so does $[\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})$. On the other hand, we just showed that $[\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})$ also has sum zero property from (5.90). Therefore, with (7.16) (case 2), we have

$$\int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ ([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}) \right)_{\mathbf{a}} du \prec \int_s^t \left(\frac{\ell_u \eta_u}{\ell_t \eta_t} \right)^n \cdot \max_{\mathbf{a}} |([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma})_{\mathbf{a}}| du. \quad (5.98)$$

By definition, we can bound

$$\begin{aligned} [\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma,\mathbf{a}} &= -[(\mathcal{P} \circ (\Theta_{u,\sigma} \circ (\mathcal{L} - \mathcal{K}))) \cdot \vartheta_u]_{\mathbf{a}} + \Theta_{u,\sigma} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K}) \cdot \vartheta_u]_{\mathbf{a}}. \\ &\prec \|\vartheta_u\|_{\max} \cdot \eta_u^{-1} \max_a \left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \right]_a. \end{aligned} \quad (5.99)$$

Substituting it into (5.98) and using (5.96), we obtain that

$$\int_s^t (\mathcal{U}_{u,t,\sigma} \circ ([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}))_{\mathbf{a}} du \prec (W\ell_t\eta_t)^{-n} \max_{u \in [s,t]} \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}. \quad (5.100)$$

Furthermore, we can write the left hand side of (5.94) as

$$\mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} = (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} - \mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} \cdot \vartheta_{t,\mathbf{a}} = (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} + O_{\prec} \left((W\ell_t\eta_t)^{-n} \Xi_{t,n-1}^{(\mathcal{L}-\mathcal{K})} \right). \quad (5.101)$$

Inserting (5.97), (5.100) and (5.101) into (5.94), we obtain

$$\begin{aligned} (W\ell_t\eta_t)^n \cdot (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &\prec 1 + \max_{u \in [s,t]} \left(\max_{k < n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-1} + \Xi_{u,n+1}^{(\mathcal{L})} \right) \\ &\quad + (W\ell_t\eta_t)^n \cdot \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} du. \end{aligned} \quad (5.102)$$

For the martingale term, one can easily derive that

$$\begin{aligned} \left[\int \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} \right]_t &= \int_s^t \sum_{\alpha} \left| \sum_{k=1}^n \mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)}(\alpha, k) \right|^2 du \\ &\leq C_n \int_s^t \sum_{\alpha} \sum_{k=1}^n \left| \mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)}(\alpha, k) \right|^2 du \\ &= C_n \int_s^t \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} du, \end{aligned} \quad (5.103)$$

where

$$\begin{aligned} ((\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ \mathcal{A})_{\mathbf{a},\mathbf{b}} &= \mathcal{A}_{\mathbf{a},\mathbf{b}} - \sum_{\mathbf{a}'} \delta_{a'_1,a_1} \mathcal{A}_{\mathbf{a}',\mathbf{b}} \cdot \vartheta_{u,\mathbf{a}} - \sum_{\mathbf{b}'} \delta_{b'_1,b_1} \mathcal{A}_{\mathbf{a},\mathbf{b}'} \cdot \vartheta_{u,\mathbf{b}} \\ &\quad + \sum_{\mathbf{a}'} \sum_{\mathbf{b}'} \delta_{a'_1,a_1} \delta_{b'_1,b_1} \mathcal{A}_{\mathbf{a}',\mathbf{b}'} \cdot \vartheta_{u,\mathbf{a}} \cdot \vartheta_{u,\mathbf{b}}. \end{aligned} \quad (5.104)$$

Equivalently, since $\mathcal{Q}_u = I - \vartheta_u \mathcal{P}$,

$$\mathcal{Q}_u \otimes \mathcal{Q}_u = I \otimes I - (\vartheta_u \mathcal{P}) \otimes I - I \otimes (\vartheta_u \mathcal{P}) + (\vartheta_u \mathcal{P}) \otimes (\vartheta_u \mathcal{P}).$$

where $\mathbf{a} = (a_1, \dots, a_n)$, $\mathbf{a}' = (a'_1, \dots, a'_n)$ and similarly for $\mathbf{b}$ and $\mathbf{b}'$. Recall $\mathcal{P} \circ \vartheta_{u,\mathbf{a}} = 1$, i.e.

$$\sum_{a_2, \dots, a_n} \vartheta_{u,\mathbf{a}} = 1, \quad \sum_{b_2, \dots, b_n} \vartheta_{u,\mathbf{b}} = 1$$

Consequently,

$$(\mathcal{P} \otimes I) \circ ((\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ \mathcal{A}) = 0, \quad (I \otimes \mathcal{P}) \circ ((\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ \mathcal{A}) = 0.$$

Using (5.93) on the fast decay tensor $\mathcal{E} \otimes \mathcal{E}$, we have

$$\left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}} \prec \left(\frac{\ell_u \eta_u}{\ell_t \eta_t} \right)^{2n} \max_{\mathbf{a},\mathbf{a}'} \left| \left((\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{\mathbf{a},\mathbf{a}'} \right|$$

$$\prec \left(\frac{\ell_u \eta_u}{\ell_t \eta_t} \right)^{2n} \max_{\mathbf{a}, \mathbf{a}'} \left| \left((\mathcal{E} \otimes \mathcal{E})_{u, \sigma} \right)_{\mathbf{a}, \mathbf{a}'} \right| \prec (W \ell_t \eta_t)^{-2n} \cdot \eta_u^{-1} \cdot \Xi_{u, 2n+2}^{\mathcal{L}}, \quad (5.105)$$

where we have used (5.77). Substituting it into (5.103) and using BDG inequality and the assumption $\Xi_{u, 2n+2}^{\mathcal{L}} \prec \Lambda$, we obtain that

$$(W \ell_t \eta_t)^n \cdot \int_s^t \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u, \sigma}^{(\mathbf{M})} \right)_{\mathbf{a}} \prec \Lambda^{1/2}.$$

This completes the proof of Lemma 5.14. □

5.6 Proof of Theorem 2.21, step 3

Recall that in this step, in addition to the assumptions in Theorem 2.21, we also have the local law (2.75) and the decay property for $\mathcal{L} - \mathcal{K}$ (2.76). With these inputs, we obtained (5.92) in Lemma 5.14. In this step, we aim to prove the correct bound (2.77) on G -loop. We have proved in Step 1 the following bounds

$$\Xi_{u, n}^{(\mathcal{L})} \prec (\ell_u / \ell_s)^{n-1}, \quad \Xi_{u, n}^{(\mathcal{L} - \mathcal{K})} \prec (W \ell_u \eta_u) \cdot (\ell_u / \ell_s)^{n-1}. \quad (5.106)$$

Using the bound (2.59) on $\mathcal{K}$, we can relate $\Xi^{(\mathcal{L})}$ with $\Xi^{(\mathcal{L} - \mathcal{K})}$ as follows:

$$\Xi_{u, n}^{(\mathcal{L})} \prec 1 + (W \ell_u \eta_u)^{-1} \cdot \Xi_{u, n}^{(\mathcal{L} - \mathcal{K})}, \quad \Xi_{u, n}^{(\mathcal{L} - \mathcal{K})} \prec (W \ell_u \eta_u) \left(\Xi_{u, n}^{(\mathcal{L})} + 1 \right). \quad (5.107)$$

Denote by

$$\Psi(n, k, s, u, t) := (W \ell_s \eta_s)^{1/2} + (\ell_t / \ell_s)^{n-1} \times \begin{cases} (W \ell_u \eta_u)^{1-k/4} & k = 0 \\ (W \ell_s \eta_s)^{1-k/4} & k \geq 1 \end{cases}, \quad u \in [s, t]. \quad (5.108)$$

We say that the estimate $\mathcal{S}(n, k, s, u, t)$ holds if $\Xi_{u, n}^{(\mathcal{L} - \mathcal{K})} \prec \Psi(n, k, s, u, t)$. We will prove that for any $n \geq 3$ and $k \geq 1$

$$\mathcal{S}(m, l, s, u, t) \text{ holds for } \{(m, l) : l = k \text{ and } m \leq n-1, \text{ or } l = k-1, m \leq n+2\} \implies \mathcal{S}(n, k, s, u, t) \text{ holds.} \quad (5.109)$$

Suppose the last statement holds. We now prove (2.77).

After the detailed proof, we provide a high-level, intuitive explanation of the argument.

Proof of (2.77). With the bound (2.73) for $\mathcal{L}$ and (3.46) for $\mathcal{K}$, $\mathcal{S}(m, l, s, u, t)$ holds for $l = 0$ and any $m \geq 1$. By (2.75), (2.76), and a standard continuity argument based on Step 1, (5.92), and the initial bound (5.110), $\mathcal{S}(m, l, s, u, t)$ holds for every l and $m = 1, 2$. By (5.109), $\mathcal{S}(m, l, s, u, t)$ holds for $(3, 1)$. Then we apply (5.109) again and $\mathcal{S}(m, l, s, u, t)$ holds for $(4, 1)$. We can continue this procedure until $\mathcal{S}(m, l, s, u, t)$ holds for $(n, 1)$ for any n . We can now repeat this process and prove that $\mathcal{S}(m, l, s, u, t)$ holds for $(n, 2)$ for any n . Eventually, the induction implies that $\mathcal{S}(n, k, s, u, t)$ holds for any n, k . By condition (2.72), for any fixed n there is a large enough k so that

$$(\ell_t / \ell_s)^{n-1} (W \ell_s \eta_s)^{1-k/4} \ll 1.$$

Thus the fact that $\mathcal{S}(n, k, s, u, t)$ holds for such n, k implies that $\Xi_{u, n}^{(\mathcal{L} - \mathcal{K})} \prec (W \ell_s \eta_s)^{1/2}$. Together with (5.107), we have proved (2.77).

Now we only need to prove (5.109). Recall at the time s that the following initial bound holds:

$$\Xi_{s, n}^{(\mathcal{L} - \mathcal{K})} \prec 1. \quad (5.110)$$

We claim that

$$\Xi_{u, 2n+2}^{(\mathcal{L})} \prec \Psi(n, k, s, u, t)^2. \quad (5.111)$$

Using this bound as an input, from (5.92) we obtain that

$$\Xi_{t',n}^{(\mathcal{L}-\mathcal{K})} \prec \max_{v \in [s,t']} \left(\Psi(n, k, s, v, t') + \max_{m < n} \Xi_{v,m}^{(\mathcal{L}-\mathcal{K})} + \max_{m: 2 \leq m \leq n} \Xi_{v,m}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{v,n-m+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W \ell_v \eta_v)^{-1} + \Xi_{v,n+1}^{(\mathcal{L})} \right) \quad (5.112)$$

for any $s \leq t' \leq t$. We aim to prove the conclusion of (5.109) for a fixed (n, k) . Using (5.107), the last term $\Xi_{v,n+1}^{(\mathcal{L})}$ can be bounded by $\Xi_{v,n+1}^{(\mathcal{L}-\mathcal{K})}$ with a small prefactor. Inserting this bound and the assumption (5.109) into the last display, we have

$$\Xi_{t',n}^{(\mathcal{L}-\mathcal{K})} \prec \max_{v \in [s,t']} \Psi(n, k, s, v, t'). \quad (5.113)$$

Here we have used $k \geq 1$. Setting $t' = u$ and using the monotonicity of Ψ in the t variable, we have proved the conclusion of (5.109).

We now prove (5.111) for $\Xi_{u,2n+2}^{(\mathcal{L})}$ under the assumption in (5.109). First, the long loop can be bounded with short chains. Let

$$\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n), \quad \mathbf{a} = (a_1, a_2, \dots, a_{n-1}), \quad \sigma_k \in \{+, -\}, \quad a_k \in \mathbb{Z}_L.$$

We temporarily define a G -chain as

$$\mathcal{C}_{t,\sigma,\mathbf{a}} = G_t(\sigma_1) E_{a_1} G_t(\sigma_2) E_{a_2} \cdots E_{a_{n-1}} G_t(\sigma_n). \quad (5.114)$$

Given an arbitrary loop $\mathcal{L}_{u,\sigma,\mathbf{a}}^{(2n+2)}$ of length $2n+2$, with

$$\sigma = (\sigma, \sigma_1, \dots, \sigma_n, \sigma', \sigma'_1, \dots, \sigma'_n), \quad \mathbf{a} = (b, a_1, \dots, a_n, b', a'_1, \dots, a'_n),$$

we can write it as

$$\mathcal{L}_{u,\sigma,\mathbf{a}}^{(2n+2)} = W^{-2} \sum_{i \in \mathcal{I}_b, j \in \mathcal{I}_{b'}} \left(\mathcal{C}_{u,\sigma_1,\mathbf{a}_1}^{(n+1)} \right)_{ij} \left(\mathcal{C}_{u,\sigma_2,\mathbf{a}_2}^{(n+1)} \right)_{ji},$$

where $\sigma_1 = (\sigma_1, \dots, \sigma_n, \sigma')$, $\sigma_2 = (\sigma'_1, \dots, \sigma'_n, \sigma)$, $\mathbf{a}_1 = (a_1, \dots, a_n)$, and $\mathbf{a}_2 = (a'_1, \dots, a'_n)$. Applying the Cauchy-Schwarz inequality, we obtain that

$$\max_{\mathbf{a}} \max_{\sigma \in \{+,-\}^{2n+2}} \left| \mathcal{L}_{u,\sigma,\mathbf{a}}^{(2n+2)} \right| \leq \max_{\mathbf{a}_1} \max_{\sigma_1 \in \{+,-\}^{n+1}} W^{-2} \sum_{i \in \mathcal{I}_b, j \in \mathcal{I}_{b'}} \left| \left(\mathcal{C}_{u,\sigma_1,\mathbf{a}_1}^{(n+1)} \right)_{ij} \right|^2. \quad (5.115)$$

We now split the $(n+1)$ chain further into an l_1 - G chain and an l_2 - G chain, where $l_1 = \lfloor (n+1)/2 \rfloor$ and $l_2 = n+1 - l_1$. Using the Cauchy-Schwarz inequality again, we get that

$$W^{-2} \sum_{i \in \mathcal{I}_b, j \in \mathcal{I}_{b'}} \left| \left(\mathcal{C}_{u,\sigma_1,\mathbf{a}_1}^{(n+1)} \right)_{ij} \right|^2 \leq W^{-2} \sum_{i \in \mathcal{I}_b, j \in \mathcal{I}_{b'}} \left(\mathcal{C}_{u,\sigma'_1,\mathbf{a}'_1}^{(2l_1)} \right)_{ii} \left(\mathcal{C}_{u,\sigma'_2,\mathbf{a}'_2}^{(2l_2)} \right)_{jj} = \mathcal{L}_{u,\sigma'_1,\tilde{\mathbf{a}}_1}^{(2l_1)} \cdot \mathcal{L}_{u,\sigma'_2,\tilde{\mathbf{a}}_2}^{(2l_2)}, \quad (5.116)$$

where $\sigma'_1, \mathbf{a}'_1, \sigma'_2, \mathbf{a}'_2$ are defined by

$$\begin{aligned} \sigma'_1 &= (\sigma_1, \dots, \sigma_{l_1}, -\sigma_{l_1}, \dots, -\sigma_1), & \mathbf{a}'_1 &= (a_1, \dots, a_{l_1-1}, a_{l_1}, a_{l_1-1}, \dots, a_1), \\ \sigma'_2 &= (-\sigma_{n+1}, \dots, -\sigma_{l_1+1}, \sigma_{l_1+1}, \dots, \sigma_{n+1}), & \mathbf{a}'_2 &= (a_n, \dots, a_{l_1+1}, a_{l_1}, a_{l_1+1}, \dots, a_n), \end{aligned}$$

$\tilde{\mathbf{a}}_1$ is obtained by adding b to $\mathbf{a}'_1$, and $\tilde{\mathbf{a}}_2$ is obtained by adding b' to $\mathbf{a}'_2$. Plugging (5.116) into (5.115) and using that $l_1 + l_2 = n+1$, we obtain

$$\max_{\mathbf{a}} \max_{\sigma \in \{+,-\}^{2n+2}} \left| \mathcal{L}_{u,\sigma,\mathbf{a}}^{(2n+2)} \right| \leq \left(\max_{\mathbf{a}} \max_{\sigma \in \{+,-\}^{2l_1}} \left| \mathcal{L}_{u,\sigma,\mathbf{a}}^{(2l_1)} \right| \right) \cdot \left(\max_{\mathbf{a}} \max_{\sigma \in \{+,-\}^{2l_2}} \left| \mathcal{L}_{u,\sigma,\mathbf{a}}^{(2l_2)} \right| \right), \quad (5.117)$$

i.e.

$$\Xi_{u,2n+2}^{(\mathcal{L})} \leq \Xi_{u,2l_1}^{(\mathcal{L})} \cdot \Xi_{u,2l_2}^{(\mathcal{L})} \cdot (W\ell_u\eta_u). \quad (5.118)$$

Since $\max(2l_1, 2l_2) \leq n+2$, with the assumption of (5.109) and (5.118), we can bound $\Xi_{u,2n+2}^{(\mathcal{L})}$ as follows. Here $m \in \{1, 2\}$ is a dummy product index.

$$\Xi_{u,2n+2}^{(\mathcal{L})} \prec (W\ell_u\eta_u) \cdot \prod_{m=1}^2 (1 + (W\ell_u\eta_u)^{-1} \cdot \Psi(2l_m, k-1, s, u, t)). \quad (5.119)$$

Then with $l_1 + l_2 = n+1$, and $|l_1 - l_2| \leq 1$, we obtain that

$$\Xi_{u,2n+2}^{(\mathcal{L})} \prec \left((W\ell_s\eta_s)^{1/2} + (\ell_t/\ell_s)^{n-1} (W\ell_s\eta_s)^{1-k/4} \right)^2 = \Psi(n, k, s, u, t)^2. \quad (5.120)$$

Hence we have proved (5.111). This completes the proof of (2.77), i.e., Step 3 in the proof of Theorem 2.21. $\square$

An intuitive explanation for the proof of (2.77): Roughly speaking, an $n - \mathcal{L}_u$ loop is of order A_u^{-n+1} with $A_u \sim W\ell_u\eta_u$. Assuming n is odd for simplicity, from (5.118) we have that

$$\Xi_{u,2n+2}^{(\mathcal{L})} \leq (\Xi_{u,n+1}^{(\mathcal{L})})^2 A_u \quad (5.121)$$

Hence (5.92) takes the form

$$\Xi_{t',n}^{(\mathcal{L}-\mathcal{K})} \prec \max_{v \in [s,t']} \left(\Xi_{v,n+1}^{(\mathcal{L})} A_v^{1/2} + \max_{m < n} \Xi_{v,m}^{(\mathcal{L}-\mathcal{K})} + \max_{m: 2 \leq m \leq n} \Xi_{v,m}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{v,n-m+2}^{(\mathcal{L}-\mathcal{K})} A_v^{-1} + \Xi_{v,n+1}^{(\mathcal{L})} \right) \quad (5.122)$$

The third term on the right hand side is of lower order while the second term is bounded by induction on n . We are left with the first and the last terms involving $n+1$ (and $n+2$ in case n is even) $\mathcal{L}$ loops. By (5.107),

$$\Xi_{v,n+1}^{(\mathcal{L})} \prec 1 + A_v^{-1} \Xi_{v,n+1}^{(\mathcal{L}-\mathcal{K})}. \quad (5.123)$$

Therefore, up to terms which are either negligible or can be bounded by induction, we have

$$\Xi_{t',n}^{(\mathcal{L}-\mathcal{K})} \prec \max_{v \in [s,t']} \left(A_v^{1/2} + A_v^{-1/2} \Xi_{v,n+1}^{(\mathcal{L}-\mathcal{K})} + A_v^{-1} \Xi_{v,n+1}^{(\mathcal{L}-\mathcal{K})} \right) \quad (5.124)$$

Although we cannot bound $\Xi_n^{(\mathcal{L}-\mathcal{K})}$ by $\Xi_{n+1}^{(\mathcal{L}-\mathcal{K})}$ by induction, we can derive rough bounds on $\Xi_n^{(\mathcal{L}-\mathcal{K})}$ for all n and then bootstrap the argument. This is basically our procedure to arrive at $\Xi_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec A_t^{1/2}$.

5.7 Proof of Theorem 2.21, Steps 4 and 5

Step 4: Proof of (2.78). Recall the key inequalities (5.111) and (5.112) in the proof of step 3. We can now use (2.77) to bound the $\Xi_{2n+2}^{(\mathcal{L})}$ in (5.111) and $\Xi_{n+1}^{(\mathcal{L})}$ in (5.112). Then we obtain

$$\Xi_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec 1 + \max_{u \in [s,t]} \left(\max_{k < n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot (W\ell_u\eta_u)^{-1} \right). \quad (5.125)$$

By (2.76), (4.5) and Step 3 for $(\mathcal{L} - \mathcal{K})$ -loops of length 1 and 2 and the condition (2.72), we have

$$\Xi_{t,1}^{(\mathcal{L}-\mathcal{K})} \prec 1, \quad \Xi_{t,2}^{(\mathcal{L}-\mathcal{K})} \prec (W\ell_t\eta_t)^{1/4}.$$

Using (5.125) and induction on n , one can easily prove $\Xi_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec 1$ for any fixed n . Indeed, for $n = 2$ the quadratic term in (5.125) is $(\Xi_{u,2}^{(\mathcal{L}-\mathcal{K})})^2 (W\ell_u\eta_u)^{-1} \prec (W\ell_u\eta_u)^{-1/2}$, so (5.125), applied at every $u \in [s,t]$, gives $\Xi_{u,2}^{(\mathcal{L}-\mathcal{K})} \prec 1$. For $n \geq 3$, the terms $k = 2$ and $k = n$ of the quadratic sum in (5.125) are linear in $\Xi_{u,n}^{(\mathcal{L}-\mathcal{K})}$, and they are $\prec 1$ by $\Xi_{u,2}^{(\mathcal{L}-\mathcal{K})} \prec 1$ and the a priori bound $\Xi_{u,n}^{(\mathcal{L}-\mathcal{K})} \prec W\ell_u\eta_u$, which follows from (2.77) and (5.107). This completes the proof of (2.78). Notice that the prefactor $(\eta_s/\eta_t)^2$ in Step 2 was eliminated in Lemma 5.14 partly by using the sum zero property. It is important that this factor is eliminated so it will not accumulate from the time splitting argument in the proof of Theorem 2.21. $\square$

Step 5: Proof of (2.79). Our goal is to prove $\mathcal{J}_{t,D}^* \prec 1$. To this end, note that we have proved in Step 4 that

$$(\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} \prec (W\ell_t\eta_t)^{-2}$$

for the $\mathcal{L} - \mathcal{K}$ -loop of length 2. It implies that

$$(\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|) \prec 1, \quad \text{for } |a_1 - a_2| = O(\ell_t^*).$$

It remains to consider the case $|a_1 - a_2| > 6\ell_t^*$. But (2.79) is a consequence of the bound for $|a_1 - a_2| > 6\ell_v^*$ proved after (5.48), with $v = t$. This completes the proof of Step 5. $\square$

5.8 Proof of Theorem 2.21, step 6

In this step, we have the estimates in the assumption (2.71) and all results from step 4 and 5, i.e., (2.78) and (2.79). Our goal is to prove (2.80). We first prove the following bound on $\mathbb{E}(\mathcal{L} - \mathcal{K})$ which improves the 1 loop bound on $(\mathcal{L} - \mathcal{K})$.

Lemma 5.15.

$$\max_a |\mathbb{E}(\mathcal{L} - \mathcal{K})_{u, (+), (a)}| = \max_a |\mathbb{E}\langle (G_u - m)E_a \rangle| \prec (W\ell_u\eta_u)^{-2}. \quad (5.126)$$

Proof of Lemma 5.15. We will denote G_u by G in this proof. Recall that $G - m = m(-H - um)G$. Using Gaussian integration by parts, we have

$$\begin{aligned} \mathbb{E}\langle (G - m)E_a \rangle &= m\mathbb{E}\langle (-H - um)GE_a \rangle \\ &= um \sum_b \mathbb{E}\langle (G - m)E_b \rangle W \cdot S_{bb'}^{(B)} \langle E_{b'}GE_a \rangle \\ &= um \sum_b \mathbb{E}\langle (G - m)E_b \rangle S_{ba}^{(B)} \langle GE_a \rangle. \end{aligned} \quad (5.127)$$

Writing $\langle GE_a \rangle = m + \langle (G - m)E_a \rangle$, we obtain

$$\begin{aligned} \mathbb{E}\langle (G - m)E_a \rangle &= \sum_{a'} \left((1 - um^2 S^{(B)})^{-1} \right)_{aa'} \mathbb{E} \left(um \sum_b \langle (G - m)E_b \rangle S_{ba'}^{(B)} \langle (G - m)E_{a'} \rangle \right) \\ &= \sum_{a'} \left(\Theta_{um^2}^{(B)} \right)_{aa'} \cdot \mathbb{E} \left(um \sum_b \langle (G - m)E_b \rangle S_{ba'}^{(B)} \langle (G - m)E_{a'} \rangle \right). \end{aligned} \quad (5.128)$$

From (2.78), we know that $\langle (G - m)E_{a'} \rangle \prec (W\ell_u\eta_u)^{-1}$. Substituting this into the above estimate, we obtain (5.126) and have completed the proof of Lemma 5.15. $\square$

Proof of (2.80). Taking expectation of both sides of (5.20) for the case $n = 2$, we have

$$\mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} = (\mathcal{U}_{s,t,\sigma} \circ \mathbb{E}(\mathcal{L} - \mathcal{K})_{s,\sigma})_{\mathbf{a}} \quad (5.129)$$

$$+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathbb{E} \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \right)_{\mathbf{a}} du \quad (5.130)$$

$$+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathbb{E} \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_{\mathbf{a}} du. \quad (5.131)$$

By assumption, we have

$$\mathbb{E}(\mathcal{L} - \mathcal{K})_{s,\sigma,\mathbf{a}} \prec (W\ell_s\eta_s)^{-3}. \quad (5.132)$$

Similarly $\mathcal{E}_{u,\sigma}^{((L-K)\times(L-K))}$ can be bounded as the product of two $(\mathcal{L} - \mathcal{K})$, and with the fast decay property, we obtain

$$\mathcal{E}_{u,\sigma}^{((L-K)\times(L-K))} \prec W\ell_u(W\ell_u\eta_u)^{-4} \prec (\eta_u)^{-1}(W\ell_u\eta_u)^{-3} \quad (5.133)$$

Similarly to the arguments used in proving (5.52), we can bound $\mathbb{E}\mathcal{E}^{(\tilde{G})}$ as the product of a 1- $(\mathcal{L} - \mathcal{K})$ loop and a 3- $\mathcal{L}$ loop. Writing $\mathcal{L} = \mathcal{K} + (\mathcal{L} - \mathcal{K})$, we obtain

$$\begin{aligned} \mathbb{E}\mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} &\prec W \cdot \ell_u \cdot \max_a \max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^3} \left| \mathbb{E} \left[\langle (G - m)E_a \rangle \cdot \mathcal{L}_{u,\sigma,\mathbf{a}} \right] \right| \\ &\prec W \cdot \ell_u \cdot \max_a \left| \mathbb{E} \langle (G - m)E_a \rangle \right| \cdot \max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^3} |\mathcal{K}_{u,\sigma,\mathbf{a}}| \\ &+ W \cdot \ell_u \cdot \max_a |(\mathcal{L} - \mathcal{K})_{u,(+),(a)}| \cdot \max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^3} |(\mathcal{L} - \mathcal{K})_{u,\sigma,\mathbf{a}}|. \end{aligned} \quad (5.134)$$

Using (5.126) for $\mathbb{E}\langle (G - m)E_a \rangle$, (2.59) for $\mathcal{K}$, and (2.78) for $\mathcal{L} - \mathcal{K}$, we have

$$\mathbb{E}\mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} \prec (\eta_u)^{-1}(W\ell_u\eta_u)^{-3}. \quad (5.135)$$

One can easily see that all tensors discussed above have fast decay. Applying (7.14) to bound $\max \rightarrow \max$ norm of $\mathcal{U}_{u,t}$ and using (5.132), (5.133) and (5.135), we can estimate the right hand side of (5.129) by

$$(W\ell_t\eta_t)^3 \cdot \mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} \prec (\ell_t/\ell_s) \frac{\ell_t\eta_t}{\ell_s\eta_s} + \int_s^t (\ell_t/\ell_u) \frac{\ell_t\eta_t}{\ell_u\eta_u} \cdot \eta_u^{-1} du \prec 1, \quad (5.136)$$

where we have used

$$u \leq t \implies \frac{\ell_t^2\eta_t}{\ell_u^2\eta_u} \leq 1.$$

This completes the proof of (2.80). $\square$

6 Continuity Estimates on Loops

In this section, we prove Lemma 5.1. Recall that $t_1 \leq t_2 \leq 1$, $G_{t_1} = (H_{t_1} - z_{t_1})^{-1}$, and $G_{t_2} = (H_{t_2} - z_{t_2})^{-1}$. Throughout this section, we use the notation

$$\tilde{G}_{t_1} := (H_{t_2} - \tilde{z}_{t_1})^{-1}, \quad \tilde{z}_{t_1} := \left(\frac{t_2}{t_1} \right)^{1/2} z_{t_1}$$

and denote by $\tilde{\mathcal{L}}_{t_1}$ the loops with resolvent $\tilde{G}_{t_1}$. By scaling, we have

$$\tilde{\mathcal{L}}_{t_1} \sim \left(\frac{t_1}{t_2} \right)^{n/2} \mathcal{L}_{t_1}, \quad \tilde{G}_{t_1} \sim \left(\frac{t_1}{t_2} \right)^{1/2} \cdot G_{t_1} \quad \text{in distribution.} \quad (6.1)$$

One can check that $|z_{t_2} - \tilde{z}_{t_1}| \leq C(1 - t_1)$ for $t_1 > c > 0$. From now on, we drop the t subscript and use the following notations

$$G_{t_2} \longleftrightarrow G, \quad \tilde{G}_{t_1} \longleftrightarrow \tilde{G}, \quad \mathcal{L}_{t_2} \longleftrightarrow \mathcal{L}, \quad \tilde{\mathcal{L}}_{t_1} \longleftrightarrow \tilde{\mathcal{L}}, \quad z_{t_2} \longleftrightarrow z, \quad \tilde{z}_{t_1} \longleftrightarrow \tilde{z}. \quad (6.2)$$

With these notations and the resolvent formula,

$$G = \tilde{G} + (z - \tilde{z}) \cdot G \cdot \tilde{G}. \quad (6.3)$$

We first recall a basic linear algebra fact.

Lemma 6.1. *Let v and w_i ($1 \leq i \leq m$) be vectors in a Hilbert space H and $A_{ij} = (w_i, w_j)$. Then for any $p \geq 1$,*

$$\sum_i |(v, w_i)|^2 \leq \|v\|_2^2 \cdot (\text{tr } A^p)^{1/p}.$$

Proof. Define a linear operator $T : H \rightarrow \mathbb{C}^m$ by $w \rightarrow \sum_i (w, w_i) e_i$. Then

$$\frac{\sum_i |(v, w_i)|^2}{\|v\|_2^2} = \frac{\|Tv\|^2}{\|v\|^2} = \|T^*T\|_{l_2 \rightarrow l_2} = \|TT^*\|_{l_2 \rightarrow l_2} = \|A\|_{l_2 \rightarrow l_2} \leq (\text{tr } A^p)^{1/p}.$$

□

Proof of Lemma 5.1. Using the Cauchy-Schwarz inequality, we know that the G loop with length $2m + 1$ ($m \geq 1$) can be bounded by the G loop with lengths $2m$ and $2m + 2$, i.e.,

$$\max_{\mathbf{a}, \boldsymbol{\sigma}} \left| \mathcal{L}_{t, \boldsymbol{\sigma}, \mathbf{a}}^{(\text{length}=2m+1)} \right|^2 \leq \left(\max_{\mathbf{a}, \boldsymbol{\sigma}} \left| \mathcal{L}_{t, \boldsymbol{\sigma}, \mathbf{a}}^{(\text{length}=2m)} \right| \right) \cdot \left(\max_{\mathbf{a}, \boldsymbol{\sigma}} \left| \mathcal{L}_{t, \boldsymbol{\sigma}, \mathbf{a}}^{(\text{length}=2m+2)} \right| \right). \quad (6.4)$$

Therefore, we will only prove (5.7) for even loop lengths. Using the Cauchy-Schwarz inequality again, we can assume that the loop is symmetric (recall G chain $\mathcal{C}$ defined in (5.114)), i.e.,

$$\begin{aligned} \mathcal{L}_{t, \boldsymbol{\sigma}', \mathbf{a}'} &= \langle E_{a_0} \cdot \mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}} \cdot E_{a_m} \cdot \mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}}^\dagger \rangle, \quad \boldsymbol{\sigma} = (\sigma_1, \sigma_2, \dots, \sigma_m), \quad \mathbf{a} = (a_1, a_2, \dots, a_{m-1}) \\ \boldsymbol{\sigma}' &= (\sigma_1, \sigma_2, \dots, \sigma_m, \bar{\sigma}_m, \bar{\sigma}_{m-1}, \dots, \bar{\sigma}_1), \quad \mathbf{a}' = (a_1, a_2, \dots, a_{m-1}, a_m, a_{m-1}, \dots, a_1, a_0). \end{aligned} \quad (6.5)$$

We only need to prove that for any fixed $m \geq 1$, the following holds

$$\mathbf{1}_\Omega \cdot \mathcal{L}_{t_2, \boldsymbol{\sigma}', \mathbf{a}'} \prec (W \cdot \ell_{t_1} \cdot \eta_{t_2})^{-2m+1}, \quad (6.6)$$

under the induction hypothesis that this bound holds for $m' \leq m - 1$ for all $\mathbf{a}$ and $\boldsymbol{\sigma}$. Recall the identity

$$\prod_{k=1}^m (a_k + b_k) = \prod_{k=1}^m a_k + \sum_{l=1}^m \left(\prod_{j=1}^{l-1} (a_j + b_j) \right) b_l \left(\prod_{j=l+1}^m a_j \right). \quad (6.7)$$

With the notations $G_i = G_{t_2}(\sigma_i)$, $\tilde{G}_i = \tilde{G}_{t_1}(\sigma_i)$, we apply this identity with $a_i + b_i \rightarrow G_i$, $a \rightarrow \tilde{G}_i$. Together with the resolvent identity (6.3), we have

$$\begin{aligned} G_1 E_{a_1} \cdots G_{m-1} E_{a_{m-1}} G_m &= \tilde{G}_1 E_{a_1} \cdots \tilde{G}_{m-1} E_{a_{m-1}} \tilde{G}_m \\ &+ (z - \tilde{z}) \cdot \sum_{l=1}^m G_1 E_{a_1} \cdots G_{l-1} E_{a_{l-1}} \cdot \left(G_l \cdot \tilde{G}_l \right) E_{a_l} \tilde{G}_{l+1} \cdots \tilde{G}_m. \end{aligned} \quad (6.8)$$

In the last term, there are l instances of G and $m - l + 1$ instances of $\tilde{G}$ and there is no matrix E between G_l and $\tilde{G}_l$. It implies that for any i and j ,

$$\begin{aligned} \left| (G_1 E_{a_1} \cdots G_{m-1} E_{a_{m-1}} G_m)_{ij} \right|^2 &\leq C_m \left| \left(\tilde{G}_1 E_{a_1} \cdots \tilde{G}_{m-1} E_{a_{m-1}} \tilde{G}_m \right)_{ij} \right|^2 \\ &+ C_m |z_{t_2} - \tilde{z}_{t_1}|^2 \cdot \sum_{l=1}^m \left| \left(G_1 E_{a_1} \cdots G_l \cdot \tilde{G}_l E_{a_l} \cdots \tilde{G}_m \right)_{ij} \right|^2. \end{aligned} \quad (6.9)$$

For any l, j , denote by $v^{(l)} \in \mathbb{C}^N$, $w_j^{(l)} \in \mathbb{C}^N$ the vectors with components given by

$$v^{(l)}(k) = (G_1 E_{a_1} \cdots E_{a_{l-1}} G_l)_{ik}, \quad \overline{w_j^{(l)}(k)} = \left(\tilde{G}_l E_{a_l} \cdots E_{a_{m-1}} \tilde{G}_m \right)_{kj}.$$

Here i is treated as a given index and it will not play any active role in the following proof. Denote by $A^{(l)}$ the matrix with elements given by

$$A_{jj'}^{(l)} = \left(w_j^{(l)}, w_{j'}^{(l)} \right), \quad j, j' \in \mathcal{I}_{a_m}.$$

Applying Lemma 6.1 to the last term in (6.9), we have

$$\sum_{j \in \mathcal{I}_{a_m}} \left| \left(G_1 E_{a_1} \cdots G_l \cdot \tilde{G}_l E_{a_l} \cdots \tilde{G}_m \right)_{ij} \right|^2 = \sum_{j \in \mathcal{I}_{a_m}} \left| \left(v^{(l)}, w_j^{(l)} \right) \right|^2 \leq \|v^{(l)}\|_2^2 \left(\text{tr} \left(A^{(l)} \right)^p \right)^{1/p}. \quad (6.10)$$

With the identity $\tilde{G} \cdot \tilde{G}^\dagger = \frac{\tilde{G} - \tilde{G}^\dagger}{2i \text{Im} \tilde{z}} A_{jj'}^{(l)}$ can be written as $\tilde{\mathcal{C}}$ chain of length $2m - 2l + 1$, namely,

$$A_{jj'}^{(l)} = \frac{1}{2i \text{Im} \tilde{z}} \left[\left(\tilde{G}_m E_{a_{m-1}} \cdots E_{a_l} \tilde{G}_l \overline{E_{a_l} \cdots E_{a_{m-1}} \tilde{G}_m} \right)_{j'j} - \left(\tilde{G}_m E_{a_{m-1}} \cdots E_{a_l} \tilde{G}_l E_{a_l} \cdots E_{a_{m-1}} \tilde{G}_m \right)_{j'j} \right].$$

Hence we can write $\left(\frac{2i \text{Im} \tilde{z}}{W} \right)^p \cdot \text{tr} \left(A^{(l)} \right)^p$ as a sum of 2^p $\tilde{\mathcal{L}}_{t_1}$ -loops of length $2p(m - l) + p$. Using (6.1), we can bound $\tilde{\mathcal{L}}_{t_1}$ with $\mathcal{L}_{t_1}$. Using the induction hypothesis for $\mathcal{L}_{t_1}$, we have

$$\left(\text{tr} \left(A^{(l)} \right)^p \right)^{1/p} \prec \frac{W}{\text{Im} \tilde{z}} (W \ell_{t_1} \eta_{t_1})^{-2(m-l)-1+1/p}, \quad \text{Im} \tilde{z}_{t_1} \sim \text{Im} z_{t_1}.$$

Inserting this bound into (6.10) and (6.9) and averaging over $i \in \mathcal{I}_{a_0}$ and $j \in \mathcal{I}_{a_m}$, we can bound (6.5) by

$$\begin{aligned} \mathcal{L}_{t_2, \sigma', \mathbf{a}'} &= \left\langle E_{a_0} \cdot \mathcal{C}_{t, \sigma, \mathbf{a}} \cdot E_{a_m} \cdot \mathcal{C}_{t, \sigma, \mathbf{a}}^\dagger \right\rangle \\ &\prec \tilde{\mathcal{L}}_{t_1, \sigma', \mathbf{a}'} + \frac{|z_{t_2} - \tilde{z}_{t_1}|^2}{W \text{Im} \tilde{z}_{t_1}} \sum_{l=1}^m \left(\sum_{i \in \mathcal{I}_{a_0}} \|v^{(l)}\|_2^2 \right) \cdot (W \ell_{t_1} \eta_{t_1})^{-2(m-l)-1}. \end{aligned} \quad (6.11)$$

Here we have taken p to be large enough so that $1/p$ can be absorbed in the $\prec$ notation. By definition, one can easily check that (here we use the assumption that $t_1 > c > 0$ for some c)

$$|z_{t_2} - \tilde{z}_{t_1}|^2 \leq C |t_2 - t_1|^2 \leq C (\eta_{t_1})^2, \quad \frac{|z_{t_2} - \tilde{z}_{t_1}|^2}{\text{Im} \tilde{z}_{t_1}} \leq C \eta_{t_1}.$$

Using Ward's identity $G \cdot G^\dagger = \frac{G - G^\dagger}{2i \text{Im} z}$ again, we have

$$\begin{aligned} \frac{1}{W} \sum_{i \in \mathcal{I}_{a_0}} \|v^{(l)}\|_2^2 &= \frac{1}{2i \text{Im} z} \left(\mathcal{L}_{t_2, \sigma_l^{(1)}, \mathbf{a}_l} - \mathcal{L}_{t_2, \sigma_l^{(2)}, \mathbf{a}_l} \right), \quad \mathbf{a}_l = (a_0, a_1, \dots, a_{l-1}, a_{l-1}, \dots, a_1), \\ \sigma_l^{(1)} &= (\sigma_1, \sigma_2, \dots, \sigma_{l-1}, \sigma_l, \bar{\sigma}_{l-1}, \dots, \bar{\sigma}_2, \bar{\sigma}_1), \quad \sigma_l^{(2)} = (\sigma_1, \sigma_2, \dots, \sigma_{l-1}, \bar{\sigma}_l, \bar{\sigma}_{l-1}, \dots, \bar{\sigma}_2, \bar{\sigma}_1). \end{aligned} \quad (6.12)$$

Here the $\mathcal{L}$ loops are length $2l - 1$. We can bound these loops by induction so that

$$\mathbf{1}_\Omega \cdot \frac{1}{W} \sum_{i \in \mathcal{I}_{a_0}} \|v^{(l)}\|_2^2 \prec \frac{1}{\text{Im} z} (W \ell_{t_1} \eta_{t_2})^{-2l+2}, \quad l < m, \quad z = z_{t_2}.$$

Inserting this bound to (6.11), we have proved that

$$\mathbf{1}_\Omega \cdot \mathcal{L}_{t_2, \sigma', \mathbf{a}'} \prec (W \ell_{t_1} \eta_{t_2})^{-2m+1} + \frac{\eta_{t_1}}{\eta_{t_2}} \cdot \left(\left| \mathcal{L}_{t_2, \sigma_m^{(1)}, \mathbf{a}_m} - \mathcal{L}_{t_2, \sigma_m^{(2)}, \mathbf{a}_m} \right| \right) (W \ell_{t_1} \eta_{t_1})^{-1}. \quad (6.13)$$

Here $\mathcal{L}_{t_2, \sigma_m^{(1)}, \mathbf{a}_m}$ and $\mathcal{L}_{t_2, \sigma_m^{(2)}, \mathbf{a}_m}$ are G loops with the length $2m - 1$ and $\sigma_m^{(1)}, \sigma_m^{(2)} \in \{+, -\}^{2m-1}$.

For $m = 1$, by definition (5.6),

$$\mathbf{1}_\Omega \cdot \mathcal{L}_{t_2, \sigma_m^{(1)}, \mathbf{a}_m}, \quad \mathbf{1}_\Omega \cdot \mathcal{L}_{t_2, \sigma_m^{(2)}, \mathbf{a}_m} \prec 1, \quad m = 1, \quad \sigma_m^{(1)}, \sigma_m^{(2)} \in \{+, -\}.$$

Inserting these bounds to (6.13), we obtain (6.6) in the case $m = 1$. For $m > 1$, applying (6.4) again, we can bound the $(2m - 1)$ - G -loop by a G -loop of length $2m - 2$ and a G -loop of length $2m$. Furthermore, we can use assumption (6.6) to bound this G loop with length $2m - 2$. Then we obtain that

$$\mathbf{1}_\Omega \cdot \mathcal{L}_{t_2, \sigma_m^{(1,2)}, \mathbf{a}_m} \prec \left[(W \ell_{t_1} \eta_{t_2})^{-2m+3} \max_{\sigma, \mathbf{a}} \left| \mathcal{L}_{t_2, \sigma, \mathbf{a}}^{\text{length}=2m} \right| \right]^{1/2}.$$

Inserting this bound to (6.13), we obtain

$$\mathbf{1}_\Omega \cdot \max_{\sigma', \mathbf{a}'} \left| \mathcal{L}_{t_2, \sigma', \mathbf{a}'}^{\text{length}=2m} \right| \prec (W \ell_{t_1} \eta_{t_2})^{-2m+1} + (W \ell_{t_1} \eta_{t_2})^{-m+1/2} \max_{\sigma', \mathbf{a}'} \left| \mathcal{L}_{t_2, \sigma', \mathbf{a}'}^{\text{length}=2m} \right|^{1/2}.$$

This implies (6.6) and we have thus proved Lemma 5.1. $\square$

7 Auxiliary proofs and technical estimates

7.1 Evolution kernel estimates

We first prove a simple $L_{\max \rightarrow \max}$ bound for $\mathcal{U}$ 5.17 in the following lemma.

Lemma 7.1 ($\|\mathcal{U}\|_{\max \rightarrow \max}$ estimate). *Let $\mathcal{A}$ be a tensor $\mathbb{Z}_L^n \rightarrow \mathbb{C}$ and $n \geq 2$. Then*

$$\|\mathcal{U}_{s,t,\sigma} \circ \mathcal{A}\|_{\max} \prec \|\mathcal{A}\|_{\max} \cdot (\eta_s/\eta_t)^n, \quad s < t < 1. \quad (7.1)$$

Proof. Recall that $\mathcal{U}_{s,t,\sigma}$ was defined in 5.17. We write

$$(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_{\mathbf{a}} = \sum_{\mathbf{b}} \prod_{i=1}^n \left(\Theta_{t\xi_i}^{(B)} (\Theta_{s\xi_i}^{(B)})^{-1} \right)_{a_i b_i} \mathcal{A}_{\mathbf{b}}, \quad \xi_i = m_i m_{i+1}.$$

Using

$$\Theta_{t\xi}^{(B)} (\Theta_{s\xi}^{(B)})^{-1} = 1 - (t-s)\xi \cdot S^{(B)} \cdot \Theta_{t\xi}^{(B)},$$

we obtain

$$\|\mathcal{U}_{s,t,\sigma} \circ \mathcal{A}\|_{\max} \prec \|\mathcal{A}\|_{\max} \cdot \left(1 + |1-s| \cdot \max_{\xi: |\xi| \leq 1} \|\Theta_{t\xi}^{(B)}\|_{\max \rightarrow \max} \right)^n.$$

The claim now follows from $|1-s| \sim \eta_s$ and

$$\|\Theta_{t\xi}^{(B)}\|_{\max \rightarrow \max} = O((1-t)^{-1}) = O(\eta_t^{-1}),$$

which are consequences of the entry bounds in (2.52). $\square$

Next, we prove that if $\mathcal{A}$ decays on the scale ℓ_s , then $\mathcal{U}_{s,t,(+,-)} \circ \mathcal{A}$ decays on the scale ℓ_t .

Lemma 7.2 (Tail Estimates). *Recall*

$$\mathcal{T}_t(\ell) := \frac{1}{(W\ell_t\eta_t)^2} \exp\left(-|\ell/\ell_t|^{1/2}\right).$$

For $\sigma = (+, -)$, assume that for $\mathcal{A}_{\mathbf{a}}$, $\mathbf{a} \in \mathbb{Z}_L^n$ and some large $D > 0$, we have

$$\mathcal{A}_{\mathbf{a}} \leq \mathcal{T}_s(a_1 - a_2) + W^{-D}.$$

Recalling $\mathcal{U}$ defined in Definition 5.17, we have

$$(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_{\mathbf{a}} \prec \mathcal{T}_t(a_1 - a_2) + W^{-D} \cdot (\eta_s/\eta_t)^2, \quad \text{if } |a_1 - a_2| \geq \ell_t^* := (\log W)^{3/2} \ell_t. \quad (7.2)$$

We first give a heuristic argument for the proof. Notice that the key term in $\mathcal{U}_{s,t,\sigma}$ is

$$(t-s)^2 \Theta_{t,a_1 b_1} \Theta_{t,a_2 b_2} \sim (\eta_s/\eta_t)^2 \omega_t(a_1 - b_1) \omega_t(a_2 - b_2); \quad \omega_t(a - b) = \eta_t \Theta_{t,ab}.$$

With this definition, we have the normalization $\sum_b \omega_t(b) = O(1)$. The corresponding term in $(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_{\mathbf{a}}$ is bounded by

$$(t-s)^2 \sum_{b_1, b_2} \Theta_{t,a_1 b_1} \Theta_{t,a_2 b_2} \mathcal{T}_s(b_1 - b_2) \sim (\eta_s/\eta_t)^2 \sum_{b_1, b_2} \omega_t(a_1 - b_1) \omega_t(a_2 - b_2) \mathcal{T}_s(b_1 - b_2). \quad (7.3)$$

Using $\omega_t * \omega_t \sim \omega_t$ and denoting $a_1 - a_2 = x$, the last line is bounded by

$$\begin{aligned} & (\eta_s/\eta_t)^2 (W\ell_s\eta_s)^{-2} \ell_t^{-1} \sum_b \exp\left(-|x-b|/\ell_t - \sqrt{|b|/\ell_s}\right) \\ & \leq (W\ell_t\eta_t)^{-2} (\ell_t\ell_s^{-2}) \sum_b \exp\left(-|x-b|/\ell_t - \sqrt{|b|/\ell_s}\right). \end{aligned} \quad (7.4)$$

We can also assume that $|b| \leq \ell_s(\log W)^{2+1/4}$. Otherwise, the last line is exponentially small. By assumption, $x \geq (\log W)^{3/2} \ell_t$. If $(\log W)\ell_s \leq \ell_t$, then $|x-b| \geq (\log W)^{3/2} \ell_t/2$ and the last line is exponentially small. Hence we can assume that $(\log W)\ell_s \geq \ell_t$. In this case, we use the trivial bound $-\sqrt{|b|/\ell_s} \leq -\sqrt{|b|/\ell_t}$ and perform the b summation to bound (7.4) by

$$(\log W)^C (W\ell_t\eta_t)^{-2} \exp\left(-\sqrt{x/\ell_t}\right). \quad (7.5)$$

Proof of Lemma 7.2. We split $\mathcal{A}$ into two parts:

$$\mathcal{A} = \mathcal{A}^T + \mathcal{A}^D, \quad \mathcal{A}_{\mathbf{a}}^T \leq \mathcal{T}_s(a_1 - a_2), \quad \mathcal{A}^D \leq W^{-D}.$$

Since $\mathcal{U}$ is linear, we only need to prove that

$$(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A}^T)_{\mathbf{a}} \prec \mathcal{T}_t(a_1 - a_2) + W^{-D}, \quad (\mathcal{U}_{s,t,\sigma} \circ \mathcal{A}^D)_{\mathbf{a}} \prec W^{-D} \cdot (\eta_s/\eta_t)^2. \quad (7.6)$$

One can easily use the $\mathcal{U}_{\max \rightarrow \max}$ bound in Lemma 7.1 to prove the estimate on $\mathcal{A}^D$. It remains to prove the $\mathcal{A}^T$ part in (7.6). Recall the assumption $\sigma = (+, -)$, the $\mathcal{U}_{s,t,\sigma}$ defined in 5.17, and the bound

$$\|\Theta_t^{(B)}\|_{\max} \prec (\ell_t \eta_t)^{-1}. \quad (7.7)$$

Due to the decay of Θ_t and $\mathcal{A}^T$, we can restrict $\mathbf{b}$ in $\sum_{\mathbf{b}}$ to

$$(\mathcal{U}_{s,t} \circ \mathcal{A}^T)_{\mathbf{a}} = \sum_{b_1, b_2}^* \prod_{i=1}^2 (\Theta_t \Theta_s^{-1})_{a_i b_i} \mathcal{A}_{\mathbf{b}} + W^{-D}, \quad (7.8)$$

where

$$\sum_{b_1, b_2}^* = \sum_{b_1, b_2} \prod_{i=1}^2 \mathbf{1} \left(|a_i - b_i| \leq \frac{1}{4} \ell_t^* \right) \cdot \mathbf{1} (|b_1 - b_2| \leq \log^3 W \cdot \ell_s). \quad (7.9)$$

By assumption $|a_1 - a_2| \geq \ell_t^* = (\log W)^{3/2} \cdot \ell_t$, the above summation is nontrivial only if

$$\ell_s \geq \ell_t \cdot (\log W)^{-2}.$$

In other words, if $|a_1 - a_2| \geq \ell_t^*$, then for any $D > 0$,

$$\ell_s \leq \ell_t \cdot (\log W)^{-2} \implies (\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_{\mathbf{a}} \leq W^{-D}.$$

From now on, we assume that

$$\ell_t \prec \ell_s, \quad \sum_{b_1, b_2}^* \mathbf{1} \prec \ell_t \ell_s, \quad \sum_{b_1, b_2}^* \delta_{a_1 b_1} \prec \ell_s.$$

We drop the superscript (B) for simplicity, write

$$\begin{aligned} \prod_{i=1}^2 (\Theta_t \Theta_s^{-1})_{a_i b_i} &= \prod_{i=1}^2 \delta_{a_i b_i} - (t-s) \delta_{a_1 b_1} \cdot \left(S^{(B)} \cdot \Theta_t \right)_{a_2 b_2} - (t-s) \delta_{a_2 b_2} \cdot \left(S^{(B)} \cdot \Theta_t \right)_{a_1 b_1} \\ &\quad + (t-s)^2 \left(S^{(B)} \cdot \Theta_t \right)_{a_1 b_1} \left(S^{(B)} \cdot \Theta_t \right)_{a_2 b_2} := \mathcal{V}_{\mathbf{a}, \mathbf{b}}^{(1)} + \mathcal{V}_{\mathbf{a}, \mathbf{b}}^{(2)} + \mathcal{V}_{\mathbf{a}, \mathbf{b}}^{(3)} + \mathcal{V}_{\mathbf{a}, \mathbf{b}}^{(4)}. \end{aligned} \quad (7.10)$$

Together with (7.7), assumption $\mathcal{A}_{b_1 b_2}^T \prec \mathcal{T}_s(b_1 - b_2)$, and $t - s = O(\eta_s)$, we have

$$\left(\sum_{\mathbf{b}}^* \mathcal{V}^{(1,2,3,4)} \circ \mathcal{A}_{\mathbf{b}}^T \right) \Bigg/ \mathcal{T}_t(a_1 - a_2) \prec \max_{\mathbf{b}}^* \exp \left(\ell_t^{-1/2} \left(|a_1 - a_2|^{1/2} - |b_1 - b_2|^{1/2} \right) \right), \quad (7.11)$$

where $\max_{\mathbf{b}}^*$ satisfies the condition in (7.9). Under this condition, we have

$$|a_1 - a_2| - |b_1 - b_2| \leq \sum_i |a_i - b_i| \leq \frac{1}{2} (\log W)^{3/2} \ell_t.$$

Note that there is no absolute values on the left hand side. It is easy to prove that

$$\max_{\mathbf{b}}^* \left(|a_1 - a_2|^{1/2} - |b_1 - b_2|^{1/2} \right) \leq C \log^{3/4} W \cdot \ell_t^{1/2}. \quad (7.12)$$

Together with (7.11) and (7.8), we have proved Lemma 7.2. $\square$

Lemma 7.3 ($\mathcal{U}_{s,t}$ on fast decay tensor). *Let $\mathcal{A}$ be a tensor $\mathbb{Z}_L^n \rightarrow \mathbb{R}$, $n \geq 2$. We say $\mathcal{A}$ is (τ, D) decay at time s if for some fixed small $\tau > 0$ and large $D > 0$,*

$$\max_{i,j} \|a_i - a_j\| \geq \ell_s W^\tau \implies \mathcal{A}_a = O(W^{-D}), \quad \mathbf{a} = (a_1, a_2, \dots, a_n). \quad (7.13)$$

Then we have the following $\max \rightarrow \max$ norm for $\mathcal{U}_{s,t,\sigma}$ for $n \geq 2$:

$$|(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_a| \leq C_n W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_t}{\ell_s}\right) \cdot \left(\frac{\ell_s \eta_s}{\ell_t \eta_t}\right)^n + W^{-D+C_n}. \quad (7.14)$$

Suppose either of the following two assumptions holds: Case 1: For some $k : 1 \leq k \leq n$,

$$\sigma_k = \sigma_{k-1}, \quad \sigma = (\sigma_1, \dots, \sigma_n).$$

Case 2: $\mathcal{A}_b$ has the sum zero property in the sense that

$$\sum_{a_2, \dots, a_n} \mathcal{A}_a = 0, \quad \forall a_1. \quad (7.15)$$

Then we have the following stronger bound

$$|(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_a| \leq W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_s \eta_s}{\ell_t \eta_t}\right)^n + W^{-D+C_n}. \quad (7.16)$$

Lemma 7.3 can be understood as follows. The evolution kernel is approximately given by

$$(\eta_s/\eta_t)^n \otimes^n \omega_t$$

where ω_t is an L^1 -normalized convolution kernel of width ℓ_t . So operator norm of this kernel in L_∞ is $(\eta_s/\eta_t)^n$. The decay length of A is $\ell_s < \ell_t$. Hence we gain a factor ℓ_s/ℓ_t for each summation restricted by A . Since there are $n-1$ summations restricted by A , we gain $(\ell_s/\ell_t)^{n-1}$ and this explains (7.14). For the sum zero case, we gain an extra (ℓ_s/ℓ_t) as we can sum by parts once and the ratio of smoothness between ω_t and A is (ℓ_s/ℓ_t) . For the case that $\sigma_k = \sigma_{k-1}$, the Θ operator has become significantly smaller and we also gain an extra factor. The details will be given in the proof.

Proof of Lemma 7.3. We first prove (7.14). By definition of $\mathcal{U}_{s,t}$ in (5.17) and (5.18), we have

$$(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_a = \sum_{b_1, \dots, b_n} \prod_{i=1}^n \psi_i \cdot \mathcal{A}_b, \quad \mathbf{b} = (b_1, \dots, b_n), \quad (7.17)$$

$$\psi_i = \delta_{a_i b_i} + \Xi_i, \quad \Xi_i := -(s-t)\xi_i \cdot \left(S^{(B)} \cdot \Theta_{t\xi_i}^{(B)}\right)_{a_i b_i}, \quad \xi_i = m(\sigma_i)m(\sigma_{i+1}). \quad (7.18)$$

Recall the following identity for all $\mathbf{x}$ and $\mathbf{y}$:

$$\prod_i (x_i) - \prod_i (x_i - y_i) = - \sum_{\emptyset \neq A \subset [[1, n]]} \left(\prod_{j \in A^c} x_j \right) \left(\prod_{j \in A} (-y_j) \right).$$

Choosing $x_i = \psi_i$ and $y_i = \delta_{a_i b_i}$, we have $x_i - y_i = \Xi_i$ and

$$\prod_{i=1}^n \psi_i - \prod_{i=1}^n \Xi_i = \sum_{\emptyset \neq A \subset [[1, n]]} \left(\prod_{j \in A^c} \psi_j \right) (-1)^{|A|+1} \left(\prod_{j \in A} \delta_{a_j b_j} \right). \quad (7.19)$$

The terms with $|A^c| \leq 1$ (in particular all terms if $n = 2$) contain at most one factor ψ_j and are estimated directly: since $\Xi_j = O(\eta_s \ell_t^{-1} \eta_t^{-1})$ and $\Theta_t^{(B)}$ decays on the scale ℓ_t , we have $\sum_{b_j} |\psi_j| \leq C W^\tau \eta_s/\eta_t + W^{-D}$,

so these terms are bounded by $CW^\tau \|\mathcal{A}\|_{\max} \eta_s / \eta_t + W^{-D}$, which is at most the right side of (7.20). Assume that we have proved Lemma 7.3 for any $2 \leq k < n$. Then by the induction hypothesis,

$$\begin{aligned} \sum_{\mathbf{b}} \left(\prod_{i=1}^n \psi_i - \prod_{i=1}^n \Xi_i \right) \cdot \mathcal{A}_{\mathbf{b}} &\leq C_n W^{C_n \tau} \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_t}{\ell_s} \right) \cdot \left(\frac{\ell_s \eta_s}{\ell_t \eta_t} \right)^{n-1} + W^{-D+C_n} \\ &\leq C_n W^{C_n \tau} \|\mathcal{A}\|_{\max} \left(\frac{\ell_s \eta_s}{\ell_t \eta_t} \right)^n + W^{-D+C_n}, \end{aligned} \quad (7.20)$$

where we have used $\ell_t / \ell_s \leq (\ell_s \eta_s) / (\ell_t \eta_t)$.

For (7.14) we only need to bound

$$\sum_{\mathbf{b}} \left(\prod_{i=1}^n \Xi_i \right) \cdot \mathcal{A}_{\mathbf{b}} \leq C_n W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_t}{\ell_s} \right) \cdot \left(\frac{\ell_s \eta_s}{\ell_t \eta_t} \right)^n + W^{-D+C_n}. \quad (7.21)$$

By definition of $\Theta^{(B)}$, we have

$$\Xi_i = O(\eta_s \ell_t^{-1} \eta_t^{-1}).$$

By ℓ_s -decay property of $\mathcal{A}$ in (7.13),

$$\begin{aligned} \sum_{\mathbf{b}} \prod_{i=1}^n \Xi_i \cdot \mathcal{A}_{\mathbf{b}} &\leq CW^\tau \cdot \frac{\ell_s \eta_s}{\ell_t \eta_t} \max_{b_n} \left| \sum_{b_1 \dots b_{n-1}} \prod_{i=1}^{n-1} \Xi_i \cdot \mathcal{A}_{\mathbf{b}} \right| + W^{-D+C} \\ &\leq C_n W^{C_n \tau} \left(\frac{\ell_s \eta_s}{\ell_t \eta_t} \right)^{n-1} \max_{b_2, \dots, b_n} \left| \sum_{b_1} \Xi_1 \mathcal{A}_{\mathbf{b}} \right| + W^{-D+C_n}. \end{aligned} \quad (7.22)$$

By the decay of $\Theta_t^{(B)}$, we have

$$\left| \sum_{b_1} \Xi_1 \mathcal{A}_{\mathbf{b}} \right| \leq CW^\tau (\eta_s / \eta_t) \|\mathcal{A}\|_{\max} + W^{-D}. \quad (7.23)$$

Together with (7.22) and (7.20), we have proved (7.14).

Next we prove Case 1 of (7.16). Without loss of generality, we assume that $\sigma_1 = \sigma_2 = +$. In this case, $\xi_1 = m^2$, and thus $\sum_{b_1} |\Xi_1| = O(1)$. This implies that

$$\left| \sum_{b_1} \Xi_1 \mathcal{A}_{\mathbf{b}} \right| \leq C \|\mathcal{A}\|_{\max}.$$

Together with (7.22) and (7.20), we have proved (7.16) in this case.

We finally prove Case 2 of (7.16), i.e., $\mathcal{A}$ has the sum zero property (7.15). Since we have proved (7.16) for the case 1, we can now assume that $\sigma_i \neq \sigma_{i+1}$ for any $1 \leq i \leq n$. It implies $\xi_i = |m|^2 = 1$ (in (7.18)). For each Ξ_i with $1 \leq i \leq n$, we write it as $\Xi_i = \Xi_i^0 + \Xi_i^*$ where

$$\frac{\Xi_i^0}{-(s-t)\xi_i} := \left(S^{(B)} \cdot \Theta_{t\xi_i}^{(B)} \right)_{a_i b_1}, \quad \frac{\Xi_i^*}{-(s-t)\xi_i} := \left(\left(S^{(B)} \cdot \Theta_{t\xi_i}^{(B)} \right)_{a_i b_i} - \left(S^{(B)} \cdot \Theta_{t\xi_i}^{(B)} \right)_{a_i b_1} \right).$$

Note that the subscript of Ξ_i^0 is $a_i b_1$, but the one for Ξ_i is $a_i b_i$. Thus

$$\sum_{\mathbf{b}} \left(\prod_{i=1}^n \Xi_i \right) \cdot \mathcal{A}_{\mathbf{b}} = \sum_{\mathbf{b}} \left(\prod_{i=1}^n (\Xi_i^0 + \Xi_i^*) \right) \cdot \mathcal{A}_{\mathbf{b}}.$$

After expanding the $\prod_i$, the leading term disappears due to the sum zero property (and the fact that Ξ_i^0 is independent of $b_2, \dots, b_n$), namely,

$$\sum_{\mathbf{b}} \left(\prod_{i=1}^n \Xi_i^0 \right) \cdot \mathcal{A}_{\mathbf{b}} = 0.$$

For the other terms, we bound them by

$$|\Xi_i^0| \leq C \frac{\eta_s}{\ell_t \eta_t}, \quad |\Xi_i^*| \leq C \frac{\eta_s}{\ell_t \eta_t} \cdot \frac{|b_i - b_1|}{\ell_t}.$$

By the decay property of $\mathcal{A}$ (7.13), the main contribution to the last equation comes from $|b_i - b_1| \leq W^\tau \ell_s$. Therefore, we obtain an improved bound for (7.22), i.e.,

$$\sum_{\mathbf{b}} \prod_{i=1}^n \Xi_i \cdot \mathcal{A}_{\mathbf{b}} \leq C_n W^{C_n \tau} \left(\frac{\ell_s \eta_s}{\ell_t \eta_t} \right)^{n-1} \cdot \frac{\ell_s}{\ell_t} \cdot \max_{b_2, \dots, b_n} \left| \sum_{b_1} \Xi_1 \mathcal{A}_{\mathbf{b}} \right| + W^{-D+C_n}. \quad (7.24)$$

Together with (7.23), we have proved Case 2 of (7.16). This completes the proof of Lemma 7.3. $\square$

7.2 Auxiliary proofs for universality

In this subsection, we prove (2.27) and (2.26) for the process H_t defined in (2.19). The notation H_t used here is specific to the universality proof and differs from the process defined in (2.34) for the main argument. For the present process, $t = 0$ corresponds to the original band matrix. The cases $0 < t < t_U$ are handled similarly to the case $t = t_U$. We therefore present only the proof at $t = t_U$, namely the estimates for H_{t_U} . Define

$$\tilde{H} := H_{t_U} \quad (7.25)$$

By definition, $\tilde{H}$ has entry variances given by

$$\mathbb{E} |\tilde{H}_{xy}|^2 = \mathbb{E} |(H_{t_U})_{xy}|^2 = \tilde{S}_{xy} = (1 - \zeta_U) S_{xy} + \frac{\zeta_U}{N}, \quad \zeta_U \sim N^{-1+\tau_U}, \quad \zeta_U = 1 - e^{-t_U}$$

In our main proof, we use the flow

$$z_t^{(E)} = E + (1 - t)m^{(E)}, \quad 0 \leq t \leq 1.$$

Then for $z \in \mathbb{C}$, we choose t_0 and E such that

$$z = t_0^{-1/2} z_{t_0}^{(E)},$$

and study $G(z)$ via $t_0^{1/2} \cdot G_{t_0}^{(E)}$ as in (2.39). Here for fixed z , we keep the same stochastic flow for $z_t^{(E)}$, but the flow for $\tilde{H}_t$ is defined by $\tilde{H}_0 = 0$ and

$$d\tilde{H}_{t,ij} = \begin{cases} \sqrt{S_{ij}} \cdot d\mathcal{B}_{t,ij} & 0 \leq t \leq t_1 := (1 - \zeta_U)t_0, \\ N^{-1/2} \cdot d\mathcal{B}_{t,ij} & t_1 \leq t \leq t_0. \end{cases}$$

It is easy to check that for any fixed z satisfying $z = t_0^{-1/2} z_{t_0}^{(E)}$,

$$(\tilde{H} - z)^{-1} \stackrel{d}{\sim} t_0^{1/2} (\tilde{H}_{t_0} - z_{t_0}^{(E)})^{-1} \quad (7.26)$$

Define $\tilde{G}_t$ and $\tilde{\mathcal{L}}_t$ with $\tilde{H}_t$ as in Definitions 2.7 and 2.9. By definition for $t < t_1$,

$$\tilde{H}_t \stackrel{d}{\sim} H_t, \quad \tilde{\mathcal{L}}_t \stackrel{d}{\sim} \mathcal{L}_t, \quad t \leq t_1.$$

Here H_t and $\mathcal{L}_t$ are defined in (2.34) and (2.41) respectively. We aim to estimate $\tilde{\mathcal{L}}_t$ with $\tilde{\mathcal{K}}_t$ for $t \in [t_1, t_0]$. Recall $\tilde{\mathcal{K}}$ is the solution to the primitive equation in Definition 2.12. For $t \in [t_1, t_0]$ the flow equation becomes

$$\frac{d}{dt} \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} = W \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} \left(\mathcal{G}_{k,l}^{(a),L} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right) \left(S_{GUE}^{(B)} \right)_{ab} \left(\mathcal{G}_{k,l}^{(b),R} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right), \quad t \in [t_1, t_0]$$

with

$$\left(S_{GUE}^{(B)}\right)_{ab} = 1/L, \quad \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} = \mathcal{K}_{t,\sigma,\mathbf{a}}, \quad t \in [0, t_1].$$

To prove (2.27) and (2.26), we only need to focus on the following case: for small enough τ_U with

$$z = E + \eta i, \quad N^{-1+2\tau_U} \leq \eta \leq N^{-1+\epsilon/3},$$

and $t \in [t_1, t_0]$, we have

$$\max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{L}}_{t,\sigma,\mathbf{a}} \right| \prec (N\eta_t)^{-n+1}, \quad \sigma \in \{+, -\}^n \quad (7.27)$$

$$\max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{L}}_{t,\sigma,\mathbf{a}} - \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right| \prec (N\eta_t)^{-n}, \quad \sigma \in \{+, -\}^n \quad (7.28)$$

$$\max_{\mathbf{a}} \left| \mathbb{E} \tilde{\mathcal{L}}_{t,\sigma,\mathbf{a}} - \mathbb{E} \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right| \prec (N\eta_t)^{-3}, \quad \sigma \in \{+, -\}^2. \quad (7.29)$$

In this setting ($\text{Im } z = \eta$)

$$1 - t_0 \sim 1 - t_1 \sim \eta, \quad t_0 - t_1 \sim N^{-1+\tau_U}, \quad N^{-1+2\tau_U} \leq \eta \leq N^{-1+\epsilon/3}. \quad (7.30)$$

Notice that $t_0 - t_1 \ll \eta$. By Lemma 2.17 and 2.18, for $t \in [0, t_1]$,

$$\begin{aligned} \max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{L}}_{t,\sigma,\mathbf{a}} - \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right| &\prec (W\ell_t\eta_t)^{-n}, \quad \eta_t := \text{Im } z_t, \\ \max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right| &\prec (W\ell_t\eta_t)^{-n+1}, \quad \max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{L}}_{t,\sigma,\mathbf{a}} \right| \prec (W\ell_t\eta_t)^{-n+1}. \end{aligned} \quad (7.31)$$

With assumption (7.30), we have $\ell_{t_1} = L$. Then for $t = t_1$,

$$\max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{L}}_{t_1,\sigma,\mathbf{a}} - \tilde{\mathcal{K}}_{t_1,\sigma,\mathbf{a}} \right| \prec (N\eta_{t_1})^{-n}, \quad \max_{\sigma,\mathbf{a}} \left(\left| \tilde{\mathcal{K}}_{t_1,\sigma,\mathbf{a}} \right| + \left| \tilde{\mathcal{L}}_{t_1,\sigma,\mathbf{a}} \right| \right) \prec (N\eta_{t_1})^{-n+1}. \quad (7.32)$$

For $t_1 \leq t \leq t_0$, we will have a new loop hierarchy and primitive equation. The new versions of these equations are the same as the standard versions in (2.45) and (2.48) except that $S^{(B)}$ and S are replaced by $S_{GUE}^{(B)}$ and S_{GUE} respectively, i.e.,

$$S^{(B)} \rightarrow S_{GUE}^{(B)}, \quad \left(S_{GUE}^{(B)}\right)_{ab} = 1/L, \quad S \rightarrow S_{GUE}, \quad (S_{GUE})_{xy} = 1/N.$$

Thus we can bound $\tilde{\mathcal{K}}$ by

$$\left| \frac{d}{dt} \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right| \leq C_n N \max_{ab} \max_{kl} \left| \mathcal{G}_{k,l}^{(a),L} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right| \cdot \left| \mathcal{G}_{k,l}^{(b),R} \circ \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right|, \quad t \in [t_1, t_0], \quad (7.33)$$

$$\leq C_n \cdot N \sum_{2 \leq k \leq n} \left(\max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}}^{(k)} \right| \right) \cdot \left(\max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}}^{(n-k+2)} \right| \right). \quad (7.34)$$

Then for $t \in [t_1, t_0]$,

$$\left| \tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \right| \leq \left| \tilde{\mathcal{K}}_{t_1,\sigma,\mathbf{a}} \right| + C_n N \cdot (t - t_1) \cdot \sum_{2 \leq k \leq n} \max_{u \in [t_1, t]} \left(\max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{K}}_{u,\sigma,\mathbf{a}}^{(k)} \right| \right) \cdot \left(\max_{\sigma,\mathbf{a}} \left| \tilde{\mathcal{K}}_{u,\sigma,\mathbf{a}}^{(n-k+2)} \right| \right) \quad (7.35)$$

By (7.30), $t - t_1 \leq t_0 - t_1$ and

$$N(t - t_1) \leq N\eta_t \cdot N^{-\tau_U}, \quad t \in [t_1, t_0].$$

By a continuity argument and the initial estimate (7.32), we have

$$\tilde{\mathcal{K}}_{t,\sigma,\mathbf{a}} \prec (N\eta_t)^{-n+1}, \quad t \in [t_1, t_0]. \quad (7.36)$$

To bound $\mathcal{L}$, we apply the main flow estimates from Lemmas 5.3 and 5.5 with $s = t_1$ and $t \in [t_1, t_0]$. Together with Lemma 7.1 which implies that

$$\|\mathcal{U}_{t_1, t, \sigma}\|_{\max \rightarrow \max} \prec (\eta_{t_1}/\eta_t)^n \prec 1,$$

we have

$$\begin{aligned} (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{t, \sigma, \mathbf{a}} &\prec \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{t_1, \sigma, \mathbf{a}} \right| + \int_{t_1}^t \left(\mathcal{U}_{u, t, \sigma} \circ \tilde{\mathcal{E}}_{u, \sigma}^{(M)} \right)_{\mathbf{a}} \\ &\quad + \int_{t_1}^t \max_{\sigma, \mathbf{a}} \left(\sum_{l_{\mathcal{K}} > 2} \left| \left[\tilde{\mathcal{K}} \sim (\tilde{\mathcal{L}} - \tilde{\mathcal{K}}) \right]_{u, \sigma, \mathbf{a}}^{l_{\mathcal{K}}} \right| + \left| \tilde{\mathcal{E}}_{u, \sigma, \mathbf{a}}^{((\mathcal{L} - \mathcal{K}) \times (\mathcal{L} - \mathcal{K}))} \right| + \left| \tilde{\mathcal{E}}_{u, \sigma, \mathbf{a}}^{(\tilde{G})} \right| \right) du \end{aligned} \quad (7.37)$$

and

$$\mathbb{E} \left| \int_{t_1}^{\tau} \left(\mathcal{U}_{u, t, \sigma} \circ \tilde{\mathcal{E}}_{u, \sigma}^{(M)} \right)_{\mathbf{a}} \right|^{2p} \leq C_{n, p} \mathbb{E} \left(\int_{t_1}^{\tau} \max_{\sigma, \mathbf{a}} \left(\tilde{\mathcal{E}} \otimes \tilde{\mathcal{E}} \right)_{u, \sigma, \mathbf{a}, \mathbf{a}} du \right)^p. \quad (7.38)$$

We have the following estimates

- Similar to (5.13) and (5.14), with $S^{(B)}$ replaced by $S_{GUE}^{(B)}$, we have

$$\sum_{l_{\mathcal{K}} > 2} \left| \left[\tilde{\mathcal{K}} \sim (\tilde{\mathcal{L}} - \tilde{\mathcal{K}}) \right]_{u, \sigma, \mathbf{a}}^{l_{\mathcal{K}}} \right| \prec N \cdot \sum_{2 \leq k \leq n} (N\eta_u)^{-k+1} \cdot \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{u, \sigma, \mathbf{a}}^{(n-k+2)} \right| \quad (7.39)$$

and

$$\tilde{\mathcal{E}}_{u, \sigma, \mathbf{a}}^{((\mathcal{L} - \mathcal{K}) \times (\mathcal{L} - \mathcal{K}))} \prec N \cdot \sum_{2 \leq k \leq n} \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{u, \sigma, \mathbf{a}}^{(k)} \right| \cdot \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{u, \sigma, \mathbf{a}}^{(n-k+2)} \right| \quad (7.40)$$

where superscript (k) represents the rank, i.e., $\mathcal{L}^{(n)}$ is an n -loop.

- For $\tilde{\mathcal{E}}^{(\tilde{G})}$, similarly one can prove (4.5) for $\tilde{G}_t$. Then

$$\tilde{\mathcal{E}}_{u, \sigma, \mathbf{a}}^{(\tilde{G})} \prec N \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(2)} \right| \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(n+1)} \right| \quad (7.41)$$

By the Cauchy–Schwarz inequality, we can bound an $(n+1)$ -loop by a product of a 2-loop and a $2n$ -loop. On the other hand, with (5.117), a $2n$ -loop can be bounded by the square of an n -loop. Inserting all these bounds into the previous inequality, we have

$$\tilde{\mathcal{E}}_{u, \sigma, \mathbf{a}}^{(\tilde{G})} \prec N \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(2)} \right|^{3/2} \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(n)} \right| \quad (7.42)$$

- For $\tilde{\mathcal{E}} \otimes \tilde{\mathcal{E}}$ term, recall the loops in Definition 5.4. The $S^{(B)}$ in $(\tilde{\mathcal{E}} \otimes \tilde{\mathcal{E}})^{(k)}$ is replaced by $S_{GUE}^{(B)}$ in $(\tilde{\mathcal{E}} \otimes \tilde{\mathcal{E}})^{(k)}$. Since $S_{GUE}^{(B)}$ is a constant matrix, we can use Ward's identity to sum up the b and b' in the definition of $(\tilde{\mathcal{E}} \otimes \tilde{\mathcal{E}})^{(k)}$ and obtain that

$$\left(\tilde{\mathcal{E}} \otimes \tilde{\mathcal{E}} \right)_{u, \sigma, \mathbf{a}, \mathbf{a}} \prec N^{-1} \cdot \eta_u^{-2} \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(2n)} \right| \quad (7.43)$$

Then, using (5.117), we can bound the $2n$ -loop with n -loops.

$$\left(\tilde{\mathcal{E}} \otimes \tilde{\mathcal{E}} \right)_{u, \sigma, \mathbf{a}, \mathbf{a}} \prec N^{-1} \cdot \eta_u^{-2} \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(n)} \right|^2 \quad (7.44)$$

Combining (7.32), (7.37), (7.38), (7.39), (7.40), (7.42) and (7.44) we obtain that

$$(\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{t, \sigma, \mathbf{a}} \prec (N|t - t_1|) \cdot \sup_{u \in [t_1, t]} \sum_{2 \leq k \leq n} \left((N\eta_u)^{-k+1} + \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{u, \sigma, \mathbf{a}}^{(k)} \right| \right) \cdot \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{u, \sigma, \mathbf{a}}^{(n-k+2)} \right|$$

$$\begin{aligned}
& + (N\eta_t)^{-n} + (N|t - t_1|) \cdot \sup_{u \in [t_1, t]} \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(2)} \right|^{3/2} \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(n)} \right| \\
& + |t - t_1|^{1/2} \cdot \sup_{u \in [t_1, t]} (N^{-1}\eta_u^{-2})^{1/2} \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(n)} \right|
\end{aligned} \tag{7.45}$$

By writing $\tilde{\mathcal{L}} = (\tilde{\mathcal{L}} - \tilde{\mathcal{K}}) + \tilde{\mathcal{K}}$ and using the bound (7.36) on $\tilde{\mathcal{K}}$, the last inequality becomes an inequality on $\tilde{\mathcal{L}}$. By assumption (7.30), we have

$$\eta_u \sim 1 - u \geq N^{-1+2\tau_U} \gg N^{-1+\tau_U} \sim |t_0 - t_1| \geq |t - t_1|.$$

With a continuity argument and an induction on $n \geq 2$, this inequality thus implies the estimate (7.27) on $\tilde{\mathcal{L}}$.

We emphasize one point concerning the form of the estimate in (7.45). For the purpose of carrying out the induction, the right-hand side of (7.45) was written so that all loop lengths appearing there are no larger than the loop length on the left-hand side. Therefore, (7.45) is not the sharpest possible estimate for $\mathcal{L} - \mathcal{K}$. After the induction for $\mathcal{L}$ has been completed, however, we already have estimates for $\mathcal{L}$ at all relevant loop lengths. We may then return to (7.45) and replace its right-hand side by the optimal bound obtained from these estimates. This gives the improved form below, where the loop lengths appearing on the right-hand side may be larger than the loop length on the left-hand side.

Combining (7.32), (7.37), (7.38), (7.39), (7.40), (7.41) and (7.43), we obtain that

$$\begin{aligned}
(\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{t, \sigma, \mathbf{a}} & \prec (N|t - t_1|) \cdot \sup_{u \in [t_1, t]} \sum_{2 \leq k \leq n} \left((N\eta_u)^{-k+1} + \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{u, \sigma, \mathbf{a}}^{(k)} \right| \right) \cdot \max_{\sigma, \mathbf{a}} \left| (\tilde{\mathcal{L}} - \tilde{\mathcal{K}})_{u, \sigma, \mathbf{a}}^{(n-k+2)} \right| \\
& + (N\eta_t)^{-n} + (N|t - t_1|) \cdot \sup_{u \in [t_1, t]} \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(2)} \right| \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(n+1)} \right| \\
& + |t - t_1|^{1/2} \cdot \sup_{u \in [t_1, t]} (N^{-1}\eta_u^{-2})^{1/2} \cdot \max_{\sigma, \mathbf{a}} \left| \tilde{\mathcal{L}}_{u, \sigma, \mathbf{a}}^{(2n)} \right|^{1/2}.
\end{aligned} \tag{7.46}$$

Inserting the $\tilde{\mathcal{L}}$ bound (7.27) into the above $\tilde{\mathcal{L}} - \tilde{\mathcal{K}}$ estimate, we have proved that (7.28) holds. Finally (7.29) can be proved by following the proof of (2.80) in subsection 5.8. Since

$$\ell_u \sim L, \quad \eta_u \sim \eta_{t_0}, \quad u \in [t_1, t_0]$$

in the current setting we can prove (7.29) similarly to (2.80) in subsection 5.8 with very minor changes.

We now prove (2.26) and (2.27) for $\tilde{H}$ (defined in (7.25)) by using (7.27), (7.28), (7.29) and the identity (7.26). First, (2.26) follows from (7.26) and (7.28), together with Lemma 4.1 applied to $\tilde{G}_u$, $u \in [t_1, t_0]$, whose assumption (4.4) is carried along $[t_1, t_0]$ by a standard continuity argument. To prove (2.27), by (7.29) we have the following estimates similar to (2.8) and (2.9), namely, with the notations

$$\tilde{G} = (\tilde{H} - z)^{-1}, \quad \text{Im } z = \eta = N^{-1+\epsilon/3},$$

and

$$\tilde{S}_{xy}^{(B)} = (1 - \zeta_U) S_{xy}^{(B)} + \frac{\zeta_U}{L}, \quad \zeta_U \sim N^{-1+\tau_U}, \quad \zeta_U = 1 - e^{-t_U},$$

we have

$$\begin{aligned}
\max_{a,b} \left| \mathbb{E} \text{Tr} \tilde{G} E_a \tilde{G}^\dagger E_b - W^{-1} \left(\frac{|m|^2}{1 - |m|^2 \tilde{S}^{(B)}} \right)_{ab} \right| & \leq (N\eta)^{-3} \cdot W^\tau, \\
\max_{a,b} \left| \mathbb{E} \text{Tr} \tilde{G} E_a \tilde{G} E_b - W^{-1} \left(\frac{m^2}{1 - m^2 \tilde{S}^{(B)}} \right)_{ab} \right| & \leq (N\eta)^{-3} \cdot W^\tau.
\end{aligned} \tag{7.47}$$

We have thus proved the conclusion of Theorem 2.4 except that $S^{(B)}$ is replaced by $\tilde{S}^{(B)}$. We claim that the proof for (2.12) from (2.14) to (2.17) can be used to derive the (2.27) for $\tilde{H}$ with only very minor

modifications. By definition of $\tilde{S}^{(B)}$, we can remove the zero mode in $\tilde{S}^{(B)}$ and write

$$\left| \left(\frac{1}{1 - \xi \cdot \tilde{S}^{(B)}} \right)_{ab} - \left(\frac{1}{1 - \xi \cdot \tilde{S}^{(B)}} \right)_{a'b'} \right| = \left| \left(\frac{1}{1 - \xi \cdot (1 - \zeta_U) S^{(B)}} \right)_{ab} - \left(\frac{1}{1 - \xi \cdot (1 - \zeta_U) S^{(B)}} \right)_{a'b'} \right|.$$

For both $\xi = m^2$ and $\xi = |m|^2$, with $m = m(z)$, we have $|1 - \xi| \geq 1 - |m|^2 \geq c\eta$ by (2.16), while $\zeta_U \leq t_U = N^{-1+\tau_U} \ll \eta$ for $\tau_U < \mathfrak{c}/3$; hence

$$|1 - \xi| \sim |1 - \xi(1 - \zeta_U)|.$$

We can now follow the argument from (2.14) to (2.17) to prove (2.27) for $\tilde{H}$ (7.29). This completes the proof of (2.27).

A G -chains

In addition to loops of G -entries defined in Definition 2.9, we can also bound certain *chains* of G entries. Although the following results were not used in the proof of our main theorem, they might be useful in the future applications of the methods discussed in this paper.

Definition A.1 (G -chains). *Let*

$$\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \dots, \sigma_n), \quad \mathbf{a} = (a_1, a_2, \dots, a_{n-1}), \quad \sigma_k \in \{+, -\}, \quad a_k \in \mathbb{Z}_L.$$

Following the structure in (5.114), we define a G -chain as

$$\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}} = G_t(\sigma_1) E_{a_1} G_t(\sigma_2) E_{a_2} \cdots E_{a_{n-1}} G_t(\sigma_n). \quad (\text{A.1})$$

By definition, $\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}}$ is an $N \times N$ matrix. We classify its entries as

$$(\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}})_{ij} \sim \begin{cases} \text{off-diagonal terms of } \mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}}, & \text{if } i \neq j, \\ \text{diagonal terms of } \mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}}, & \text{if } i = j. \end{cases}$$

Before presenting the lemmas for estimating G -chains, we provide a heuristic argument for the following bounds on G -chains:

$$(\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}})_{ii} \prec (W \ell_t \eta_t)^{-n+1}, \quad (\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}})_{ij} \prec (W \ell_t \eta_t)^{-n+1/2}. \quad (\text{A.2})$$

A G -chain can be converted into a G -loop by multiplication with an E_a operator followed by taking the trace:

$$\mathcal{L}_{t, \boldsymbol{\sigma}, \mathbf{a}'} = \langle \mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}} E_a \rangle, \quad \mathbf{a}' = (a_1, a_2, \dots, a_{n-1}, a).$$

Assuming that $(\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}})_{ii}$ is of comparable size for all $i \in \mathcal{I}_a$, we obtain from (2.77)

$$(\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}})_{ii} \sim \mathcal{L}_{t, \boldsymbol{\sigma}, \mathbf{a}'} \sim (W \ell_t \eta_t)^{-n+1}.$$

Similarly, a different G -loop can be constructed by multiplying a G -chain with two operators E_a and E_b and taking the trace:

$$\mathcal{L}_{t, \boldsymbol{\sigma}'', \mathbf{a}''} = \langle E_a \cdot \mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}} \cdot E_b \cdot \mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}}^\dagger \rangle,$$

where

$$\boldsymbol{\sigma}'' = (\sigma_1, \sigma_2, \dots, \sigma_n, \bar{\sigma}_n, \bar{\sigma}_{n-1}, \dots, \bar{\sigma}_1), \quad \mathbf{a}'' = (a_1, a_2, \dots, a_{n-1}, b, a_{n-1}, \dots, a_1, a).$$

Assuming that $(\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}})_{ij}$ has a typical size across all $i \in \mathcal{I}_a, j \in \mathcal{I}_b$, we deduce from (2.77) that

$$(\mathcal{C}_{t, \boldsymbol{\sigma}, \mathbf{a}})_{ij} \sim (\mathcal{L}_{t, \boldsymbol{\sigma}'', \mathbf{a}''})^{1/2} \sim (W \ell_t \eta_t)^{-n+1/2}.$$

Lemma A.2 (n -chain estimate). *Under the assumptions of Lemma 2.18, for any fixed $n \in \mathbb{N}$ and $\sigma \in \{+, -\}^n$, we have*

$$\max_{\mathbf{a}} \max_i |(\mathcal{C}_{t,\sigma,\mathbf{a}})_{ii}| \prec (W\ell_t\eta_t)^{-n+1}, \quad (\text{A.3})$$

$$\max_{\mathbf{a}} \max_{i \neq j} |(\mathcal{C}_{t,\sigma,\mathbf{a}})_{ij}| \prec (W\ell_t\eta_t)^{-n+1/2}. \quad (\text{A.4})$$

The proof relies on a standard decomposition widely used for Wigner matrices (see, e.g., [39]) and random band matrices (e.g., [34]).

Proof of Lemma A.2. We begin by introducing the following shorthand:

$$\Xi_n^{(\text{d})} := \Xi_{t,n}^{(\mathcal{C},\text{diag})} := \max_{\sigma,\mathbf{a}} \max_i |(\mathcal{C}_{t,\sigma,\mathbf{a}})_{ii}| \cdot (W\ell_t\eta_t)^{n-1} \cdot \mathbf{1}(\sigma \in \{+, -\}^n), \quad (\text{A.5})$$

$$\Xi_n^{(\text{o})} := \Xi_{t,n}^{(\mathcal{C},\text{off})} := \max_{\sigma,\mathbf{a}} \max_{i \neq j} |(\mathcal{C}_{t,\sigma,\mathbf{a}})_{ij}| \cdot (W\ell_t\eta_t)^{n-1/2} \cdot \mathbf{1}(\sigma \in \{+, -\}^n). \quad (\text{A.6})$$

For the base case $n = 1$, Theorem 2.3 implies

$$\Xi_1^{(\text{d})} + \Xi_1^{(\text{o})} \prec 1.$$

Applying the Cauchy–Schwarz inequality to the diagonal part of a 2-chain yields

$$\Xi_2^{(\text{d})} \prec \left(\Xi_1^{(\text{d})}\right)^2 + \left(\Xi_1^{(\text{o})}\right)^2 \prec 1.$$

We prove by induction that, for each $n \geq 2$, the following implication holds:

$$\left(\max_{k \leq n-1} \Xi_k^{(\text{o})} + \max_{k \leq 2n-2} \Xi_k^{(\text{d})} \right) \prec 1 \implies \left(\max_{k \leq n} \Xi_k^{(\text{o})} + \max_{k \leq 2n} \Xi_k^{(\text{d})} \right) \prec 1. \quad (\text{A.7})$$

This inductive bound immediately implies the estimates (A.3) and (A.4). To proceed, define the n -chain

$$\mathcal{C}_n := G_1 E_{a_1} G_2 \cdots G_{n-1} E_{a_{n-1}} G_n. \quad (\text{A.8})$$

Let $\mathcal{C}_n^{(i)}$ denote the n -chain where all G_k with $k \geq 2$ are replaced by $G_k^{(i)}$ (defined in (4.6)):

$$\mathcal{C}_n^{(i)} := G_1 E_{a_1} G_2^{(i)} \cdots G_{n-1}^{(i)} E_{a_{n-1}} G_n^{(i)}. \quad (\text{A.9})$$

Similarly, let $\mathcal{C}_n^{(ii)}$ be the n -chain with all G_k replaced by $G_k^{(i)}$, including the first:

$$\mathcal{C}_n^{(ii)} := G_1^{(i)} E_{a_1} G_2^{(i)} \cdots G_{n-1}^{(i)} E_{a_{n-1}} G_n^{(i)}. \quad (\text{A.10})$$

Under the assumption of (A.7), we claim the following perturbation estimates hold for $n \geq 2$:

$$i \neq j, \quad (\mathcal{C}_n)_{ij} - (\mathcal{C}_n^{(i)})_{ij} \prec (W\ell\eta)^{-n+1/2}, \quad (\text{A.11})$$

and

$$\left\langle \mathcal{C}_n^{(ii)} E_a \mathcal{C}_n^{(ii)\dagger} E_b \right\rangle - \langle \mathcal{C}_n E_a \mathcal{C}_n^\dagger E_b \rangle \prec \left(\Xi_{2n}^{(\text{d})} + 1 \right) \cdot (W\ell\eta)^{-2n+1/2}. \quad (\text{A.12})$$

Moreover, if in addition $\Xi_{2n-1}^{(\text{d})} + \Xi_{2n}^{(\text{d})} \prec 1$, then we also have

$$i \neq j, \quad \left(\mathcal{C}_n^{(ii)} E_a \mathcal{C}_n^{(ii)\dagger} \right)_{jj} - (\mathcal{C}_n E_a \mathcal{C}_n^\dagger)_{jj} \prec \left(\left(\Xi_n^{(\text{o})} \right)^2 + 1 \right) \cdot (W\ell\eta)^{-2n+1/2}. \quad (\text{A.13})$$

We postpone the proofs of these perturbative estimates to the end of this section.

We now prove

$$\Xi_{2n-1}^{(\text{d})} + \Xi_{2n}^{(\text{d})} \prec 1, \quad (\text{A.14})$$

using (A.11) and (A.12). By the Cauchy–Schwarz inequality and the assumption in (A.7), it suffices to control the case $2n$. Without loss of generality, we may assume that the $2n$ -chain $\mathcal{C}$ is symmetric and of the form

$$(\mathcal{C}_n E_a \mathcal{C}_n^\dagger)_{ii}.$$

We decompose this expression as

$$(\mathcal{C}_n E_a \mathcal{C}_n^\dagger)_{ii} = W^{-1} \mathbf{1}(i \in \mathcal{I}_a) \mathcal{C}_{n;ii} \cdot \mathcal{C}_{n;ii}^\dagger + \sum_{j \neq i} \mathcal{C}_{n;ij} \cdot E_a(j) \cdot \mathcal{C}_{n;j i}^\dagger,$$

and similarly for $\mathcal{C}_n^{(i)} E_a \left(\mathcal{C}_n^{(i)} \right)^\dagger$. Then,

$$\begin{aligned} (\mathcal{C}_n E_a \mathcal{C}_n^\dagger)_{ii} - \left(\mathcal{C}_n^{(i)} E_a \left(\mathcal{C}_n^{(i)} \right)^\dagger \right)_{ii} &\prec (W\ell\eta)^{-2n+1} \left(\Xi_n^{(d)} \right)^2 + \max_{j \neq i} \left| (\mathcal{C}_n)_{ij} - \left(\mathcal{C}_n^{(i)} \right)_{ij} \right|^2 \\ &\quad + \left(\sum_{j \neq i} |\mathcal{C}_{n;ij}|^2 E_a(j) \right)^{1/2} \cdot \max_{j \neq i} \left| (\mathcal{C}_n)_{ij} - \left(\mathcal{C}_n^{(i)} \right)_{ij} \right|. \end{aligned}$$

Applying (A.11) and the assumption in (A.7), we obtain

$$(\mathcal{C}_n E_a \mathcal{C}_n^\dagger)_{ii} - \left(\mathcal{C}_n^{(i)} E_a \left(\mathcal{C}_n^{(i)} \right)^\dagger \right)_{ii} \prec (W\ell\eta)^{-2n+1} \cdot \left(\Xi_{2n}^{(d)} + 1 \right)^{1/2}. \quad (\text{A.15})$$

To conclude (A.14), it remains to estimate $\mathcal{C}_n^{(i)} E_a \left(\mathcal{C}_n^{(i)} \right)^\dagger$. Recall the definition of $\mathcal{C}_n^{(ii)}$ in (A.10). Using (4.8), we write:

$$\left(\mathcal{C}_n^{(i)} E_a \left(\mathcal{C}_n^{(i)} \right)^\dagger \right)_{ii} = |G_{ii}|^2 \cdot \left(H \cdot \mathcal{C}_n^{(ii)} E_a \left(\mathcal{C}_n^{(ii)} \right)^\dagger \cdot H \right)_{ii}.$$

Since $\mathcal{C}_n^{(ii)}$ is independent of $\{H_{ik}\}_{k=1}^N$, we apply Lemma 3.3 from [39] to obtain

$$\left(\mathcal{C}_n^{(i)} E_a \left(\mathcal{C}_n^{(i)} \right)^\dagger \right)_{ii} = |G_{ii}|^2 \sum_k S_{ik} \left(\mathcal{C}_n^{(ii)} E_a \left(\mathcal{C}_n^{(ii)} \right)^\dagger \right)_{kk} + O \left(\sum_{kl} S_{ik} \left| \left(\mathcal{C}_n^{(ii)} E_a \left(\mathcal{C}_n^{(ii)} \right)^\dagger \right)_{kl} \right|^2 S_{li} \right)^{1/2}. \quad (\text{A.16})$$

The first term on the right can be bounded as $2n$ -chain loops with $G^{(i)}$ entries, while the second can be interpreted as $4n$ -chain loops with $G^{(i)}$ entries. Similar to (5.118), we can bound $4n - G^{(i)}$ loop with two $2n - G^{(i)}$ loops as follows

$$\Xi_{4n}^{\mathcal{L},(i)} \prec \left(\Xi_{2n}^{\mathcal{L},(i)} \right)^2 \cdot (W\ell\eta).$$

Using (A.12), we have

$$\Xi_{2n}^{\mathcal{L},(i)} \prec \left(\Xi_{2n}^{(d)} + (W\ell\eta)^{1/2} \right) \cdot (W\ell\eta)^{-2n+1/2}.$$

Hence,

$$\left(\mathcal{C}_n^{(i)} E_a \left(\mathcal{C}_n^{(i)} \right)^\dagger \right)_{ii} \prec \left(\Xi_{2n}^{(d)} + (W\ell\eta)^{1/2} \right) \cdot (W\ell\eta)^{-2n+1/2}.$$

Combining this with (A.15), we conclude

$$(\mathcal{C}_n E_a \mathcal{C}_n^\dagger)_{ii} \prec (W\ell\eta)^{-2n+1} \cdot \left(\left(\Xi_{2n}^{(d)} \right)^{1/2} + \Xi_{2n}^{(d)} \cdot (W\ell\eta)^{-1/2} + 1 \right),$$

which establishes (A.14).

We now prove

$$\Xi_n^{(o)} \prec 1, \quad (\text{A.17})$$

using (A.11) and (A.13). By (4.8), we express $(\mathcal{C}_n^{(i)})_{ij}$ in terms of $\mathcal{C}_n^{(ii)}$ as

$$(\mathcal{C}_n^{(i)})_{ij} = -(G_1)_{ii} \cdot (H \cdot \mathcal{C}_n^{(ii)})_{ij}. \quad (\text{A.18})$$

Since $\mathcal{C}_n^{(ii)}$ is independent of $\{H_{ik}\}_{k=1}^N$, we again apply Lemma 3.3 of [39]:

$$(H \cdot \mathcal{C}_n^{(ii)})_{ij} \prec \left(\sum_k S_{ik} \left| (\mathcal{C}_n^{(ii)})_{kj} \right|^2 \right)^{1/2} \prec \sum_a \mathbf{1}(|a - [i]| \leq 1) \cdot (\mathcal{C}_n^{(ii)\dagger} E_a \mathcal{C}_n^{(ii)})_{jj}^{1/2}. \quad (\text{A.19})$$

Combining this with (A.18), we find that for $i \neq j$,

$$(\mathcal{C}_n^{(i)})_{ij} \prec \max_a (\mathcal{C}_n^{(ii)\dagger} E_a \mathcal{C}_n^{(ii)})_{jj}^{1/2}. \quad (\text{A.20})$$

Using this estimate along with (A.13) and (A.11), we obtain

$$(\mathcal{C}_n)_{ij} \prec (W\ell\eta)^{-n+1/2} \left(1 + \Xi_n^{(o)} \cdot (W\ell\eta)^{-1/4} \right),$$

which yields the desired bound (A.17).

Proof of (A.11). Using (4.9) to express $G^{(i)}$ in (A.9) in terms of G , and bounding $1/G_{ii} = O(1)$ via Theorem 2.3, we can estimate the difference $(\mathcal{C}_n)_{ij} - (\mathcal{C}_n^{(i)})_{ij}$ in terms of products of diagonal and off-diagonal chain entries. For example, for $n = 2$, we have:

$$(\mathcal{C}_2^{(i)})_{ij} - (\mathcal{C}_2)_{ij} = -\frac{(G_1 E_{a_1} G_2)_{ii} \cdot (G_2)_{ij}}{(G_2)_{ii}} = O\left(\Xi_2^{(d)} \cdot \Xi_1^{(o)} \cdot (W\ell\eta)^{-3/2}\right). \quad (\text{A.21})$$

For $n = 3$, we similarly write:

$$(\mathcal{C}_3^{(i)})_{ij} - (\mathcal{C}_3)_{ij} = -\frac{(G_1 E_{a_1} G_2)_{ii} \cdot (G_2 E_{a_2} G_3)_{ij}}{(G_2)_{ii}} - \frac{(G_1 E_{a_1} G_2 E_{a_2} G_3)_{ii} \cdot (G_3)_{ij}}{(G_3)_{ii}} \quad (\text{A.22})$$

$$+ \frac{(G_1 E_{a_1} G_2)_{ii} \cdot (G_2 E_{a_2} G_3)_{ii} \cdot (G_3)_{ij}}{(G_2)_{ii} (G_3)_{ii}} \\ = O\left(\Xi_2^{(d)} \Xi_2^{(o)} + \Xi_3^{(d)} \Xi_1^{(o)} + \Xi_2^{(d)} \Xi_2^{(d)} \Xi_1^{(o)}\right) \cdot (W\ell\eta)^{-5/2}. \quad (\text{A.23})$$

In general, we obtain the bound:

$$(\mathcal{C}_n)_{ij} - (\mathcal{C}_n^{(i)})_{ij} \prec \sum_{k \geq 1} \sum_{\{n_i\}_{i=1}^k} \left(\prod_{i=1}^k \Xi_{n_i}^{(d)} \right) \Xi_\ell^{(o)} \cdot (W\ell\eta)^{-n+1/2} \cdot \mathbf{1}\left(\ell + \sum_{i=1}^k n_i = n + k\right), \quad (\text{A.24})$$

where $2 \leq n_i \leq n \leq 2n - 2$, and $1 \leq \ell \leq n - 1$. Applying the induction hypothesis (A.7) to all shorter chains involved, we get:

$$\Xi_{n_i}^{(d)} + \Xi_\ell^{(o)} \prec 1,$$

which implies (A.11).

Proof of (A.12). Observe that $(\mathcal{C}_n^{(ii)} E_a \mathcal{C}_n^{(ii)\dagger})_{jj}$ is a diagonal entry of a $2n$ - $G^{(i)}$ chain, while $(\mathcal{C}_n E_a \mathcal{C}_n^\dagger)_{jj}$ is the same object but with $G^{(i)}$ replaced by G . Applying (4.9) to express all $G^{(i)}$ in terms of G , and again bounding $1/G_{ii} = O(1)$, we estimate the difference:

$$\left\langle \mathcal{C}_n^{(ii)} E_a \mathcal{C}_n^{(ii)\dagger} E_b \right\rangle - \left\langle \mathcal{C}_n E_a \mathcal{C}_n^\dagger E_b \right\rangle.$$

This difference involves products of diagonal chains, and similar to (A.24), we obtain:

$$\left\langle \mathcal{C}_n^{(ii)} E_a \mathcal{C}_n^{(ii)\dagger} E_b \right\rangle - \left\langle \mathcal{C}_n E_a \mathcal{C}_n^\dagger E_b \right\rangle \prec \sum_{k \geq 1} \sum_{\{n_i\}_{i=1}^k} \left(\prod_{i=1}^k \Xi_{n_i}^{(d)} \right) \cdot (W\ell\eta)^{-2n} \cdot \mathbf{1} \left(\sum_{i=1}^k n_i = 2n + k \right), \quad (\text{A.25})$$

with $2 \leq n_i \leq 2n + 1$. By the induction hypothesis (A.7), we can bound:

$$\left(\prod_{i=1}^k \Xi_{n_i}^{(d)} \right) \cdot \mathbf{1} \left(\sum_{i=1}^k n_i = 2n + k \right) \prec 1 + \mathbf{1}_{n=2} \left(\Xi_{2n-1}^{(d)} \right)^2 + \Xi_{2n}^{(d)} + \Xi_{2n+1}^{(d)}.$$

Applying the Cauchy–Schwarz inequality yields:

$$\Xi_{2n-1}^{(d)} \prec \left(\Xi_{2n-2}^{(d)} \cdot \Xi_{2n}^{(d)} \right)^{1/2} \prec \left(\Xi_{2n}^{(d)} \right)^{1/2}.$$

Furthermore, splitting a $2n + 1$ -chain into two n -chains and one 1-chain, and using a triple-product Cauchy – Schwarz inequality (i.e., $\int f(x)g(y)h(x,y) \leq \|f\|_2 \|g\|_2 \|h\|_2$), we get:

$$\Xi_{2n+1}^{(d)} \prec \Xi_{2n}^{(d)} \cdot (W\ell\eta)^{1/2}.$$

Putting all of these bounds into (A.25), we conclude the proof of (A.12).

Proof of (A.13). Similar to the expansion in (A.25), we estimate the difference:

$$\begin{aligned} & \left(\mathcal{C}_n^{(ii)} E_a \mathcal{C}_n^{(ii)\dagger} \right)_{jj} - \left(\mathcal{C}_n E_a \mathcal{C}_n^\dagger \right)_{jj} \\ & \prec \sum_{k \geq 0} \sum_{\{n_i\}_{i=1}^k} \sum_{l_1, l_2} \left(\prod_{i=1}^k \Xi_{n_i}^{(d)} \right) \left(\prod_{j=1}^2 \Xi_{l_j}^{(o)} \right) \cdot (W\ell\eta)^{-2n} \cdot \mathbf{1} \left(l_1 + l_2 + \sum_{i=1}^k n_i = 2n + k + 1 \right), \end{aligned} \quad (\text{A.26})$$

where the index ranges satisfy:

$$2 \leq n_i \leq 2n, \quad 1 \leq l_1 \leq l_2 \leq 2n.$$

Under the assumptions on $\Xi_{n_i}^{(d)}$, it suffices to bound the product of off-diagonal terms. That is, we only need to show that:

$$l_1 + l_2 \leq 2n + 1 \implies \Xi_{l_1}^{(o)} \cdot \Xi_{l_2}^{(o)} \prec \left(\left(\Xi_n^{(o)} \right)^2 + 1 \right) \cdot (W\ell\eta)^{1/2}. \quad (\text{A.27})$$

By the induction hypothesis, we have

$$\Xi_{l_j}^{(o)} \prec \begin{cases} 1, & l_j \leq n - 1, \\ \Xi_n^{(o)}, & l_j = n, \\ (W\ell\eta)^{1/2}, & n < l_j \leq 2n, \end{cases}$$

where for $l_j > n$, the bound follows from applying the Cauchy–Schwarz inequality. These bounds clearly imply (A.27), and hence we obtain (A.13). This completes the proof of Lemma A.2. $\square$

B Proof of the properties of the propagator

Proof of Lemma 2.14. Since $S^{(B)}$ is a symmetric circulant matrix on $\mathbb{Z}_L$, the symmetry, translation invariance, and commutativity properties are immediate. Moreover, $\|S^{(B)}\| = 1$, so for every $|\xi| < 1$,

$$(1 - \xi S^{(B)})^{-1} = \sum_{n=0}^{\infty} \xi^n (S^{(B)})^n$$

in operator norm.

We prove the entrywise estimates by Fourier analysis. Put

$$\mathbb{T}_L := \frac{2\pi}{L} \mathbb{Z}_L.$$

The Fourier symbol of $S^{(B)}$ is

$$\widehat{S}(p) = \frac{1 + 2 \cos p}{3}, \quad p \in \mathbb{T}_L.$$

Consequently, with $u = x - y$,

$$(\Theta_\xi^{(B)})_{xy} = K_{\xi,L}(u) := \frac{1}{L} \sum_{p \in \mathbb{T}_L} \frac{\exp(ipu)}{1 - \xi \widehat{S}(p)}. \quad (\text{B.1})$$

Define the infinite-volume kernel

$$K_{\xi,\infty}(u) := \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\exp(ipu)}{1 - \xi \widehat{S}(p)} dp, \quad u \in \mathbb{Z}. \quad (\text{B.2})$$

By comparing Fourier coefficients (or by the Poisson summation formula), we have

$$(\Theta_\xi^{(B)})_{xy} = K_{\xi,L}(u) = \sum_{n \in \mathbb{Z}} K_{\xi,\infty}(u + nL).$$

Therefore, it remains to estimate the kernel $K_{\xi,\infty}$.

Denote by $\kappa := |1 - \xi|^{1/2}$, $\widehat{\ell} := \min(\kappa^{-1}, L)$, $|p|_* := \text{dist}(p, 2\pi\mathbb{Z})$. Define the notation

$$D_\xi(p) := 1 - \xi \widehat{S}(p) = 1 - \xi + \xi q(p), \quad q(p) := 1 - \widehat{S}(p) = \frac{2}{3}(1 - \cos p).$$

Clearly, $q(p) \asymp |p|_*^2$. The following elementary inequality can be easily verified:

$$|1 - \xi \widehat{S}(p)| \asymp \kappa^2 + q(p) \asymp \kappa^2 + |p|_*^2. \quad (\text{B.3})$$

For real p and $|\eta| \leq 1$,

$$|q(p + i\eta) - q(p)| \leq C(|\eta||p|_* + |\eta|^2).$$

Hence for $c_0 > 0$ sufficiently small and $|\eta| \leq c_0 \kappa$, we have

$$|D_\xi(p + i\eta)| \geq c(\kappa^2 + |p|_*^2). \quad (\text{B.4})$$

Thus by shifting the contour in (B.2) by $ic_0 \kappa \text{sgn}(u)$, we have

$$|K_{\xi,\infty}(u)| \leq C \exp(-c\kappa|u|) \int_{-\pi}^{\pi} \frac{dp}{\kappa^2 + |p|_*^2} \leq \frac{C}{\kappa} \exp(-c\kappa|u|). \quad (\text{B.5})$$

We now prove an exponential-decay bound for $K_{\xi,L}(u)$. Suppose first that $\kappa L \geq 1$, so that $\widehat{\ell} = \kappa^{-1}$. Using (B.5) and $\kappa L \geq 1$, we obtain

$$|K_{\xi,L}(u)| \leq \frac{C}{\kappa} \exp(-c\kappa|u|) = \frac{C \exp(-c|u|/\widehat{\ell})}{\kappa^2 \widehat{\ell}}.$$

Suppose next that $\kappa L < 1$, so that $\widehat{\ell} = L$. Separating the zero Fourier mode in (B.1) and using (B.3),

$$|K_{\xi,L}(u)| \leq \frac{1}{\kappa^2 L} + \frac{C}{L} \sum_{p \in \mathbb{T}_L \setminus \{0\}} \frac{1}{|p|_*^2} \leq \frac{1}{\kappa^2 L} + CL \leq \frac{C}{\kappa^2 L}.$$

Since $\|u\| \leq L/2$, the factor $\exp(-c\|u\|/L)$ is bounded below by a positive constant. This proves (2.52) in both regimes.

It remains to prove the two difference estimates. The zero Fourier mode is constant in u and hence cancels from both differences. Choose a smooth dyadic partition of unity $\{\chi_r\}_r$ on $[-\pi, \pi] \setminus \{0\}$, indexed by

$$r \in \{2^j L^{-1} : j \geq 0\}, \quad L^{-1} \lesssim r \lesssim 1,$$

such that χ_r is supported where $|p|_* \asymp r$ and

$$|\partial_p^j \chi_r(p)| \leq C_j r^{-j}, \quad j \geq 0.$$

For $p \neq 0$, the cutoffs satisfy $\sum_r \chi_r(p) = 1$. Define

$$K_r(u) := \frac{1}{L} \sum_{p \in \mathbb{T}_L} \frac{\chi_r(p) \exp(ipu)}{D_\xi(p)}.$$

Thus the nonzero-mode part of $K_{\xi,L}$ is $\sum_r K_r$. On the support of χ_r , (B.3) and the explicit formula for q imply

$$\left| \partial_p^j \left(\frac{\chi_r(p)}{D_\xi(p)} \right) \right| \leq \frac{C_j r^{-j}}{\kappa^2 + r^2}, \quad j \geq 0.$$

The support of χ_r contains $O(Lr)$ Fourier modes. Hence the trivial estimate gives, for $m = 0, 1, 2$,

$$|\partial_u^m K_r(u)| \leq C \frac{r^{m+1}}{\kappa^2 + r^2}.$$

For $u \neq 0$, repeated discrete summation by parts in p , using the preceding derivative bounds, gains a factor $(r\|u\|)^{-1}$ at each step. Combining this with the trivial estimate, for every fixed $M \geq 0$, we obtain

$$|\partial_u^m K_r(u)| \leq C_M \frac{r^{m+1}}{\kappa^2 + r^2} (1 + r\|u\|)^{-M}, \quad m = 0, 1, 2. \quad (\text{B.6})$$

The fundamental theorem of calculus and (B.6) yield

$$|K_{\xi,L}(u) - K_{\xi,L}(u-1)| \leq C \sum_r \frac{r^2}{\kappa^2 + r^2} \leq C \log(2+L).$$

The logarithm is harmless under $\prec$, and $\kappa\hat{\ell} \leq 1$. Therefore

$$|K_{\xi,L}(u) - K_{\xi,L}(u-1)| \prec 1 \leq \frac{1}{\kappa\hat{\ell}},$$

which is (2.53).

For the second difference, suppose first that $|u| \geq 2$. Applying the fundamental theorem of calculus twice and using (B.6),

$$\begin{aligned} |2K_{\xi,L}(u) - K_{\xi,L}(u-1) - K_{\xi,L}(u+1)| &\leq C \sum_r \frac{r^3}{\kappa^2 + r^2} (1 + r|u|)^{-M} \\ &\leq C \sum_r r (1 + r|u|)^{-M} \leq \frac{C}{|u| + 1}. \end{aligned}$$

If $|u| \leq 1$, the Fourier formula, $2 - 2\cos p = 3q(p)$, and (B.3) give directly

$$|2K_{\xi,L}(u) - K_{\xi,L}(u-1) - K_{\xi,L}(u+1)| \leq \frac{C}{L} \sum_{p \in \mathbb{T}_L} \frac{q(p)}{\kappa^2 + q(p)} \leq C.$$

This proves (2.54) and completes the proof. $\square$

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